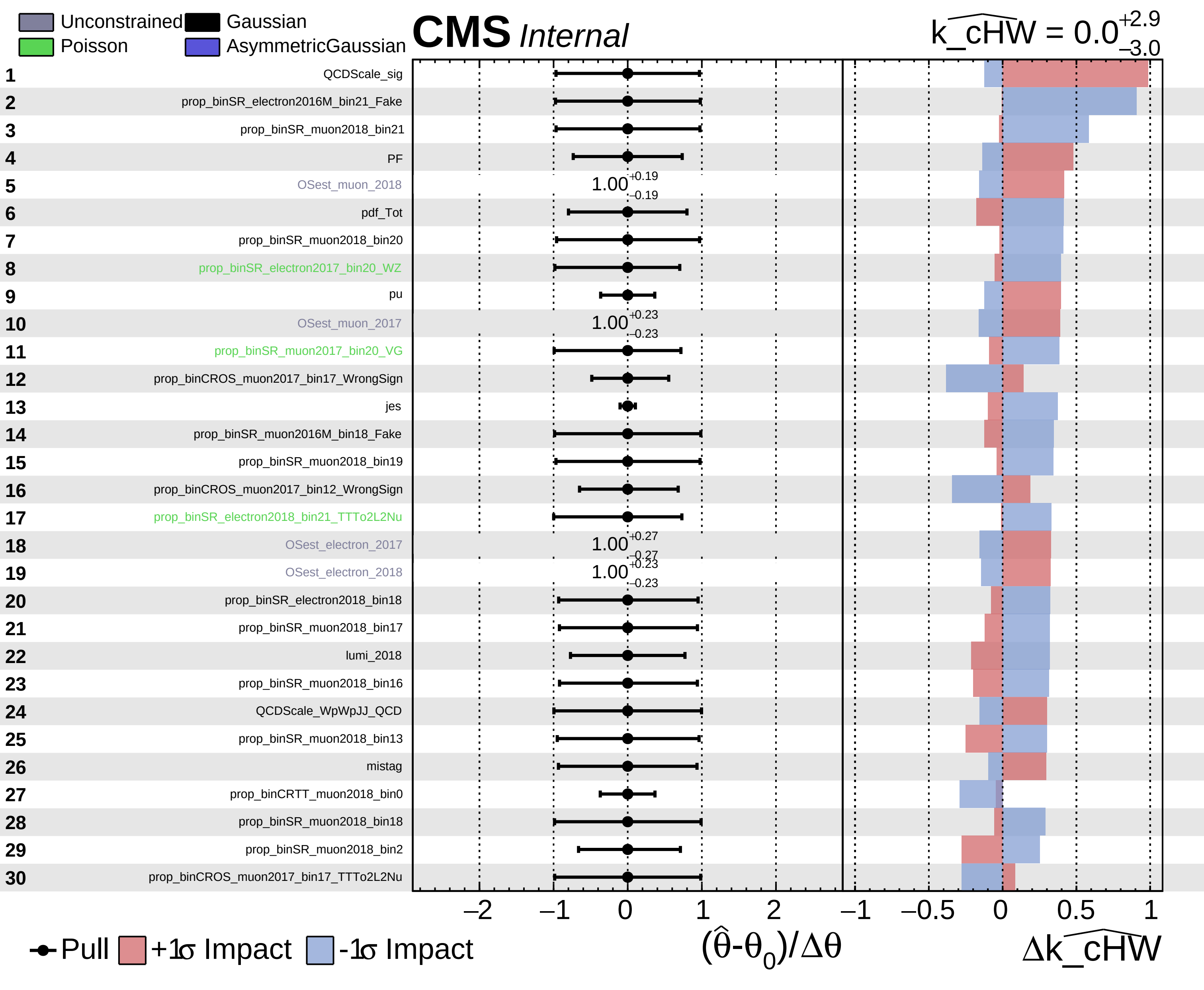
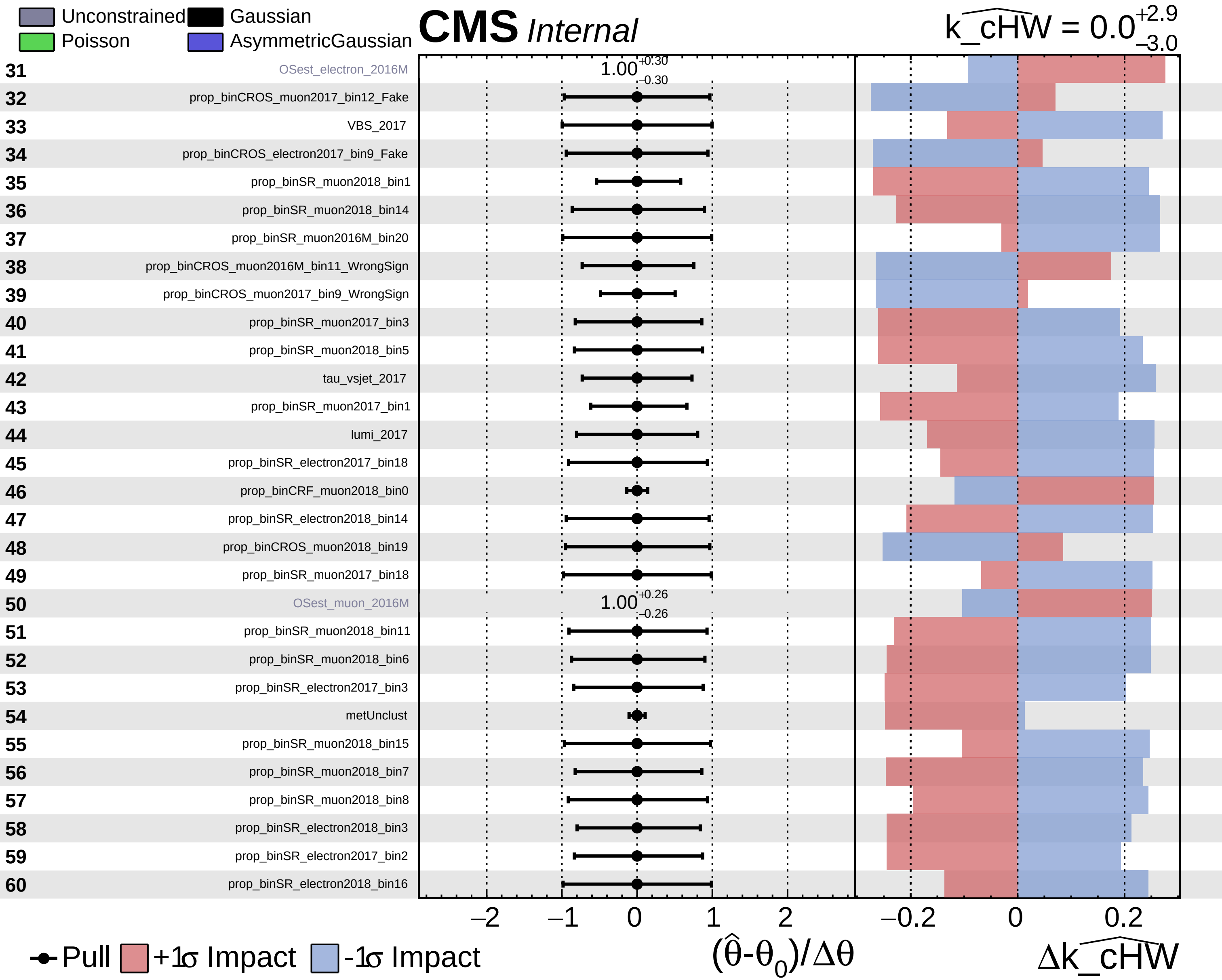
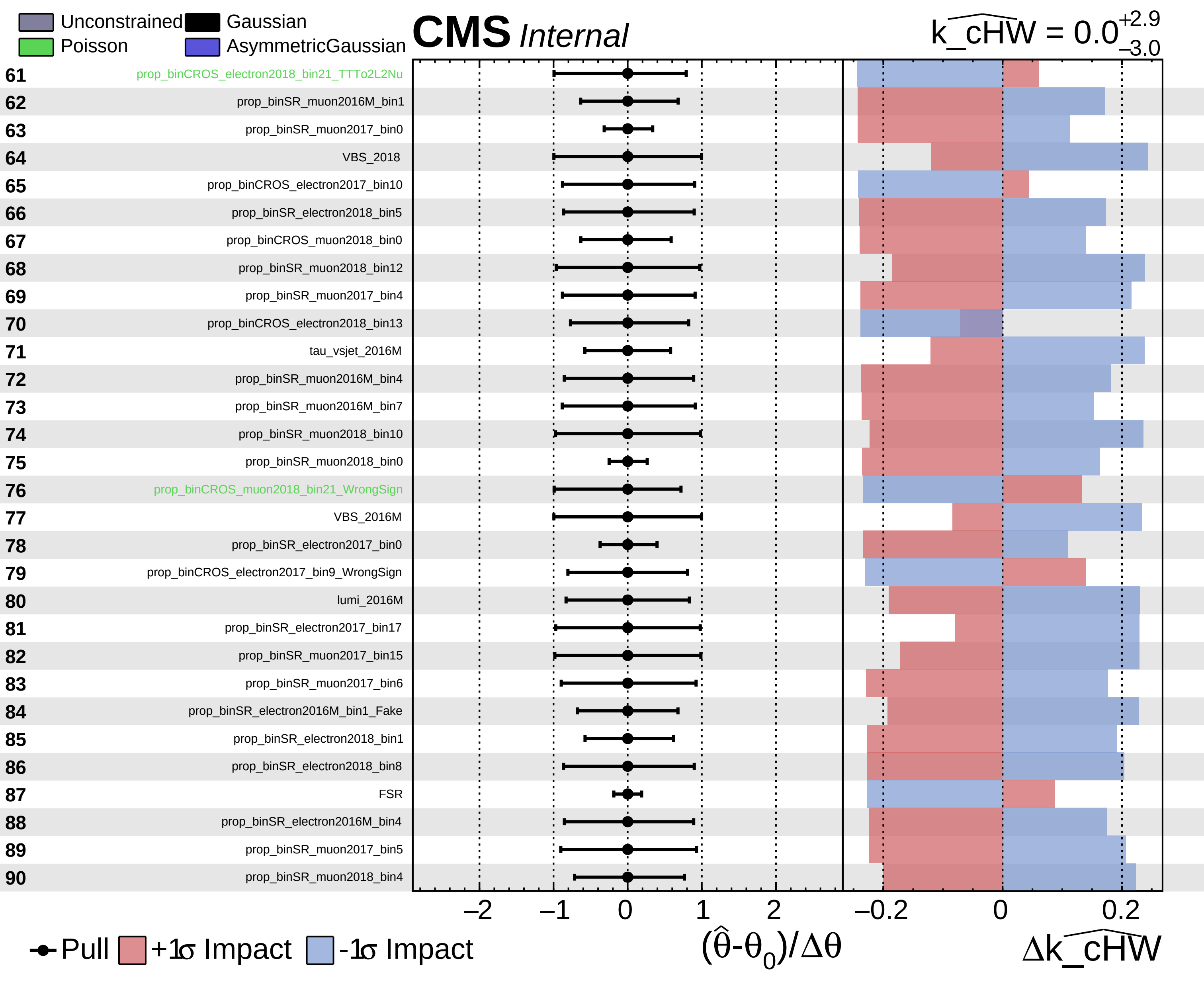
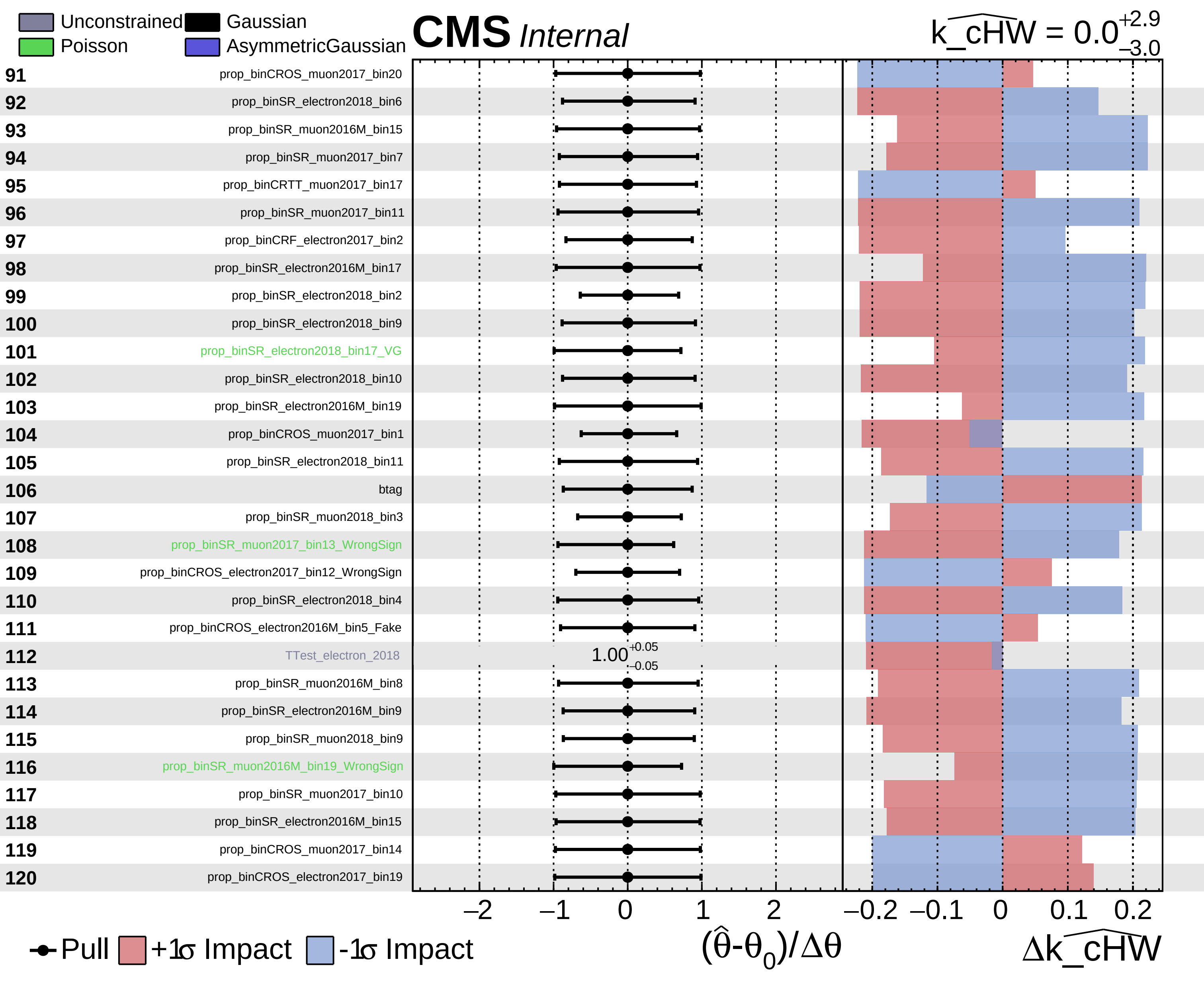


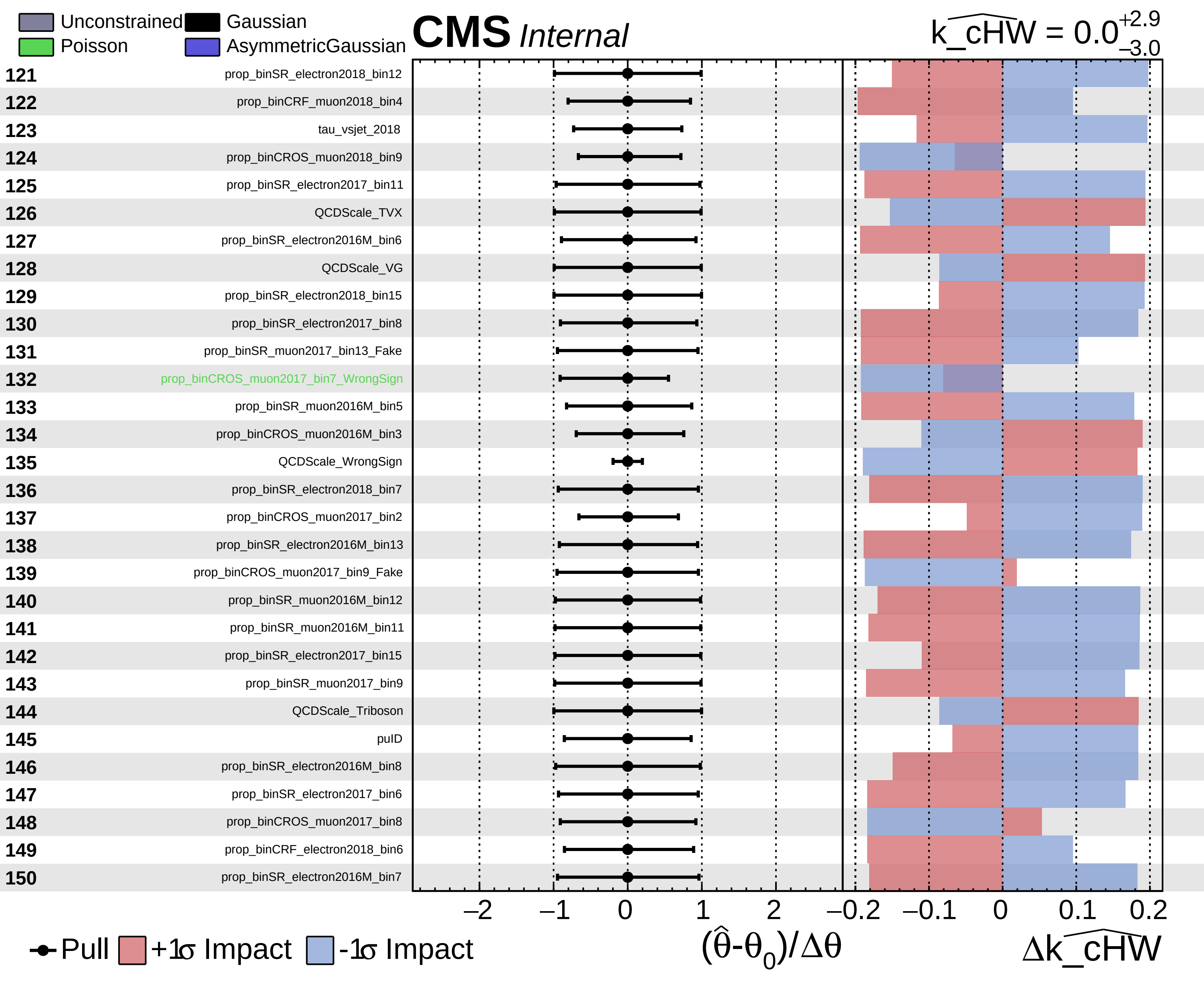
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
1 QCDScale_sig
2 prop_binSR_electron2016M_bin21_Fake
3 prop_binSR_muon2018_bin21
4 PF
5 OSest_muon_2018
6 pdf_Tot
7 prop_binSR_muon2018_bin20
8 prop_binSR_electron2017_bin20_WZ
9 pu
10 OSest_muon_2017
11 prop_binSR_muon2017_bin20_VG
12 prop_binCROS_muon2017_bin17_WrongSign
13 jes
14 prop_binSR_muon2016M_bin18_Fake
15 prop_binSR_muon2018_bin19
16 prop_binCROS_muon2017_bin12_WrongSign
17 prop_binSR_electron2018_bin21_TTTo2L2Nu
18 OSest_electron_2017
19 OSest_electron_2018
20 prop_binSR_electron2018_bin18
21 prop_binSR_muon2018_bin17
22 lumi_2018
23 prop_binSR_muon2018_bin16
24 QCDScale_WpWpJJ_QCD
25 prop_binSR_muon2018_bin13
26 mistag
27 prop_binCRTT_muon2018_bin0
28 prop_binSR_muon2018_bin18
29 prop_binSR_muon2018_bin2
30 prop_binCROS_muon2017_bin17_TTTo2L2Nu
1.00 +0.19 −0.19
1.00 +0.23 −0.23
1.00 +0.27 −0.27
1.00 +0.23 −0.23
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW



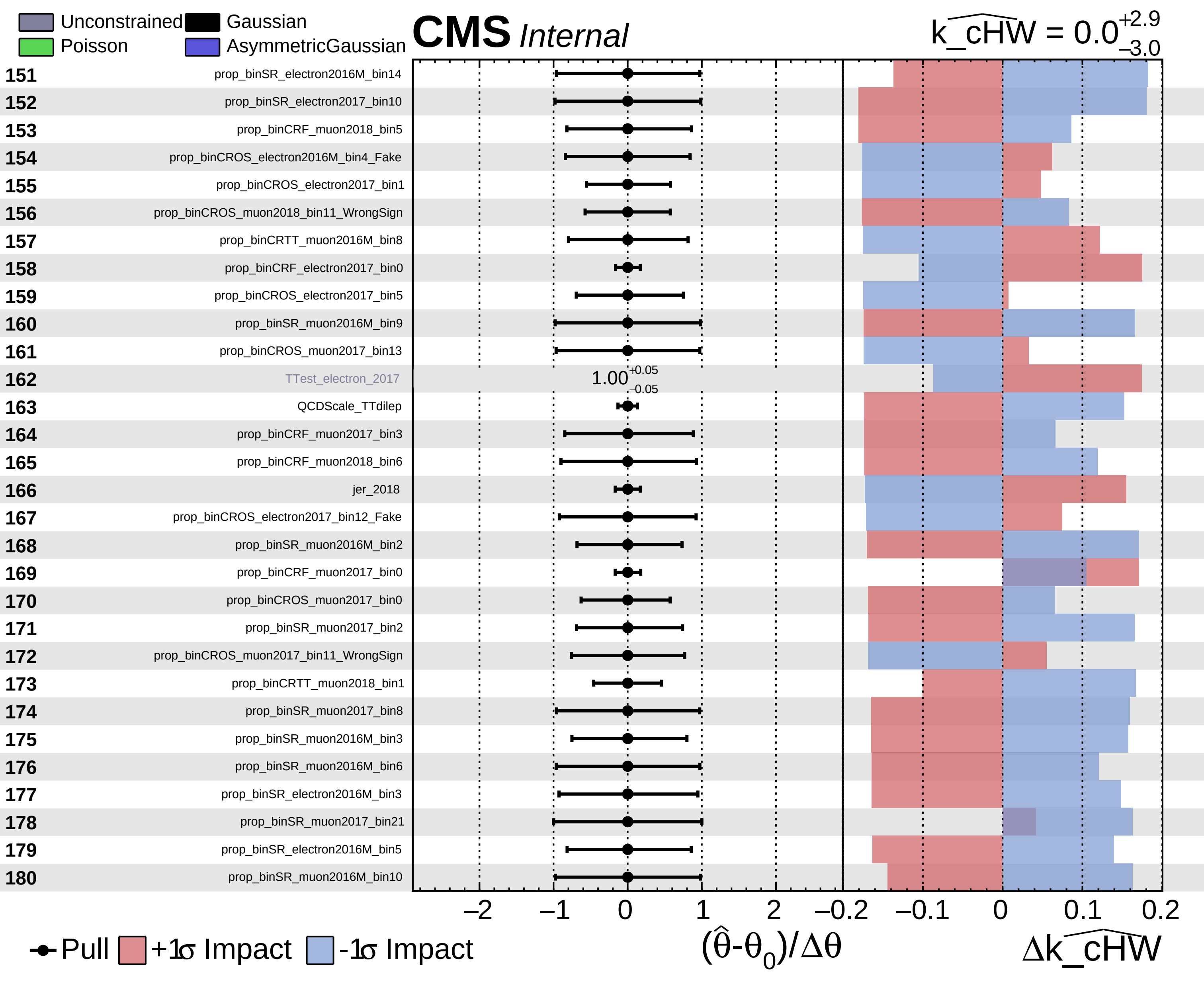
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
31 OSest_electron_2016M
$1.00^{+0.30}_{-0.30}$
32 prop_binCROS_muon2017_bin12_Fake
33 VBS_2017
34 prop_binCROS_electron2017_bin9_Fake
35 prop_binSR_muon2018_bin1
36 prop_binSR_muon2018_bin14
37 prop_binSR_muon2016M_bin20
38 prop_binCROS_muon2016M_bin11_WrongSign
39 prop_binCROS_muon2017_bin9_WrongSign
40 prop_binSR_muon2017_bin3
41 prop_binSR_muon2018_bin5
42 tau_vsjet_2017
43 prop_binSR_muon2017_bin1
44 lumi_2017
45 prop_binSR_electron2017_bin18
46 prop_binCRF_muon2018_bin0
47 prop_binSR_electron2018_bin14
48 prop_binCROS_muon2018_bin19
49 prop_binSR_muon2017_bin18
50 OSest_muon_2016M
$1.00^{+0.26}_{-0.26}$
51 prop_binSR_muon2018_bin11
52 prop_binSR_muon2018_bin6
53 prop_binSR_electron2017_bin3
54 metUnclust
55 prop_binSR_muon2018_bin15
56 prop_binSR_muon2018_bin7
57 prop_binSR_muon2018_bin8
58 prop_binSR_electron2018_bin3
59 prop_binSR_electron2017_bin2
60 prop_binSR_electron2018_bin16
−2
−1
0
1
2
−0.2
0
0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
61 prop_binCROS_electron2018_bin21_TTTo2L2Nu
62 prop_binSR_muon2016M_bin1
63 prop_binSR_muon2017_bin0
64 VBS_2018
65 prop_binCROS_electron2017_bin10
66 prop_binSR_electron2018_bin5
67 prop_binCROS_muon2018_bin0
68 prop_binSR_muon2018_bin12
69 prop_binSR_muon2017_bin4
70 prop_binCROS_electron2018_bin13
71 tau_vsjet_2016M
72 prop_binSR_muon2016M_bin4
73 prop_binSR_muon2016M_bin7
74 prop_binSR_muon2018_bin10
75 prop_binSR_muon2018_bin0
76 prop_binCROS_muon2018_bin21_WrongSign
77 VBS_2016M
78 prop_binSR_electron2017_bin0
79 prop_binCROS_electron2017_bin9_WrongSign
80 lumi_2016M
81 prop_binSR_electron2017_bin17
82 prop_binSR_muon2017_bin15
83 prop_binSR_muon2017_bin6
84 prop_binSR_electron2016M_bin1_Fake
85 prop_binSR_electron2018_bin1
86 prop_binSR_electron2018_bin8
87 FSR
88 prop_binSR_electron2016M_bin4
89 prop_binSR_muon2017_bin5
90 prop_binSR_muon2018_bin4
−2
−1
0
1
2
−0.2
0
0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta \widehat{k\_cHW}$

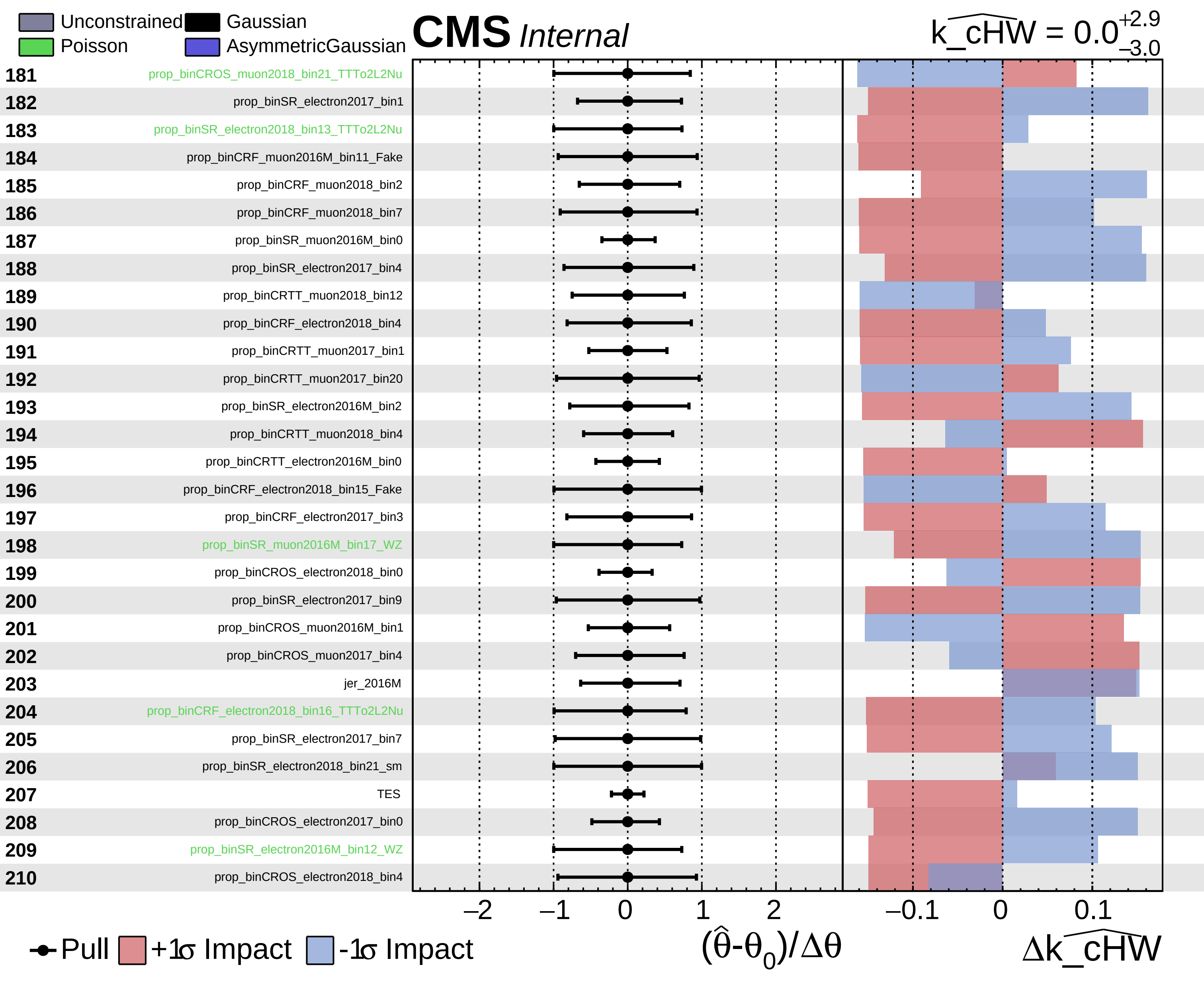

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
91 prop_binCROS_muon2017_bin20
92 prop_binSR_electron2018_bin6
93 prop_binSR_muon2016M_bin15
94 prop_binSR_muon2017_bin7
95 prop_binCRTT_muon2017_bin17
96 prop_binSR_muon2017_bin11
97 prop_binCRF_electron2017_bin2
98 prop_binSR_electron2016M_bin17
99 prop_binSR_electron2018_bin2
100 prop_binSR_electron2018_bin9
101 prop_binSR_electron2018_bin17_VG
102 prop_binSR_electron2018_bin10
103 prop_binSR_electron2016M_bin19
104 prop_binCROS_muon2017_bin1
105 prop_binSR_electron2018_bin11
106 btag
107 prop_binSR_muon2018_bin3
108 prop_binSR_muon2017_bin13_WrongSign
109 prop_binCROS_electron2017_bin12_WrongSign
110 prop_binSR_electron2018_bin4
111 prop_binCROS_electron2016M_bin5_Fake
112 TTest_electron_2018
1.00 +0.05 −0.05
113 prop_binSR_muon2016M_bin8
114 prop_binSR_electron2016M_bin9
115 prop_binSR_muon2018_bin9
116 prop_binSR_muon2016M_bin19_WrongSign
117 prop_binSR_muon2017_bin10
118 prop_binSR_electron2016M_bin15
119 prop_binCROS_muon2017_bin14
120 prop_binCROS_electron2017_bin19
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

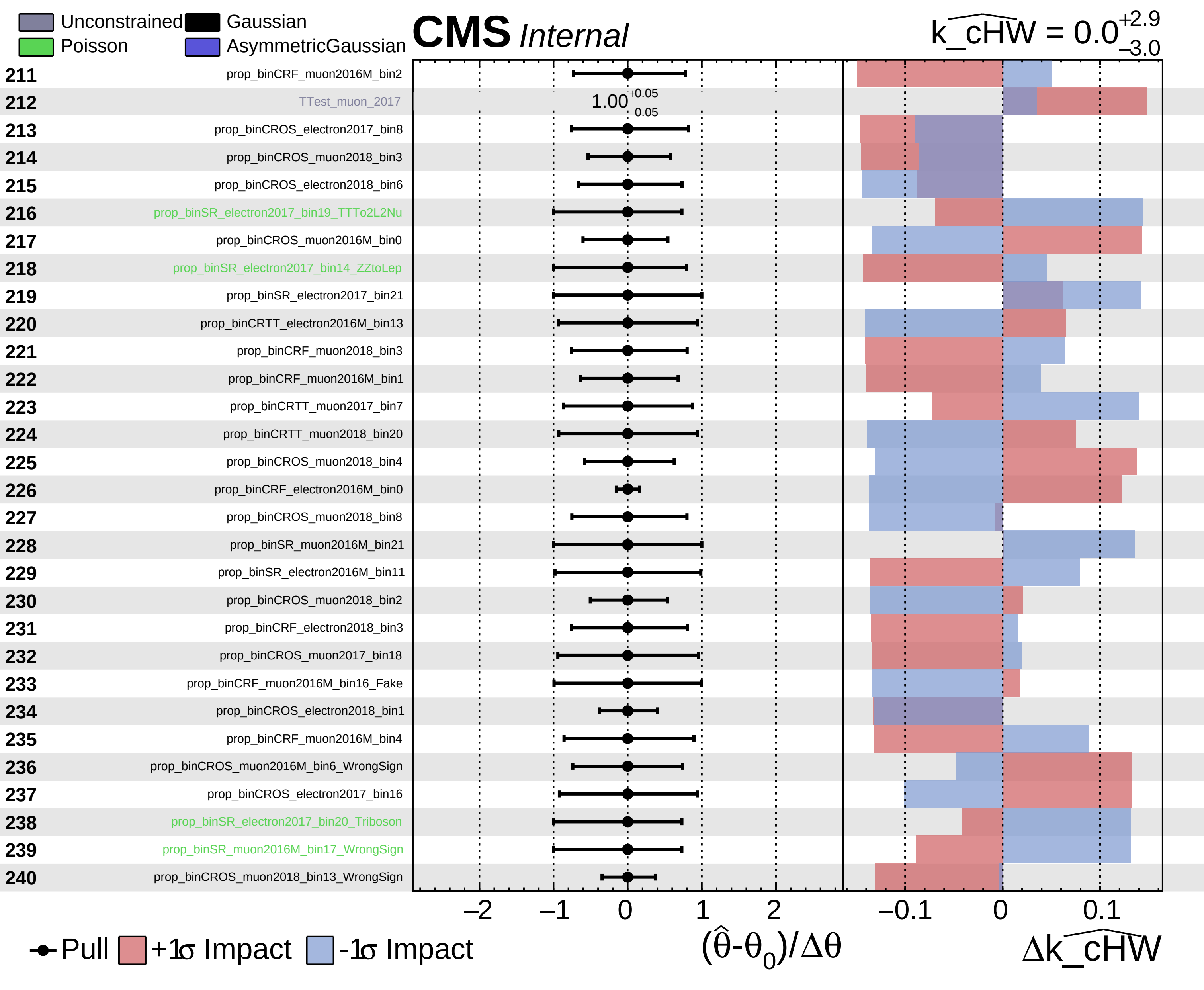


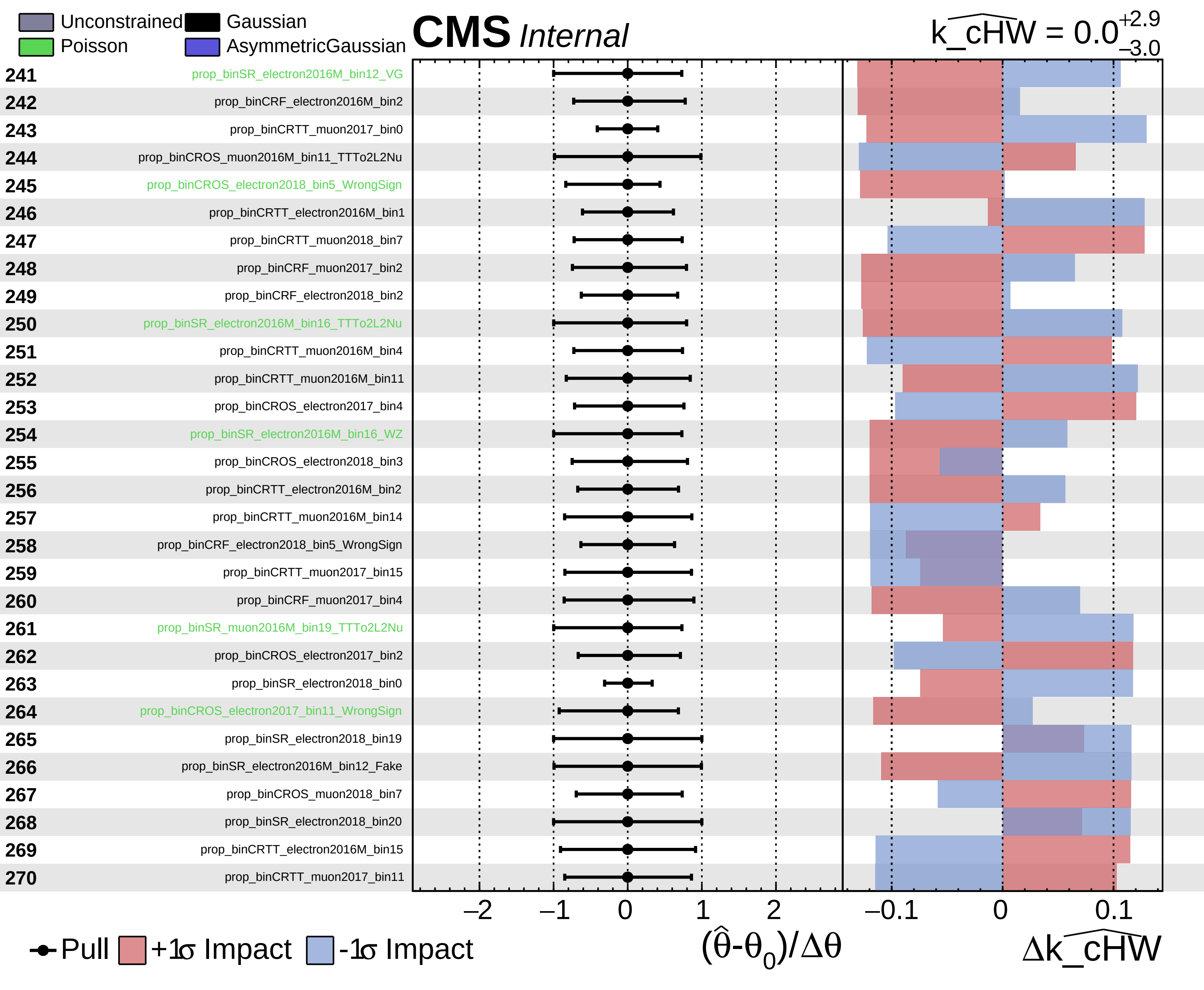
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
121 prop_binSR_electron2018_bin12
122 prop_binCRF_muon2018_bin4
123 tau_vsjet_2018
124 prop_binCROS_muon2018_bin9
125 prop_binSR_electron2017_bin11
126 QCDScale_TVX
127 prop_binSR_electron2016M_bin6
128 QCDScale_VG
129 prop_binSR_electron2018_bin15
130 prop_binSR_electron2017_bin8
131 prop_binSR_muon2017_bin13_Fake
132 prop_binCROS_muon2017_bin7_WrongSign
133 prop_binSR_muon2016M_bin5
134 prop_binCROS_muon2016M_bin3
135 QCDScale_WrongSign
136 prop_binSR_electron2018_bin7
137 prop_binCROS_muon2017_bin2
138 prop_binSR_electron2016M_bin13
139 prop_binCROS_muon2017_bin9_Fake
140 prop_binSR_muon2016M_bin12
141 prop_binSR_muon2016M_bin11
142 prop_binSR_electron2017_bin15
143 prop_binSR_muon2017_bin9
144 QCDScale_Triboson
145 puID
146 prop_binSR_electron2016M_bin8
147 prop_binSR_electron2017_bin6
148 prop_binCROS_muon2017_bin8
149 prop_binCRF_electron2018_bin6
150 prop_binSR_electron2016M_bin7
−2
−1
0
1
2
−0.2
−0.1
0
0.1
0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

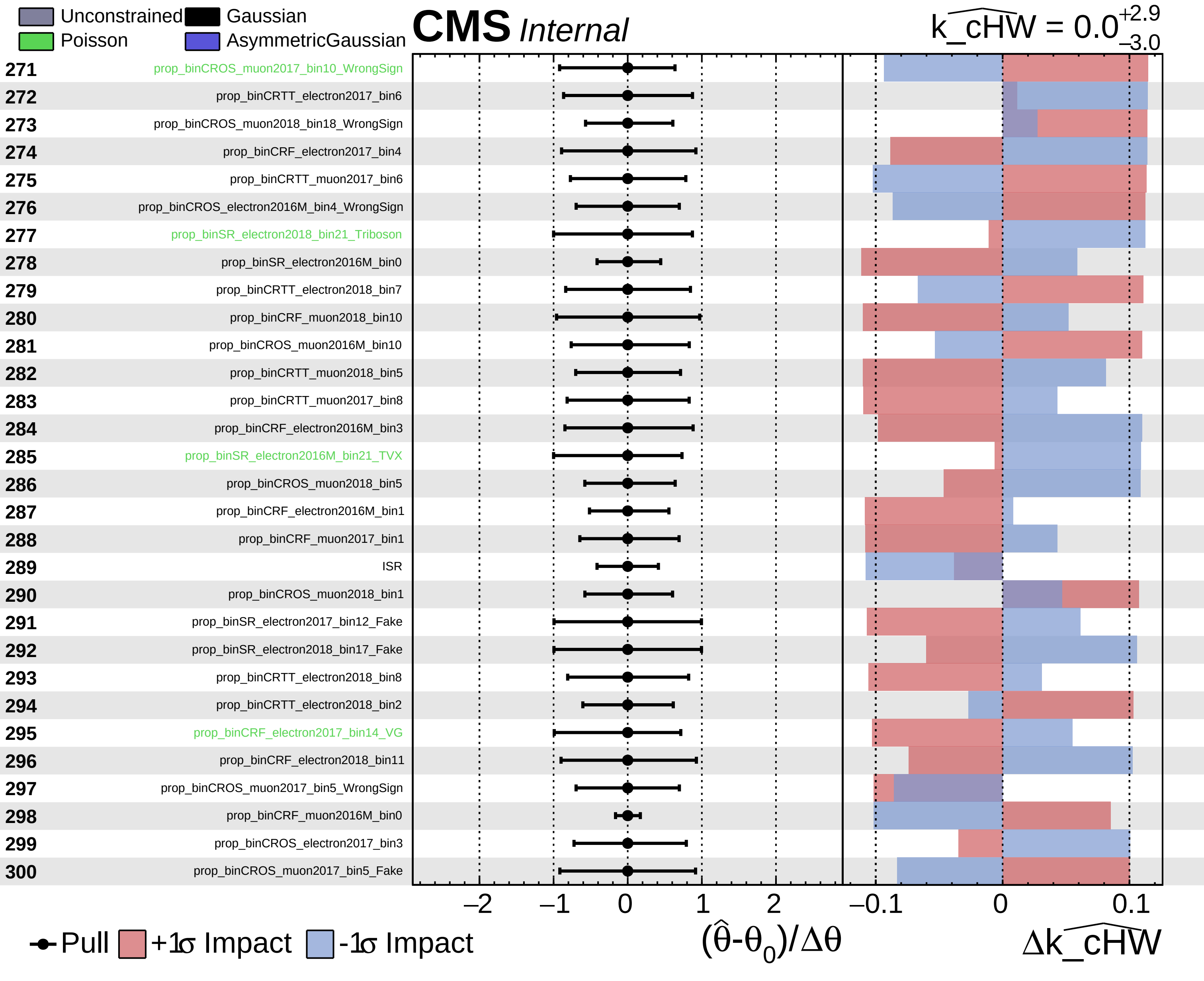


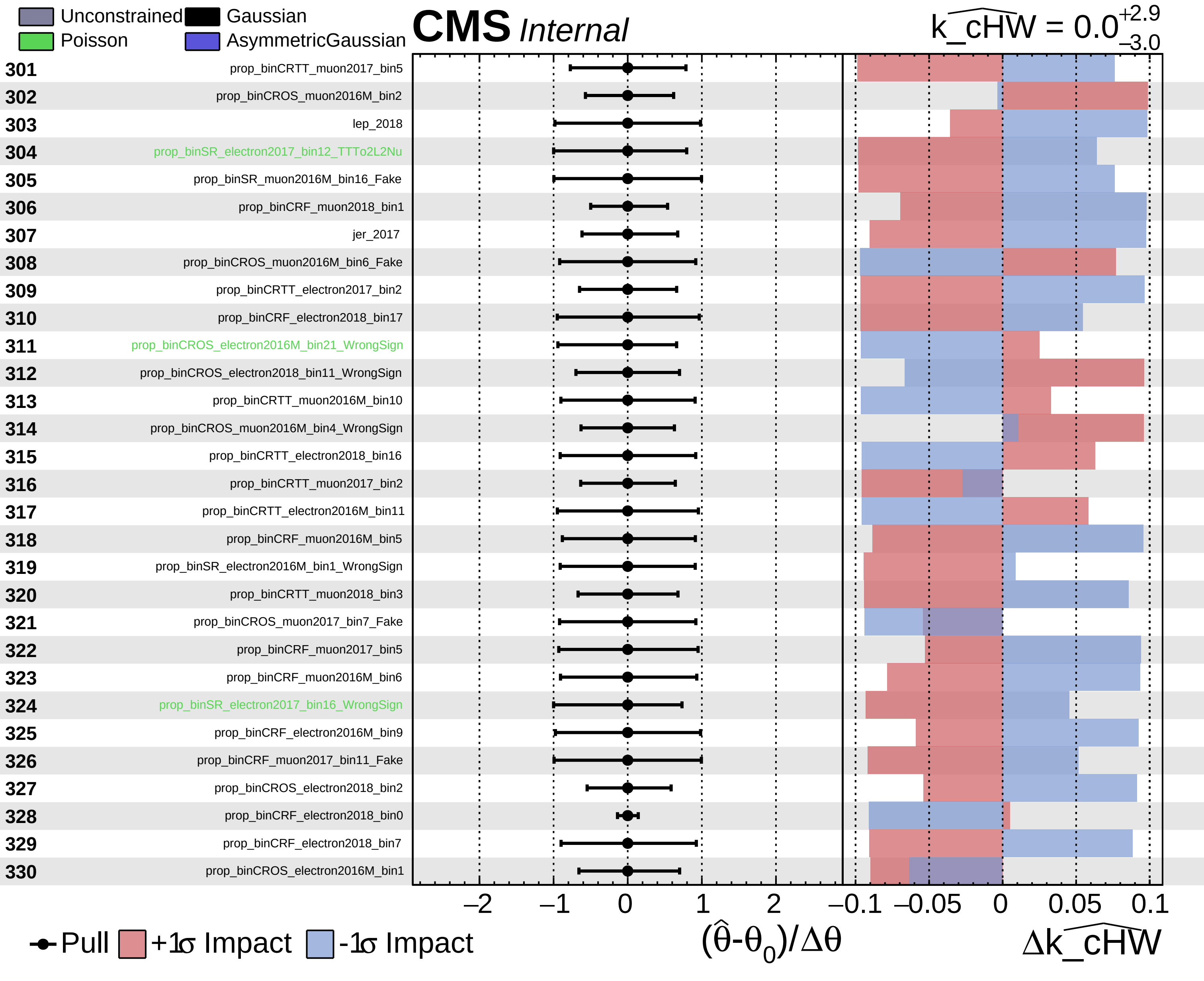
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
151 prop_binSR_electron2016M_bin14
152 prop_binSR_electron2017_bin10
153 prop_binCRF_muon2018_bin5
154 prop_binCROS_electron2016M_bin4_Fake
155 prop_binCROS_electron2017_bin1
156 prop_binCROS_muon2018_bin11_WrongSign
157 prop_binCRTT_muon2016M_bin8
158 prop_binCRF_electron2017_bin0
159 prop_binCROS_electron2017_bin5
160 prop_binSR_muon2016M_bin9
161 prop_binCROS_muon2017_bin13
162 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
163 QCDScale_TTdilep
164 prop_binCRF_muon2017_bin3
165 prop_binCRF_muon2018_bin6
166 jer_2018
167 prop_binCROS_electron2017_bin12_Fake
168 prop_binSR_muon2016M_bin2
169 prop_binCRF_muon2017_bin0
170 prop_binCROS_muon2017_bin0
171 prop_binSR_muon2017_bin2
172 prop_binCROS_muon2017_bin11_WrongSign
173 prop_binCRTT_muon2018_bin1
174 prop_binSR_muon2017_bin8
175 prop_binSR_muon2016M_bin3
176 prop_binSR_muon2016M_bin6
177 prop_binSR_electron2016M_bin3
178 prop_binSR_muon2017_bin21
179 prop_binSR_electron2016M_bin5
180 prop_binSR_muon2016M_bin10
−2
−1
0
1
2
−0.2
−0.1
0
0.1
0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
181 prop_binCROS_muon2018_bin21_TTTo2L2Nu
182 prop_binSR_electron2017_bin1
183 prop_binSR_electron2018_bin13_TTTo2L2Nu
184 prop_binCRF_muon2016M_bin11_Fake
185 prop_binCRF_muon2018_bin2
186 prop_binCRF_muon2018_bin7
187 prop_binSR_muon2016M_bin0
188 prop_binSR_electron2017_bin4
189 prop_binCRTT_muon2018_bin12
190 prop_binCRF_electron2018_bin4
191 prop_binCRTT_muon2017_bin1
192 prop_binCRTT_muon2017_bin20
193 prop_binSR_electron2016M_bin2
194 prop_binCRTT_muon2018_bin4
195 prop_binCRTT_electron2016M_bin0
196 prop_binCRF_electron2018_bin15_Fake
197 prop_binCRF_electron2017_bin3
198 prop_binSR_muon2016M_bin17_WZ
199 prop_binCROS_electron2018_bin0
200 prop_binSR_electron2017_bin9
201 prop_binCROS_muon2016M_bin1
202 prop_binCROS_muon2017_bin4
203 jer_2016M
204 prop_binCRF_electron2018_bin16_TTTo2L2Nu
205 prop_binSR_electron2017_bin7
206 prop_binSR_electron2018_bin21_sm
207 TES
208 prop_binCROS_electron2017_bin0
209 prop_binSR_electron2016M_bin12_WZ
210 prop_binCROS_electron2018_bin4
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW

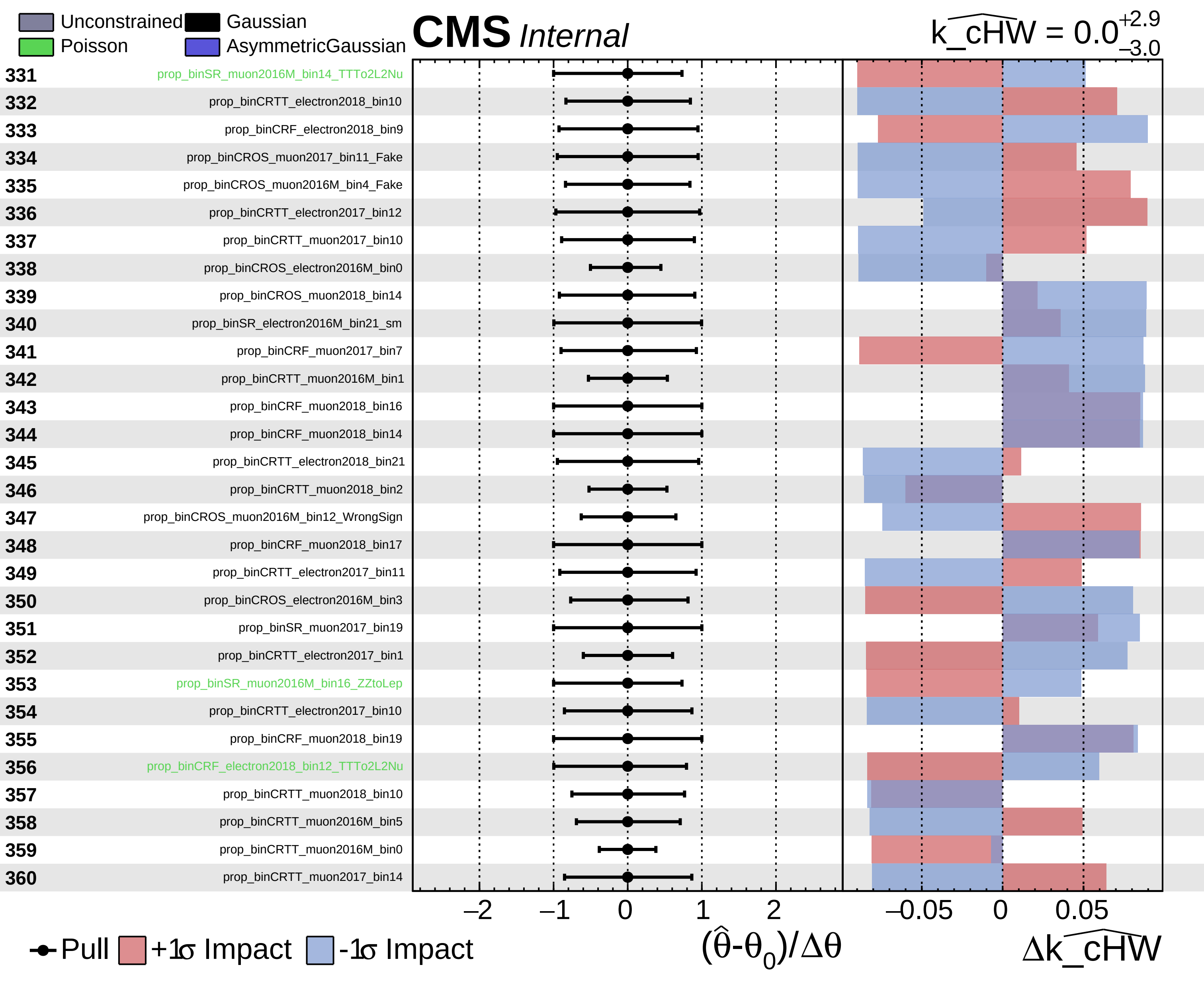

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
211 prop_binCRF_muon2016M_bin2
212 TTest_muon_2017 1.00 +0.05 −0.05
213 prop_binCROS_electron2017_bin8
214 prop_binCROS_muon2018_bin3
215 prop_binCROS_electron2018_bin6
216 prop_binSR_electron2017_bin19_TTTo2L2Nu
217 prop_binCROS_muon2016M_bin0
218 prop_binSR_electron2017_bin14_ZZtoLep
219 prop_binSR_electron2017_bin21
220 prop_binCRTT_electron2016M_bin13
221 prop_binCRF_muon2018_bin3
222 prop_binCRF_muon2016M_bin1
223 prop_binCRTT_muon2017_bin7
224 prop_binCRTT_muon2018_bin20
225 prop_binCROS_muon2018_bin4
226 prop_binCRF_electron2016M_bin0
227 prop_binCROS_muon2018_bin8
228 prop_binSR_muon2016M_bin21
229 prop_binSR_electron2016M_bin11
230 prop_binCROS_muon2018_bin2
231 prop_binCRF_electron2018_bin3
232 prop_binCROS_muon2017_bin18
233 prop_binCRF_muon2016M_bin16_Fake
234 prop_binCROS_electron2018_bin1
235 prop_binCRF_muon2016M_bin4
236 prop_binCROS_muon2016M_bin6_WrongSign
237 prop_binCROS_electron2017_bin16
238 prop_binSR_electron2017_bin20_Triboson
239 prop_binSR_muon2016M_bin17_WrongSign
240 prop_binCROS_muon2018_bin13_WrongSign
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
241 prop_binSR_electron2016M_bin12_VG
242 prop_binCRF_electron2016M_bin2
243 prop_binCRTT_muon2017_bin0
244 prop_binCROS_muon2016M_bin11_TTTo2L2Nu
245 prop_binCROS_electron2018_bin5_WrongSign
246 prop_binCRTT_electron2016M_bin1
247 prop_binCRTT_muon2018_bin7
248 prop_binCRF_muon2017_bin2
249 prop_binCRF_electron2018_bin2
250 prop_binSR_electron2016M_bin16_TTTo2L2Nu
251 prop_binCRTT_muon2016M_bin4
252 prop_binCRTT_muon2016M_bin11
253 prop_binCROS_electron2017_bin4
254 prop_binSR_electron2016M_bin16_WZ
255 prop_binCROS_electron2018_bin3
256 prop_binCRTT_electron2016M_bin2
257 prop_binCRTT_muon2016M_bin14
258 prop_binCRF_electron2018_bin5_WrongSign
259 prop_binCRTT_muon2017_bin15
260 prop_binCRF_muon2017_bin4
261 prop_binSR_muon2016M_bin19_TTTo2L2Nu
262 prop_binCROS_electron2017_bin2
263 prop_binSR_electron2018_bin0
264 prop_binCROS_electron2017_bin11_WrongSign
265 prop_binSR_electron2018_bin19
266 prop_binSR_electron2016M_bin12_Fake
267 prop_binCROS_muon2018_bin7
268 prop_binSR_electron2018_bin20
269 prop_binCRTT_electron2016M_bin15
270 prop_binCRTT_muon2017_bin11
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

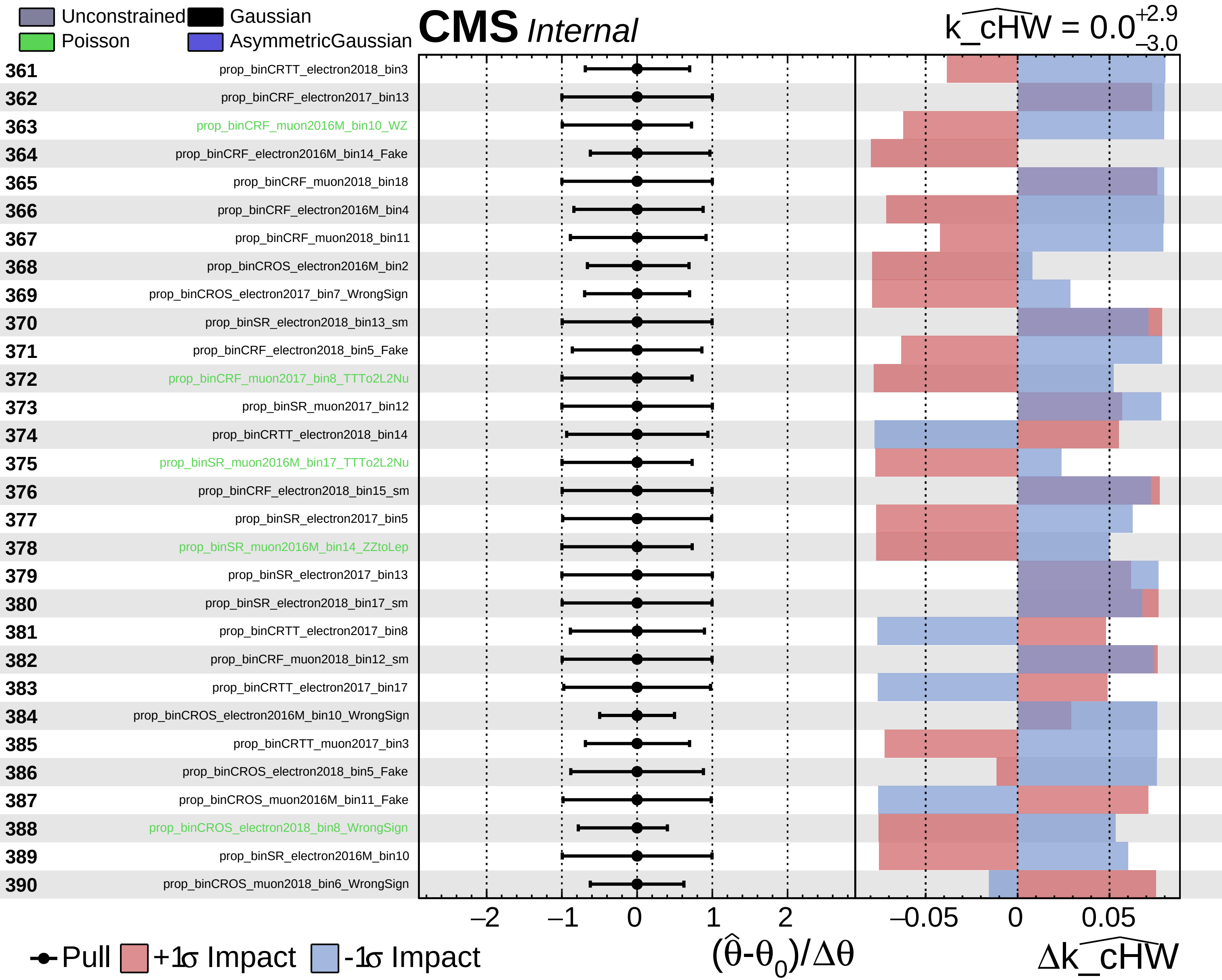

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
271 prop_binCROS_muon2017_bin10_WrongSign
272 prop_binCRTT_electron2017_bin6
273 prop_binCROS_muon2018_bin18_WrongSign
274 prop_binCRF_electron2017_bin4
275 prop_binCRTT_muon2017_bin6
276 prop_binCROS_electron2016M_bin4_WrongSign
277 prop_binSR_electron2018_bin21_Triboson
278 prop_binSR_electron2016M_bin0
279 prop_binCRTT_electron2018_bin7
280 prop_binCRF_muon2018_bin10
281 prop_binCROS_muon2016M_bin10
282 prop_binCRTT_muon2018_bin5
283 prop_binCRTT_muon2017_bin8
284 prop_binCRF_electron2016M_bin3
285 prop_binSR_electron2016M_bin21_TVX
286 prop_binCROS_muon2018_bin5
287 prop_binCRF_electron2016M_bin1
288 prop_binCRF_muon2017_bin1
289 ISR
290 prop_binCROS_muon2018_bin1
291 prop_binSR_electron2017_bin12_Fake
292 prop_binSR_electron2018_bin17_Fake
293 prop_binCRTT_electron2018_bin8
294 prop_binCRTT_electron2018_bin2
295 prop_binCRF_electron2017_bin14_VG
296 prop_binCRF_electron2018_bin11
297 prop_binCROS_muon2017_bin5_WrongSign
298 prop_binCRF_muon2016M_bin0
299 prop_binCROS_electron2017_bin3
300 prop_binCROS_muon2017_bin5_Fake
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



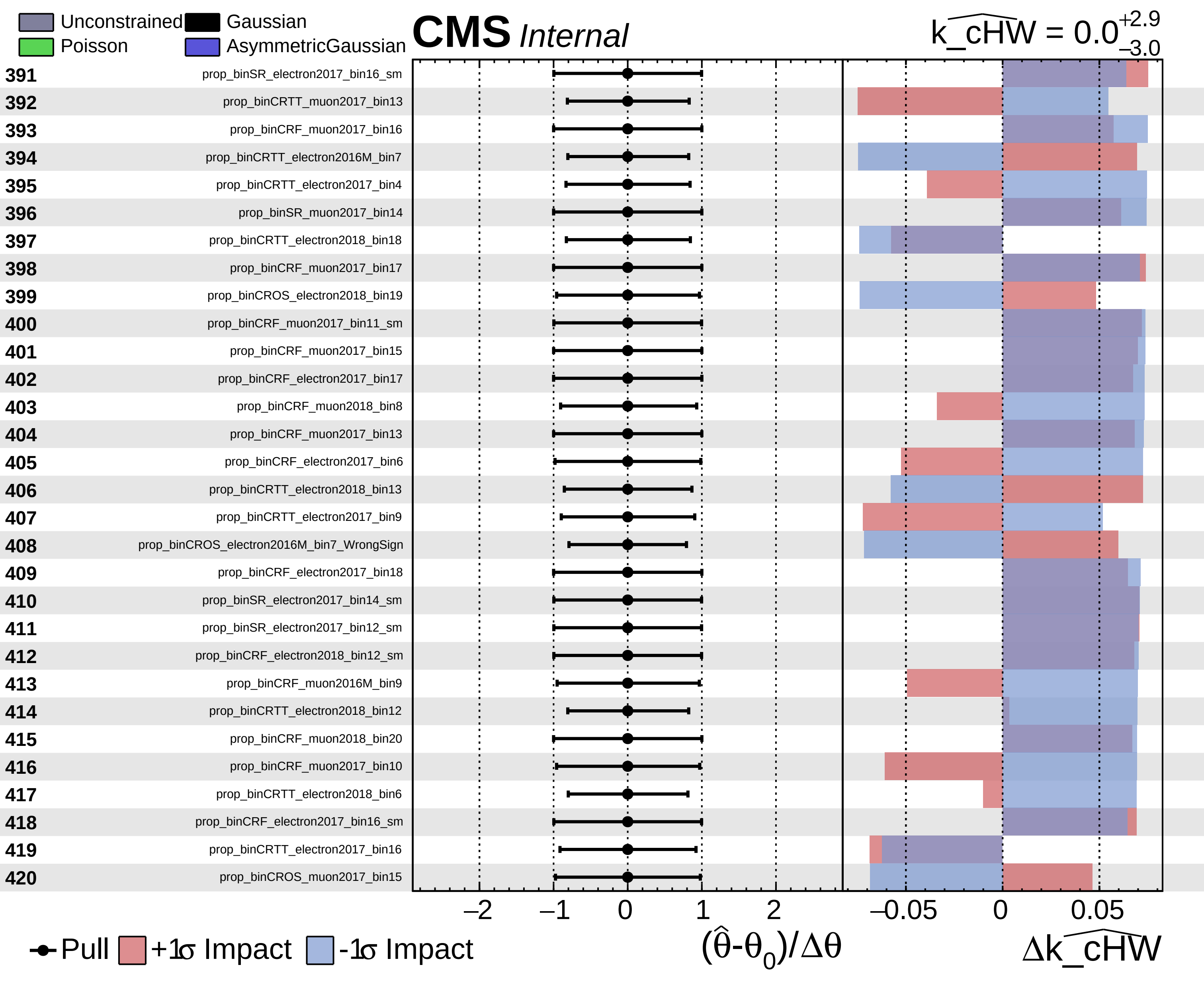
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
301 prop_binCRTT_muon2017_bin5
302 prop_binCROS_muon2016M_bin2
303 lep_2018
304 prop_binSR_electron2017_bin12_TTTo2L2Nu
305 prop_binSR_muon2016M_bin16_Fake
306 prop_binCRF_muon2018_bin1
307 jer_2017
308 prop_binCROS_muon2016M_bin6_Fake
309 prop_binCRTT_electron2017_bin2
310 prop_binCRF_electron2018_bin17
311 prop_binCROS_electron2016M_bin21_WrongSign
312 prop_binCROS_electron2018_bin11_WrongSign
313 prop_binCRTT_muon2016M_bin10
314 prop_binCROS_muon2016M_bin4_WrongSign
315 prop_binCRTT_electron2018_bin16
316 prop_binCRTT_muon2017_bin2
317 prop_binCRTT_electron2016M_bin11
318 prop_binCRF_muon2016M_bin5
319 prop_binSR_electron2016M_bin1_WrongSign
320 prop_binCRTT_muon2018_bin3
321 prop_binCROS_muon2017_bin7_Fake
322 prop_binCRF_muon2017_bin5
323 prop_binCRF_muon2016M_bin6
324 prop_binSR_electron2017_bin16_WrongSign
325 prop_binCRF_electron2016M_bin9
326 prop_binCRF_muon2017_bin11_Fake
327 prop_binCROS_electron2018_bin2
328 prop_binCRF_electron2018_bin0
329 prop_binCRF_electron2018_bin7
330 prop_binCROS_electron2016M_bin1
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



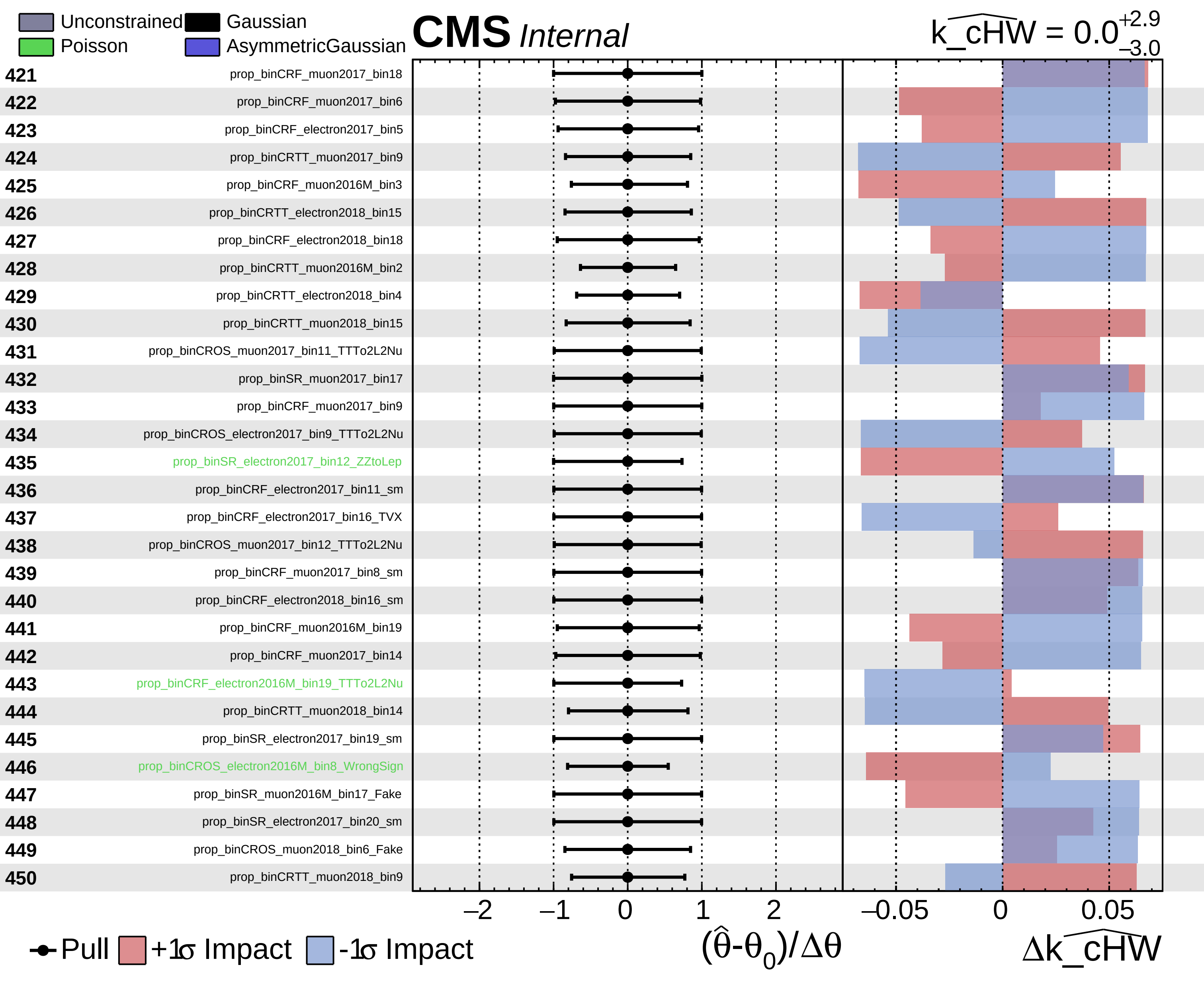
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
331 prop_binSR_muon2016M_bin14_TTTo2L2Nu
332 prop_binCRTT_electron2018_bin10
333 prop_binCRF_electron2018_bin9
334 prop_binCROS_muon2017_bin11_Fake
335 prop_binCROS_muon2016M_bin4_Fake
336 prop_binCRTT_electron2017_bin12
337 prop_binCRTT_muon2017_bin10
338 prop_binCROS_electron2016M_bin0
339 prop_binCROS_muon2018_bin14
340 prop_binSR_electron2016M_bin21_sm
341 prop_binCRF_muon2017_bin7
342 prop_binCRTT_muon2016M_bin1
343 prop_binCRF_muon2018_bin16
344 prop_binCRF_muon2018_bin14
345 prop_binCRTT_electron2018_bin21
346 prop_binCRTT_muon2018_bin2
347 prop_binCROS_muon2016M_bin12_WrongSign
348 prop_binCRF_muon2018_bin17
349 prop_binCRTT_electron2017_bin11
350 prop_binCROS_electron2016M_bin3
351 prop_binSR_muon2017_bin19
352 prop_binCRTT_electron2017_bin1
353 prop_binSR_muon2016M_bin16_ZZtoLep
354 prop_binCRTT_electron2017_bin10
355 prop_binCRF_muon2018_bin19
356 prop_binCRF_electron2018_bin12_TTTo2L2Nu
357 prop_binCRTT_muon2018_bin10
358 prop_binCRTT_muon2016M_bin5
359 prop_binCRTT_muon2016M_bin0
360 prop_binCRTT_muon2017_bin14
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



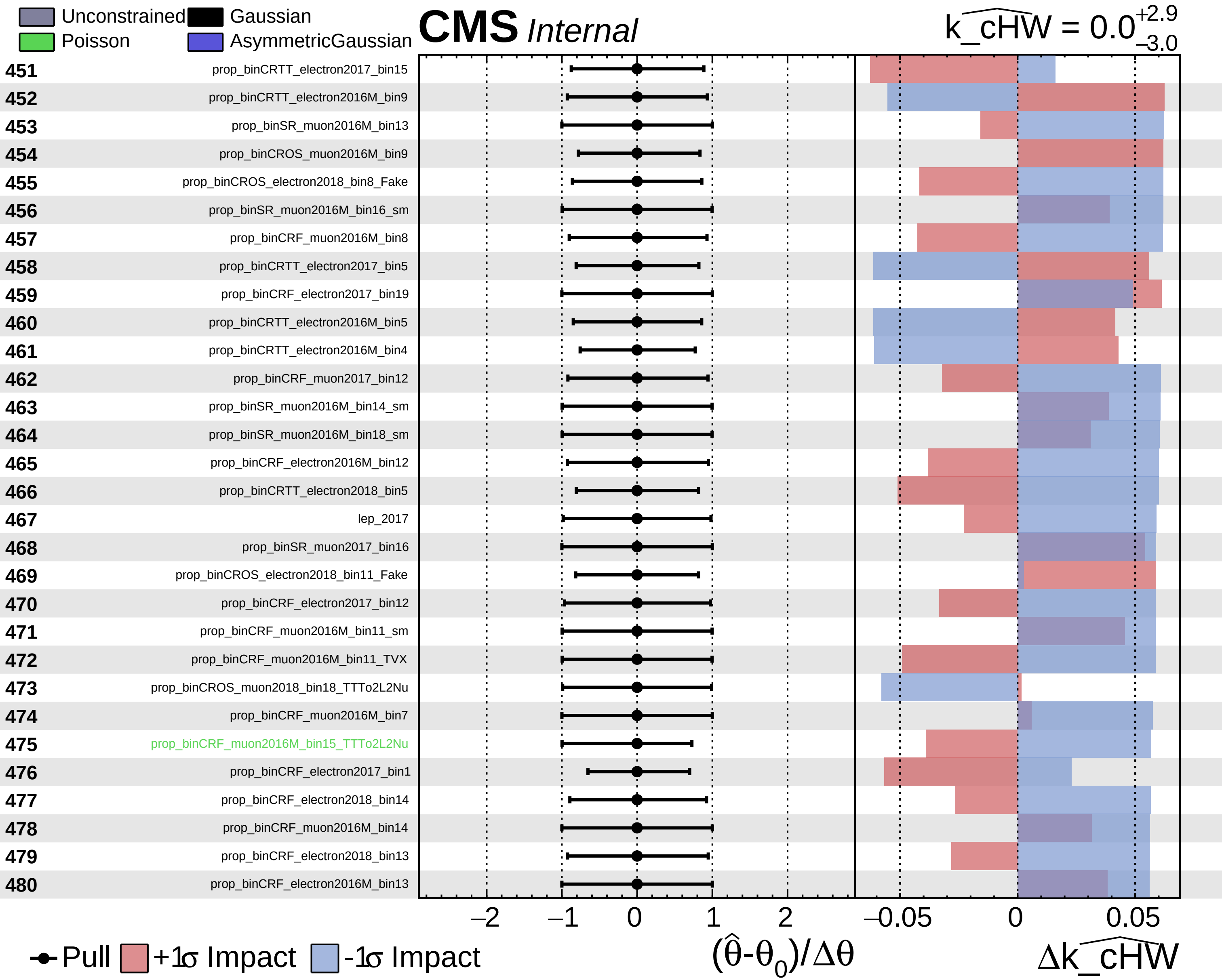
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = $0.0^{+2.9}_{-3.0}$
361 prop_binCRTT_electron2018_bin3
362 prop_binCRF_electron2017_bin13
363 prop_binCRF_muon2016M_bin10_WZ
364 prop_binCRF_electron2016M_bin14_Fake
365 prop_binCRF_muon2018_bin18
366 prop_binCRF_electron2016M_bin4
367 prop_binCRF_muon2018_bin11
368 prop_binCROS_electron2016M_bin2
369 prop_binCROS_electron2017_bin7_WrongSign
370 prop_binSR_electron2018_bin13_sm
371 prop_binCRF_electron2018_bin5_Fake
372 prop_binCRF_muon2017_bin8_TTTo2L2Nu
373 prop_binSR_muon2017_bin12
374 prop_binCRTT_electron2018_bin14
375 prop_binSR_muon2016M_bin17_TTTo2L2Nu
376 prop_binCRF_electron2018_bin15_sm
377 prop_binSR_electron2017_bin5
378 prop_binSR_muon2016M_bin14_ZZtoLep
379 prop_binSR_electron2017_bin13
380 prop_binSR_electron2018_bin17_sm
381 prop_binCRTT_electron2017_bin8
382 prop_binCRF_muon2018_bin12_sm
383 prop_binCRTT_electron2017_bin17
384 prop_binCROS_electron2016M_bin10_WrongSign
385 prop_binCRTT_muon2017_bin3
386 prop_binCROS_electron2018_bin5_Fake
387 prop_binCROS_muon2016M_bin11_Fake
388 prop_binCROS_electron2018_bin8_WrongSign
389 prop_binSR_electron2016M_bin10
390 prop_binCROS_muon2018_bin6_WrongSign
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW



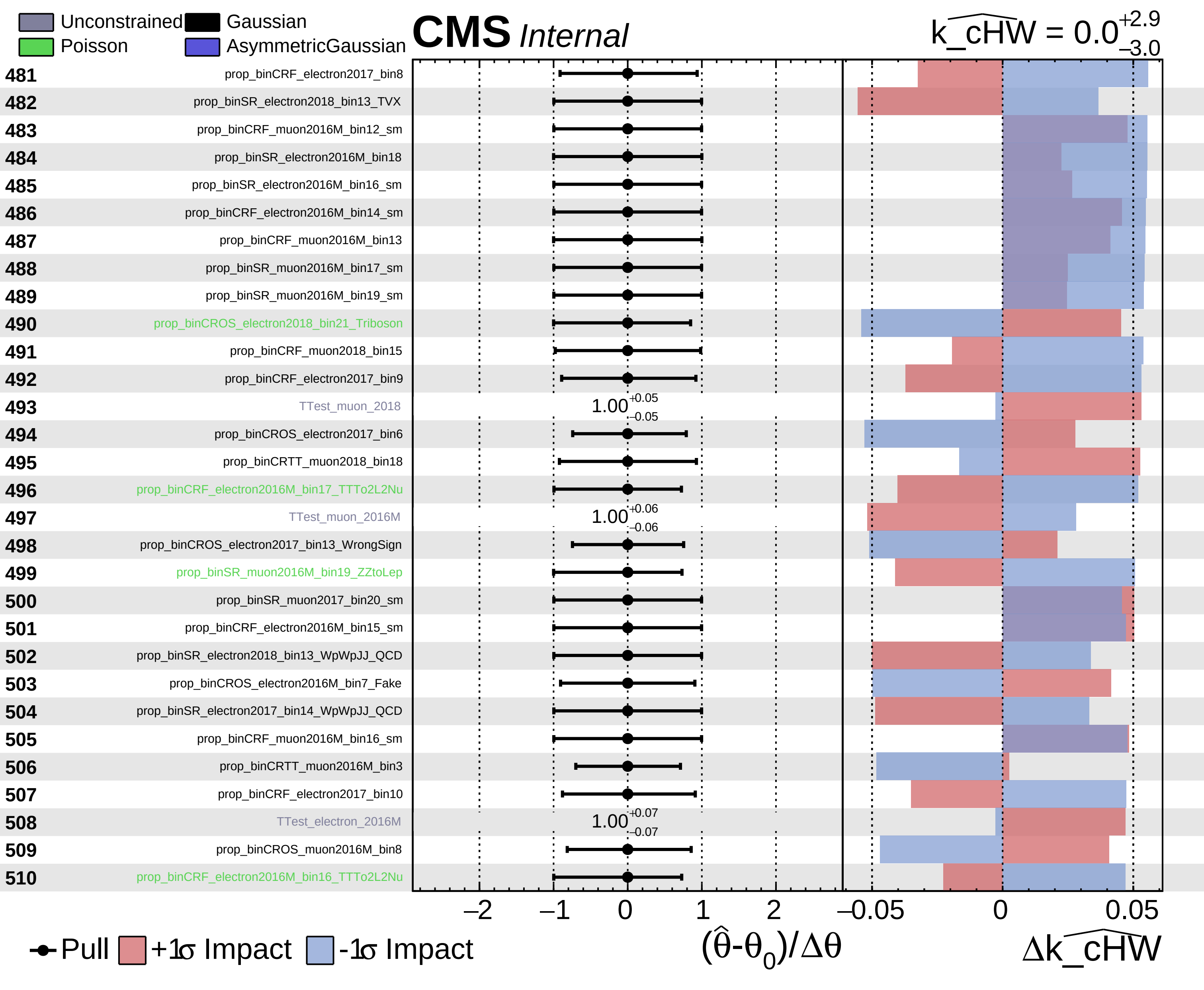
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
391 prop_binSR_electron2017_bin16_sm
392 prop_binCRTT_muon2017_bin13
393 prop_binCRF_muon2017_bin16
394 prop_binCRTT_electron2016M_bin7
395 prop_binCRTT_electron2017_bin4
396 prop_binSR_muon2017_bin14
397 prop_binCRTT_electron2018_bin18
398 prop_binCRF_muon2017_bin17
399 prop_binCROS_electron2018_bin19
400 prop_binCRF_muon2017_bin11_sm
401 prop_binCRF_muon2017_bin15
402 prop_binCRF_electron2017_bin17
403 prop_binCRF_muon2018_bin8
404 prop_binCRF_muon2017_bin13
405 prop_binCRF_electron2017_bin6
406 prop_binCRTT_electron2018_bin13
407 prop_binCRTT_electron2017_bin9
408 prop_binCROS_electron2016M_bin7_WrongSign
409 prop_binCRF_electron2017_bin18
410 prop_binSR_electron2017_bin14_sm
411 prop_binSR_electron2017_bin12_sm
412 prop_binCRF_electron2018_bin12_sm
413 prop_binCRF_muon2016M_bin9
414 prop_binCRTT_electron2018_bin12
415 prop_binCRF_muon2018_bin20
416 prop_binCRF_muon2017_bin10
417 prop_binCRTT_electron2018_bin6
418 prop_binCRF_electron2017_bin16_sm
419 prop_binCRTT_electron2017_bin16
420 prop_binCROS_muon2017_bin15
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW



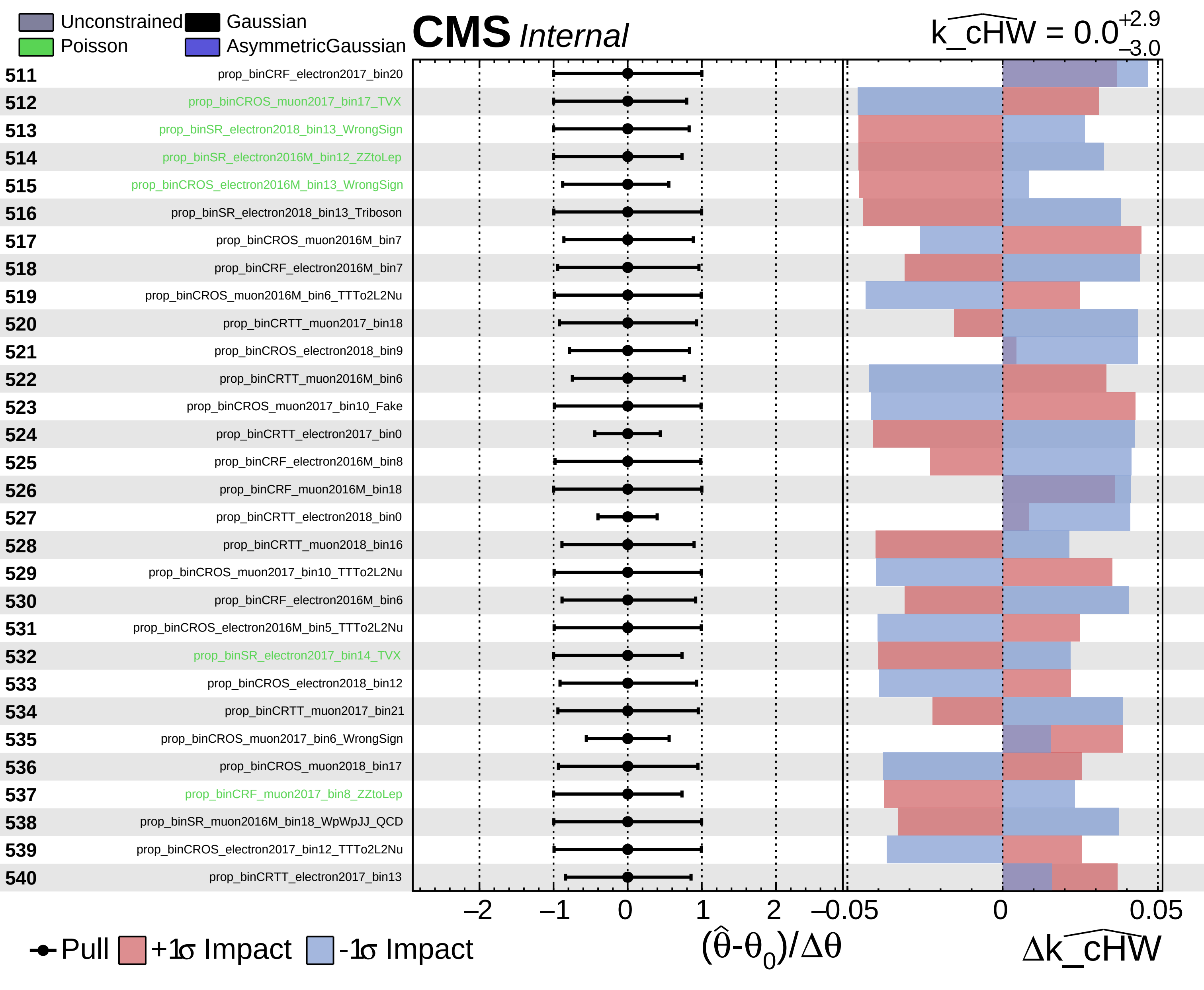
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
421 prop_binCRF_muon2017_bin18
422 prop_binCRF_muon2017_bin6
423 prop_binCRF_electron2017_bin5
424 prop_binCRTT_muon2017_bin9
425 prop_binCRF_muon2016M_bin3
426 prop_binCRTT_electron2018_bin15
427 prop_binCRF_electron2018_bin18
428 prop_binCRTT_muon2016M_bin2
429 prop_binCRTT_electron2018_bin4
430 prop_binCRTT_muon2018_bin15
431 prop_binCROS_muon2017_bin11_TTTo2L2Nu
432 prop_binSR_muon2017_bin17
433 prop_binCRF_muon2017_bin9
434 prop_binCROS_electron2017_bin9_TTTo2L2Nu
435 prop_binSR_electron2017_bin12_ZZtoLep
436 prop_binCRF_electron2017_bin11_sm
437 prop_binCRF_electron2017_bin16_TVX
438 prop_binCROS_muon2017_bin12_TTTo2L2Nu
439 prop_binCRF_muon2017_bin8_sm
440 prop_binCRF_electron2018_bin16_sm
441 prop_binCRF_muon2016M_bin19
442 prop_binCRF_muon2017_bin14
443 prop_binCRF_electron2016M_bin19_TTTo2L2Nu
444 prop_binCRTT_muon2018_bin14
445 prop_binSR_electron2017_bin19_sm
446 prop_binCROS_electron2016M_bin8_WrongSign
447 prop_binSR_muon2016M_bin17_Fake
448 prop_binSR_electron2017_bin20_sm
449 prop_binCROS_muon2018_bin6_Fake
450 prop_binCRTT_muon2018_bin9
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW



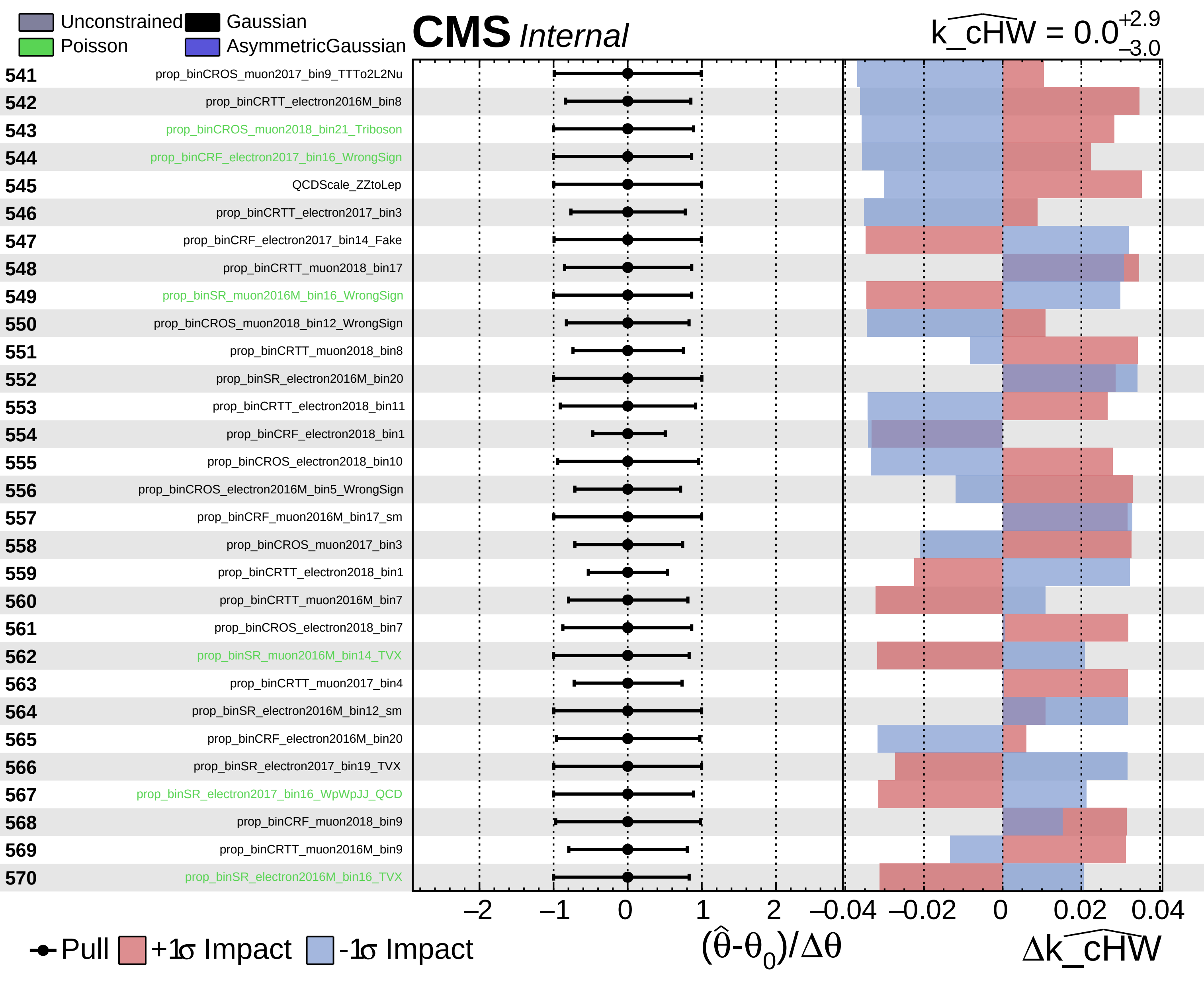
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
451 prop_binCRTT_electron2017_bin15
452 prop_binCRTT_electron2016M_bin9
453 prop_binSR_muon2016M_bin13
454 prop_binCROS_muon2016M_bin9
455 prop_binCROS_electron2018_bin8_Fake
456 prop_binSR_muon2016M_bin16_sm
457 prop_binCRF_muon2016M_bin8
458 prop_binCRTT_electron2017_bin5
459 prop_binCRF_electron2017_bin19
460 prop_binCRTT_electron2016M_bin5
461 prop_binCRTT_electron2016M_bin4
462 prop_binCRF_muon2017_bin12
463 prop_binSR_muon2016M_bin14_sm
464 prop_binSR_muon2016M_bin18_sm
465 prop_binCRF_electron2016M_bin12
466 prop_binCRTT_electron2018_bin5
467 lep_2017
468 prop_binSR_muon2017_bin16
469 prop_binCROS_electron2018_bin11_Fake
470 prop_binCRF_electron2017_bin12
471 prop_binCRF_muon2016M_bin11_sm
472 prop_binCRF_muon2016M_bin11_TVX
473 prop_binCROS_muon2018_bin18_TTTo2L2Nu
474 prop_binCRF_muon2016M_bin7
475 prop_binCRF_muon2016M_bin15_TTTo2L2Nu
476 prop_binCRF_electron2017_bin1
477 prop_binCRF_electron2018_bin14
478 prop_binCRF_muon2016M_bin14
479 prop_binCRF_electron2018_bin13
480 prop_binCRF_electron2016M_bin13
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW

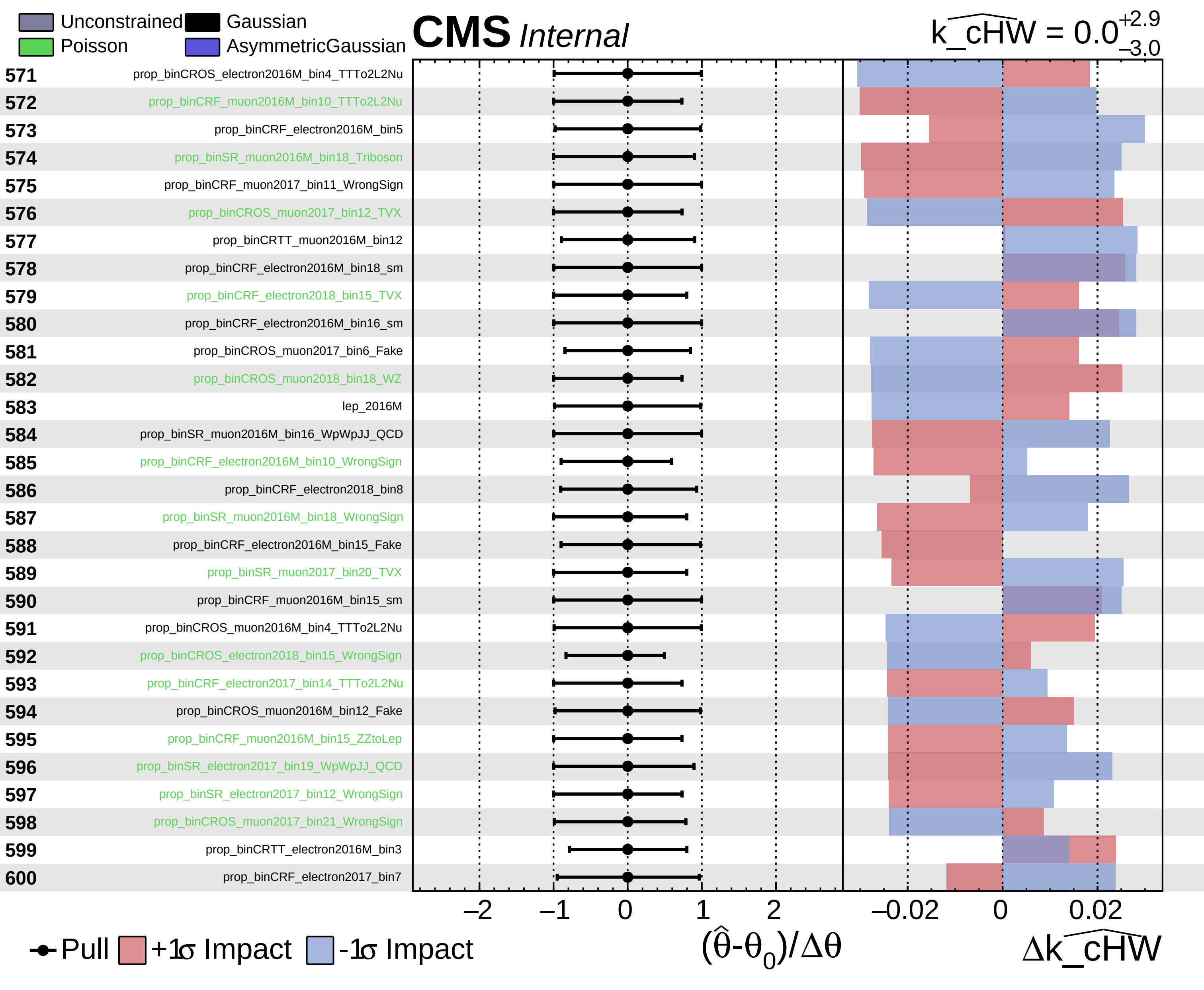


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
481 prop_binCRF_electron2017_bin8
482 prop_binSR_electron2018_bin13_TVX
483 prop_binCRF_muon2016M_bin12_sm
484 prop_binSR_electron2016M_bin18
485 prop_binSR_electron2016M_bin16_sm
486 prop_binCRF_electron2016M_bin14_sm
487 prop_binCRF_muon2016M_bin13
488 prop_binSR_muon2016M_bin17_sm
489 prop_binSR_muon2016M_bin19_sm
490 prop_binCROS_electron2018_bin21_Triboson
491 prop_binCRF_muon2018_bin15
492 prop_binCRF_electron2017_bin9
493 TTest_muon_2018
1.00 +0.05 −0.05
494 prop_binCROS_electron2017_bin6
495 prop_binCRTT_muon2018_bin18
496 prop_binCRF_electron2016M_bin17_TTTo2L2Nu
497 TTest_muon_2016M
1.00 +0.06 −0.06
498 prop_binCROS_electron2017_bin13_WrongSign
499 prop_binSR_muon2016M_bin19_ZZtoLep
500 prop_binSR_muon2017_bin20_sm
501 prop_binCRF_electron2016M_bin15_sm
502 prop_binSR_electron2018_bin13_WpWpJJ_QCD
503 prop_binCROS_electron2016M_bin7_Fake
504 prop_binSR_electron2017_bin14_WpWpJJ_QCD
505 prop_binCRF_muon2016M_bin16_sm
506 prop_binCRTT_muon2016M_bin3
507 prop_binCRF_electron2017_bin10
508 TTest_electron_2016M
1.00 +0.07 −0.07
509 prop_binCROS_muon2016M_bin8
510 prop_binCRF_electron2016M_bin16_TTTo2L2Nu
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

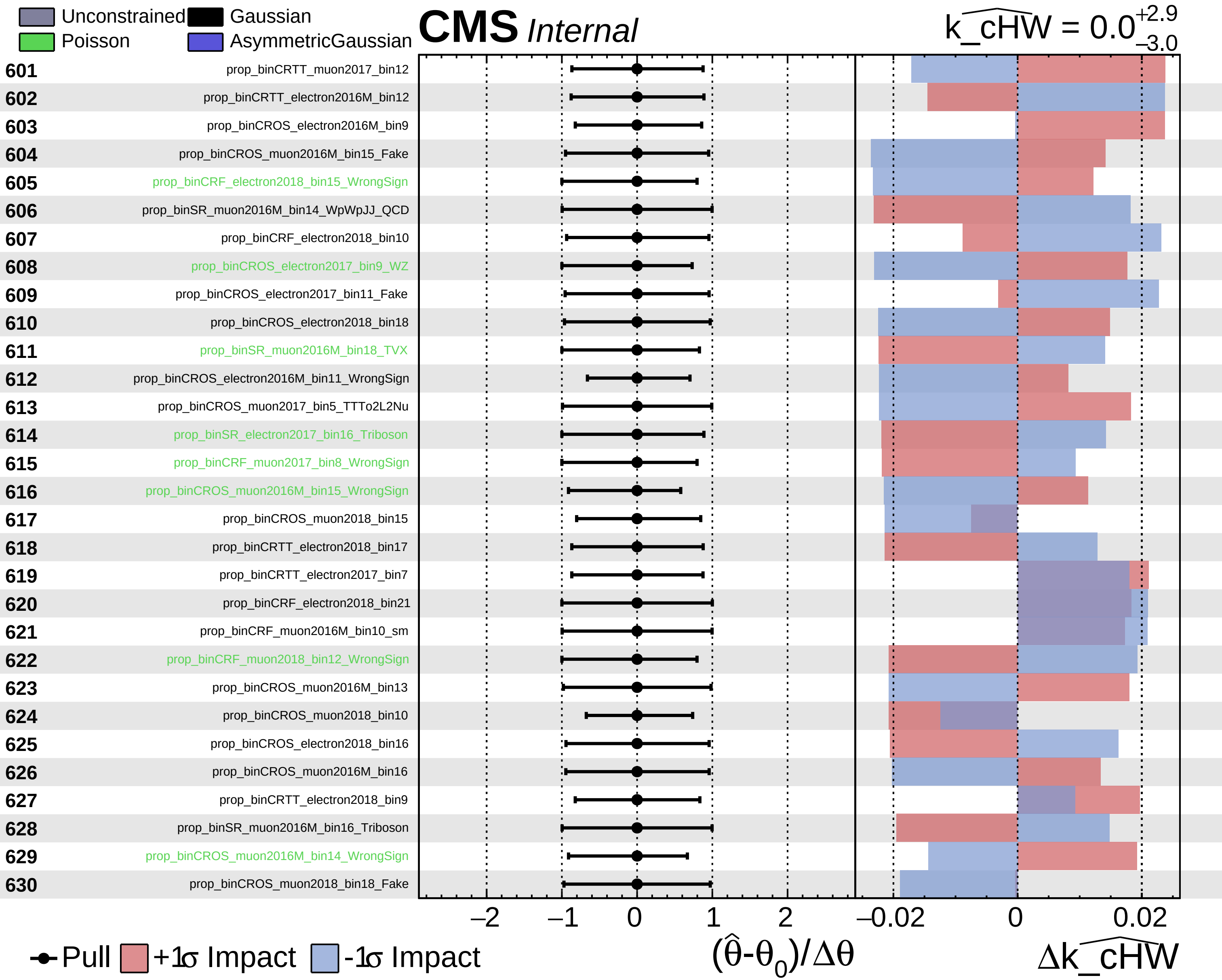


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
511 prop_binCRF_electron2017_bin20
512 prop_binCROS_muon2017_bin17_TVX
513 prop_binSR_electron2018_bin13_WrongSign
514 prop_binSR_electron2016M_bin12_ZZtoLep
515 prop_binCROS_electron2016M_bin13_WrongSign
516 prop_binSR_electron2018_bin13_Triboson
517 prop_binCROS_muon2016M_bin7
518 prop_binCRF_electron2016M_bin7
519 prop_binCROS_muon2016M_bin6_TTTo2L2Nu
520 prop_binCRTT_muon2017_bin18
521 prop_binCROS_electron2018_bin9
522 prop_binCRTT_muon2016M_bin6
523 prop_binCROS_muon2017_bin10_Fake
524 prop_binCRTT_electron2017_bin0
525 prop_binCRF_electron2016M_bin8
526 prop_binCRF_muon2016M_bin18
527 prop_binCRTT_electron2018_bin0
528 prop_binCRTT_muon2018_bin16
529 prop_binCROS_muon2017_bin10_TTTo2L2Nu
530 prop_binCRF_electron2016M_bin6
531 prop_binCROS_electron2016M_bin5_TTTo2L2Nu
532 prop_binSR_electron2017_bin14_TVX
533 prop_binCROS_electron2018_bin12
534 prop_binCRTT_muon2017_bin21
535 prop_binCROS_muon2017_bin6_WrongSign
536 prop_binCROS_muon2018_bin17
537 prop_binCRF_muon2017_bin8_ZZtoLep
538 prop_binSR_muon2016M_bin18_WpWpJJ_QCD
539 prop_binCROS_electron2017_bin12_TTTo2L2Nu
540 prop_binCRTT_electron2017_bin13
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW

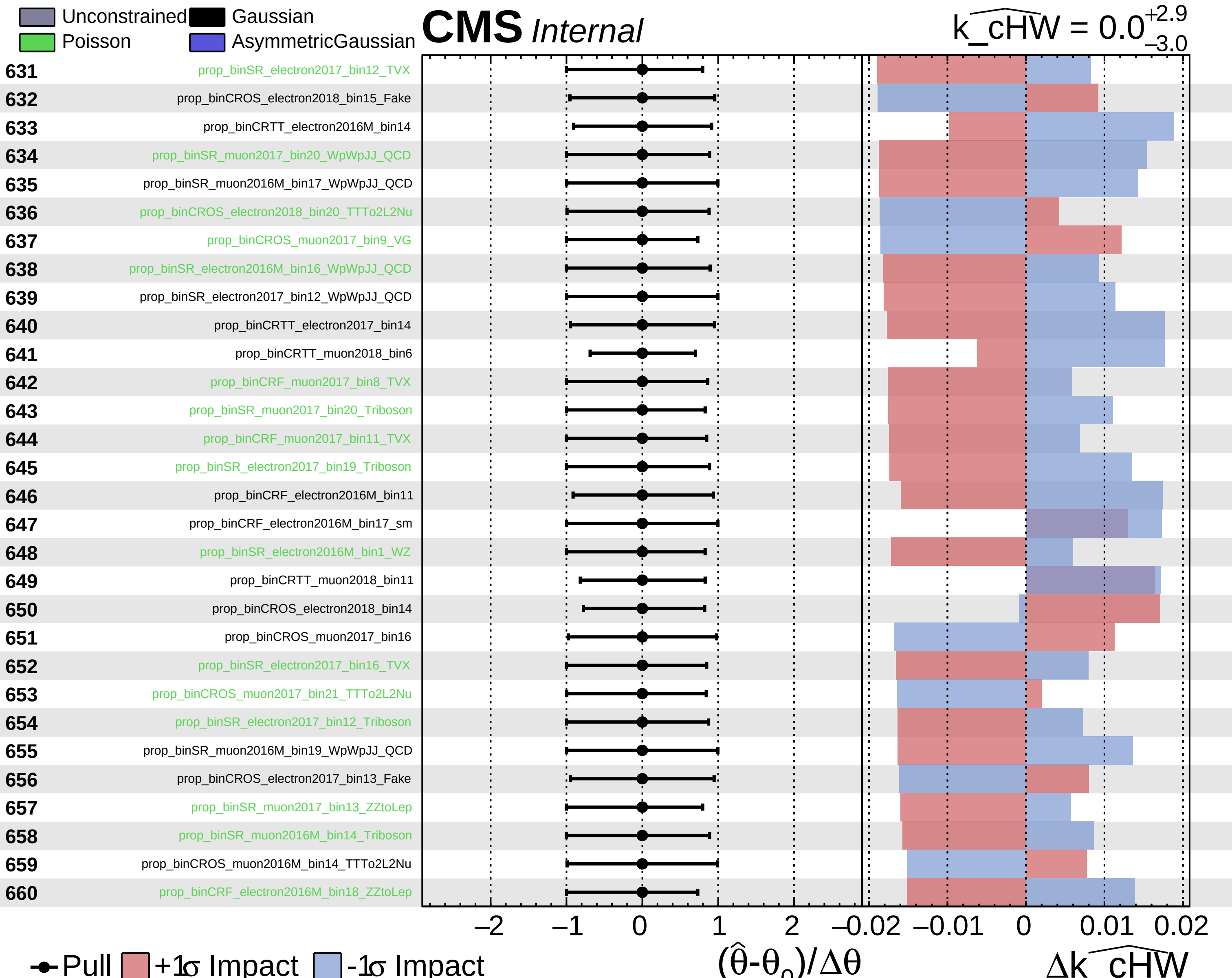

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
541 prop_binCROS_muon2017_bin9_TTTo2L2Nu
542 prop_binCRTT_electron2016M_bin8
543 prop_binCROS_muon2018_bin21_Triboson
544 prop_binCRF_electron2017_bin16_WrongSign
545 QCDScale_ZZtoLep
546 prop_binCRTT_electron2017_bin3
547 prop_binCRF_electron2017_bin14_Fake
548 prop_binCRTT_muon2018_bin17
549 prop_binSR_muon2016M_bin16_WrongSign
550 prop_binCROS_muon2018_bin12_WrongSign
551 prop_binCRTT_muon2018_bin8
552 prop_binSR_electron2016M_bin20
553 prop_binCRTT_electron2018_bin11
554 prop_binCRF_electron2018_bin1
555 prop_binCROS_electron2018_bin10
556 prop_binCROS_electron2016M_bin5_WrongSign
557 prop_binCRF_muon2016M_bin17_sm
558 prop_binCROS_muon2017_bin3
559 prop_binCRTT_electron2018_bin1
560 prop_binCRTT_muon2016M_bin7
561 prop_binCROS_electron2018_bin7
562 prop_binSR_muon2016M_bin14_TVX
563 prop_binCRTT_muon2017_bin4
564 prop_binSR_electron2016M_bin12_sm
565 prop_binCRF_electron2016M_bin20
566 prop_binSR_electron2017_bin19_TVX
567 prop_binSR_electron2017_bin16_WpWpJJ_QCD
568 prop_binCRF_muon2018_bin9
569 prop_binCRTT_muon2016M_bin9
570 prop_binSR_electron2016M_bin16_TVX
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW



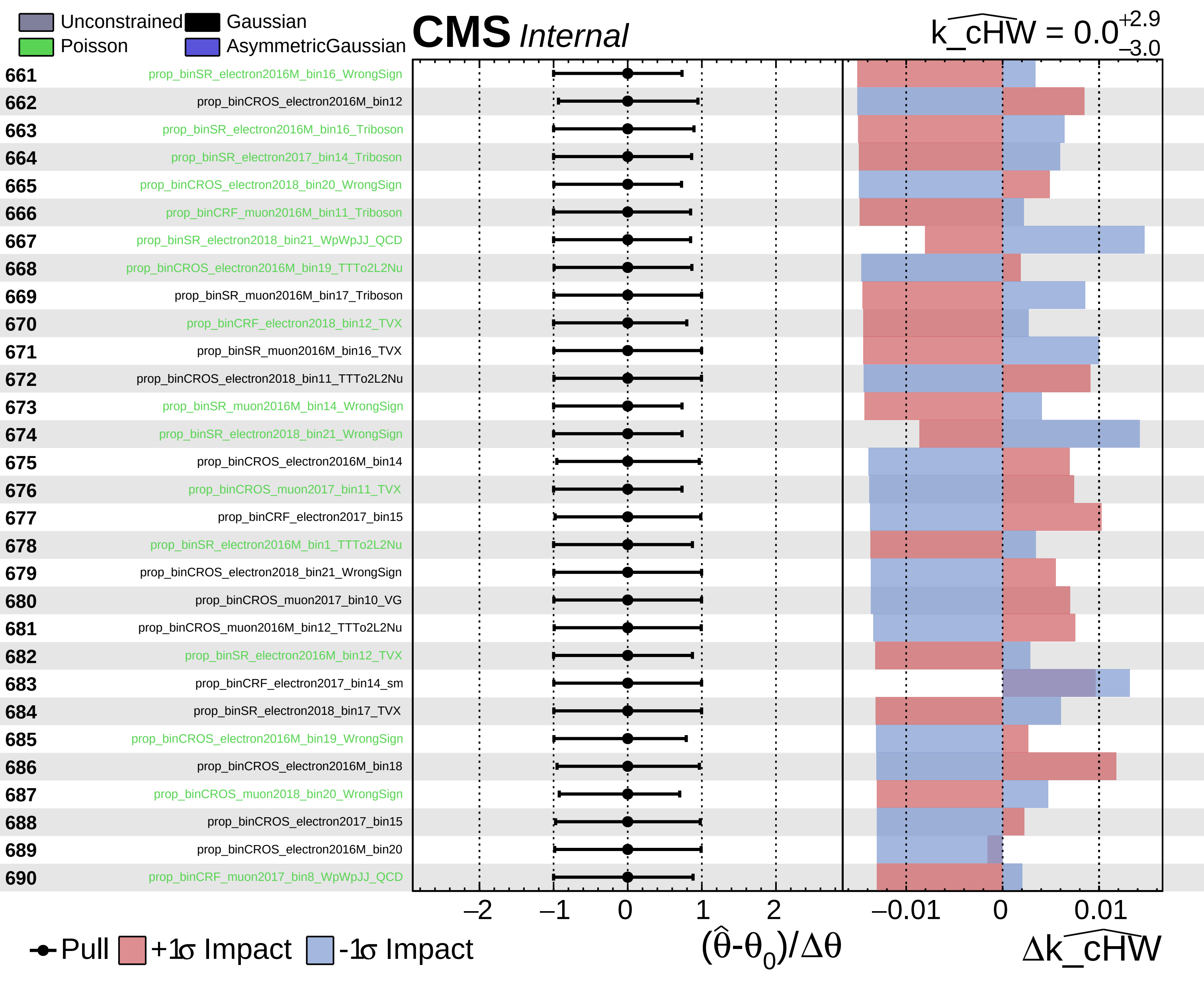
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
571 prop_binCROS_electron2016M_bin4_TTTo2L2Nu
572 prop_binCRF_muon2016M_bin10_TTTo2L2Nu
573 prop_binCRF_electron2016M_bin5
574 prop_binSR_muon2016M_bin18_Triboson
575 prop_binCRF_muon2017_bin11_WrongSign
576 prop_binCROS_muon2017_bin12_TVX
577 prop_binCRTT_muon2016M_bin12
578 prop_binCRF_electron2016M_bin18_sm
579 prop_binCRF_electron2018_bin15_TVX
580 prop_binCRF_electron2016M_bin16_sm
581 prop_binCROS_muon2017_bin6_Fake
582 prop_binCROS_muon2018_bin18_WZ
583 lep_2016M
584 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
585 prop_binCRF_electron2016M_bin10_WrongSign
586 prop_binCRF_electron2018_bin8
587 prop_binSR_muon2016M_bin18_WrongSign
588 prop_binCRF_electron2016M_bin15_Fake
589 prop_binSR_muon2017_bin20_TVX
590 prop_binCRF_muon2016M_bin15_sm
591 prop_binCROS_muon2016M_bin4_TTTo2L2Nu
592 prop_binCROS_electron2018_bin15_WrongSign
593 prop_binCRF_electron2017_bin14_TTTo2L2Nu
594 prop_binCROS_muon2016M_bin12_Fake
595 prop_binCRF_muon2016M_bin15_ZZtoLep
596 prop_binSR_electron2017_bin19_WpWpJJ_QCD
597 prop_binSR_electron2017_bin12_WrongSign
598 prop_binCROS_muon2017_bin21_WrongSign
599 prop_binCRTT_electron2016M_bin3
600 prop_binCRF_electron2017_bin7
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW



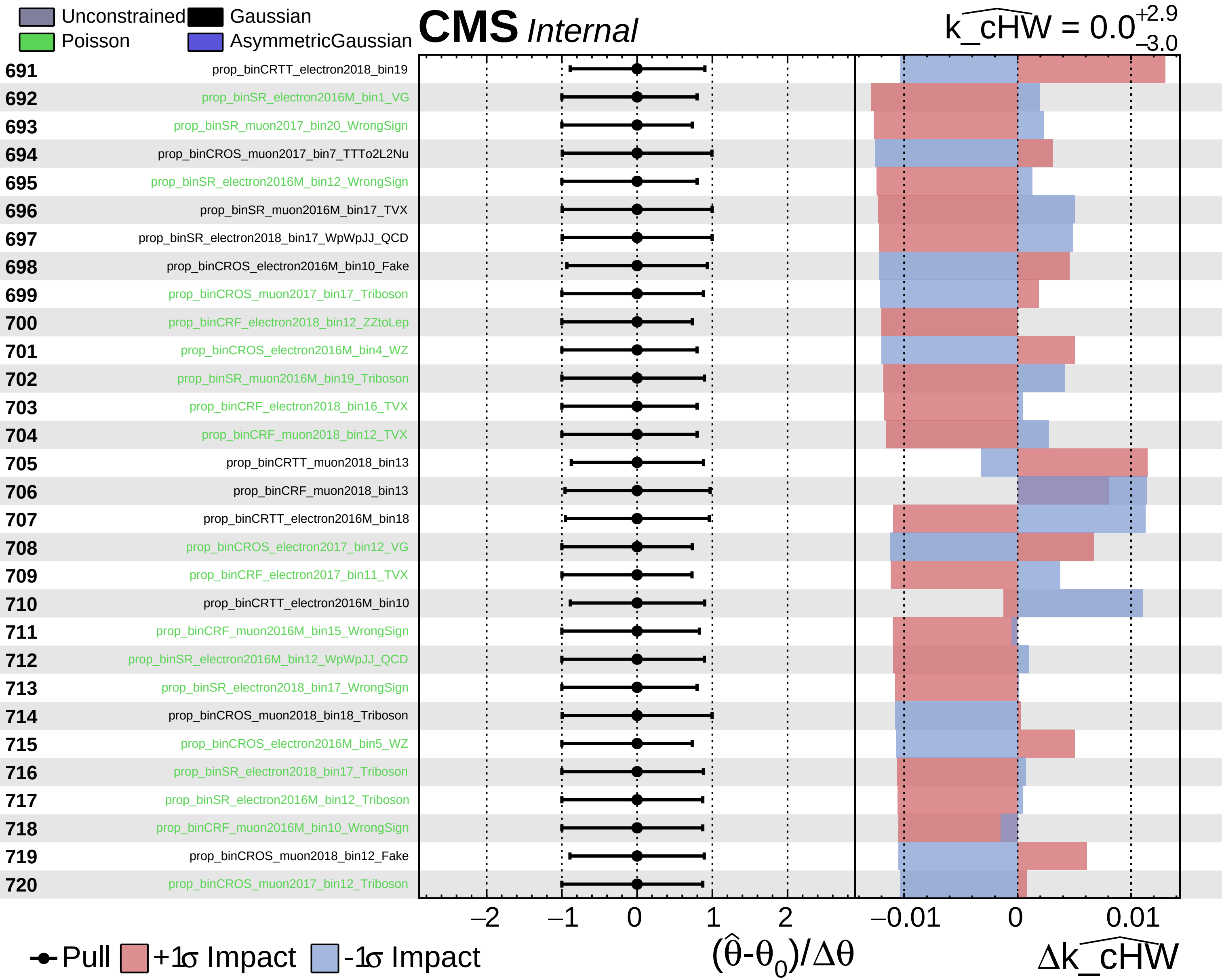
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
601 prop_binCRTT_muon2017_bin12
602 prop_binCRTT_electron2016M_bin12
603 prop_binCROS_electron2016M_bin9
604 prop_binCROS_muon2016M_bin15_Fake
605 prop_binCRF_electron2018_bin15_WrongSign
606 prop_binSR_muon2016M_bin14_WpWpJJ_QCD
607 prop_binCRF_electron2018_bin10
608 prop_binCROS_electron2017_bin9_WZ
609 prop_binCROS_electron2017_bin11_Fake
610 prop_binCROS_electron2018_bin18
611 prop_binSR_muon2016M_bin18_TVX
612 prop_binCROS_electron2016M_bin11_WrongSign
613 prop_binCROS_muon2017_bin5_TTTo2L2Nu
614 prop_binSR_electron2017_bin16_Triboson
615 prop_binCRF_muon2017_bin8_WrongSign
616 prop_binCROS_muon2016M_bin15_WrongSign
617 prop_binCROS_muon2018_bin15
618 prop_binCRTT_electron2018_bin17
619 prop_binCRTT_electron2017_bin7
620 prop_binCRF_electron2018_bin21
621 prop_binCRF_muon2016M_bin10_sm
622 prop_binCRF_muon2018_bin12_WrongSign
623 prop_binCROS_muon2016M_bin13
624 prop_binCROS_muon2018_bin10
625 prop_binCROS_electron2018_bin16
626 prop_binCROS_muon2016M_bin16
627 prop_binCRTT_electron2018_bin9
628 prop_binSR_muon2016M_bin16_Triboson
629 prop_binCROS_muon2016M_bin14_WrongSign
630 prop_binCROS_muon2018_bin18_Fake
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



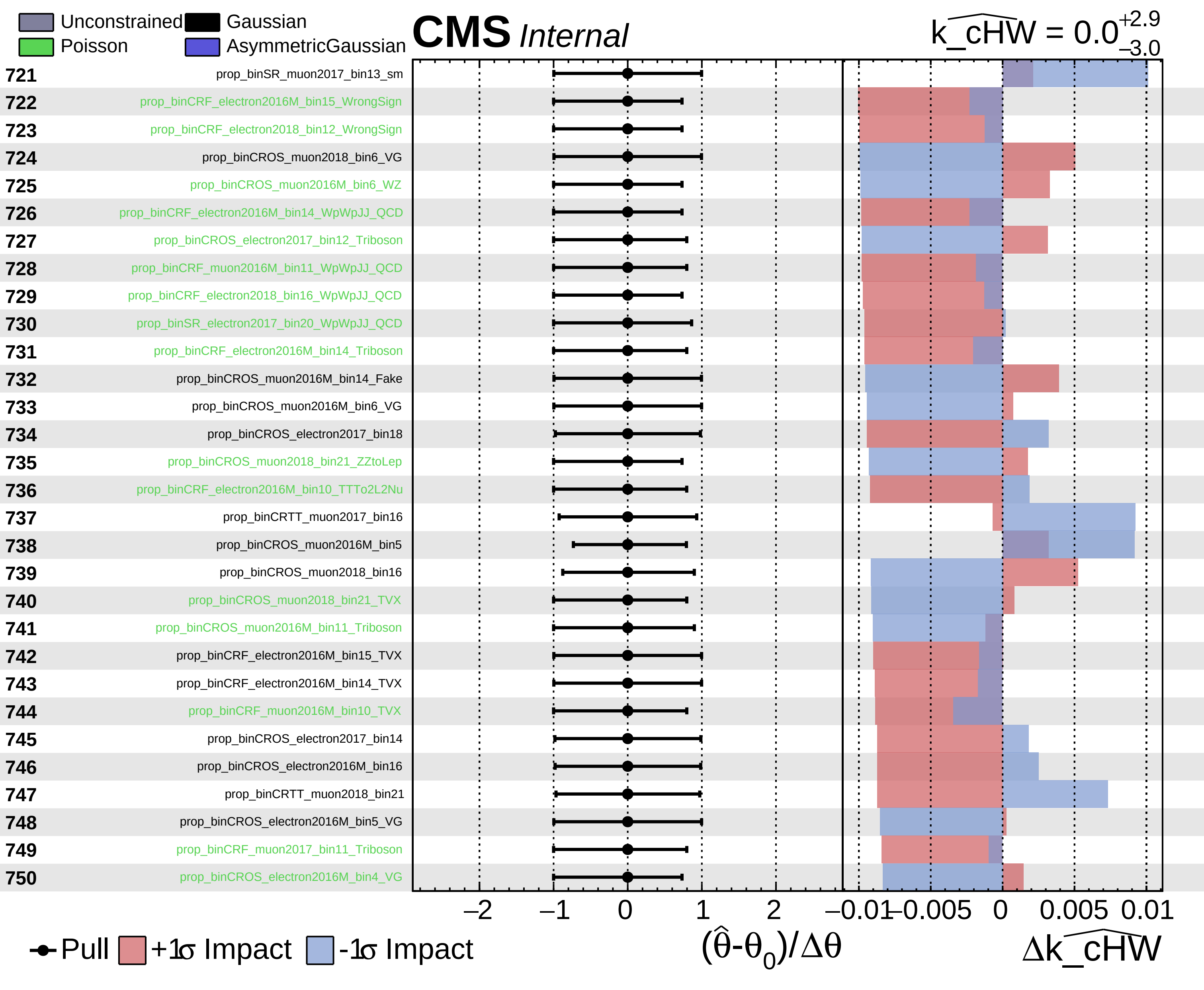
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
631 prop_binSR_electron2017_bin12_TVX
632 prop_binCROS_electron2018_bin15_Fake
633 prop_binCRTT_electron2016M_bin14
634 prop_binSR_muon2017_bin20_WpWpJJ_QCD
635 prop_binSR_muon2016M_bin17_WpWpJJ_QCD
636 prop_binCROS_electron2018_bin20_TTTo2L2Nu
637 prop_binCROS_muon2017_bin9_VG
638 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
639 prop_binSR_electron2017_bin12_WpWpJJ_QCD
640 prop_binCRTT_electron2017_bin14
641 prop_binCRTT_muon2018_bin6
642 prop_binCRF_muon2017_bin8_TVX
643 prop_binSR_muon2017_bin20_Triboson
644 prop_binCRF_muon2017_bin11_TVX
645 prop_binSR_electron2017_bin19_Triboson
646 prop_binCRF_electron2016M_bin11
647 prop_binCRF_electron2016M_bin17_sm
648 prop_binSR_electron2016M_bin1_WZ
649 prop_binCRTT_muon2018_bin11
650 prop_binCROS_electron2018_bin14
651 prop_binCROS_muon2017_bin16
652 prop_binSR_electron2017_bin16_TVX
653 prop_binCROS_muon2017_bin21_TTTo2L2Nu
654 prop_binSR_electron2017_bin12_Triboson
655 prop_binSR_muon2016M_bin19_WpWpJJ_QCD
656 prop_binCROS_electron2017_bin13_Fake
657 prop_binSR_muon2017_bin13_ZZtoLep
658 prop_binSR_muon2016M_bin14_Triboson
659 prop_binCROS_muon2016M_bin14_TTTo2L2Nu
660 prop_binCRF_electron2016M_bin18_ZZtoLep
−2
−1
0
1
2
−0.02
−0.01
0
0.01
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



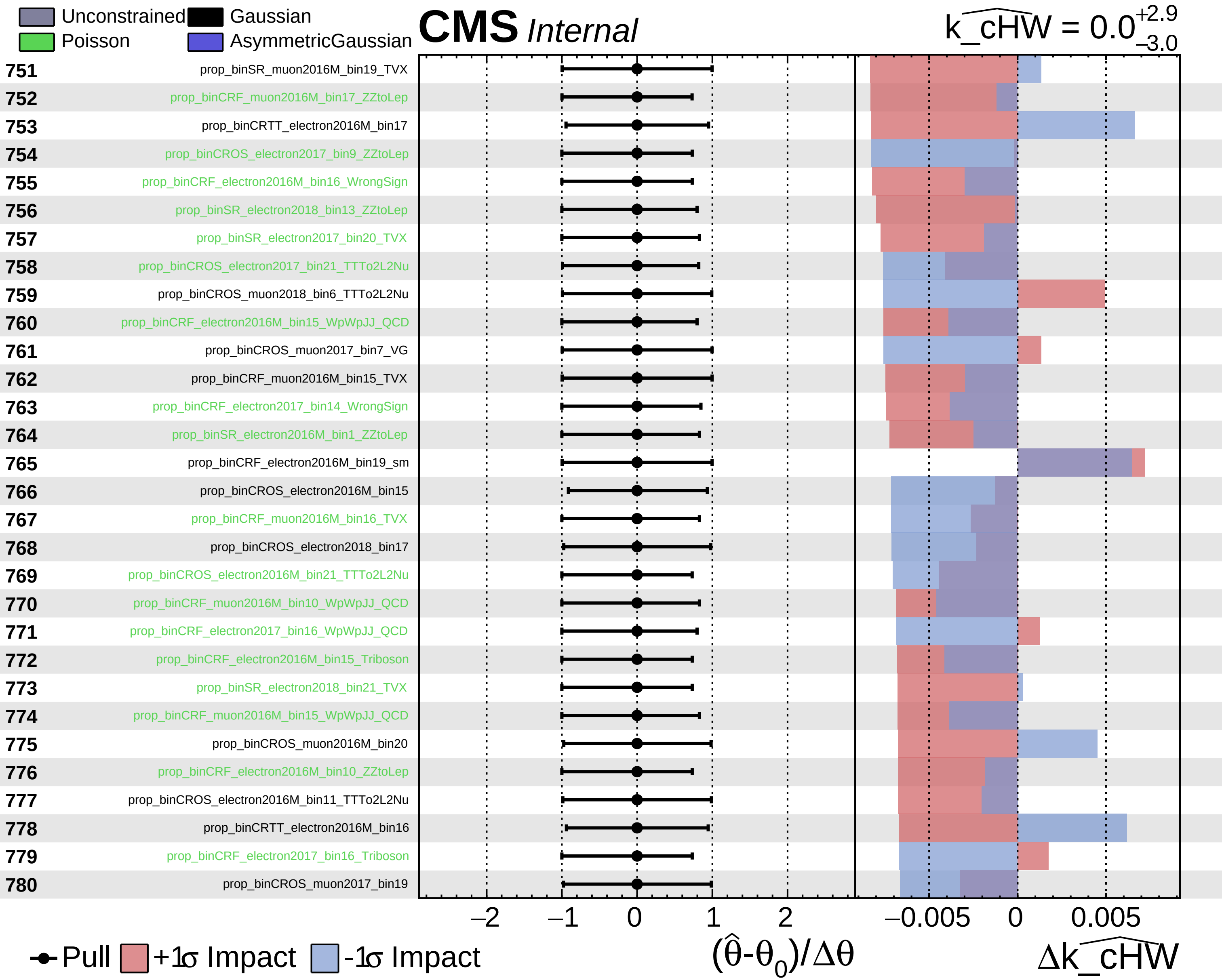
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
661 prop_binSR_electron2016M_bin16_WrongSign
662 prop_binCROS_electron2016M_bin12
663 prop_binSR_electron2016M_bin16_Triboson
664 prop_binSR_electron2017_bin14_Triboson
665 prop_binCROS_electron2018_bin20_WrongSign
666 prop_binCRF_muon2016M_bin11_Triboson
667 prop_binSR_electron2018_bin21_WpWpJJ_QCD
668 prop_binCROS_electron2016M_bin19_TTTo2L2Nu
669 prop_binSR_muon2016M_bin17_Triboson
670 prop_binCRF_electron2018_bin12_TVX
671 prop_binSR_muon2016M_bin16_TVX
672 prop_binCROS_electron2018_bin11_TTTo2L2Nu
673 prop_binSR_muon2016M_bin14_WrongSign
674 prop_binSR_electron2018_bin21_WrongSign
675 prop_binCROS_electron2016M_bin14
676 prop_binCROS_muon2017_bin11_TVX
677 prop_binCRF_electron2017_bin15
678 prop_binSR_electron2016M_bin1_TTTo2L2Nu
679 prop_binCROS_electron2018_bin21_WrongSign
680 prop_binCROS_muon2017_bin10_VG
681 prop_binCROS_muon2016M_bin12_TTTo2L2Nu
682 prop_binSR_electron2016M_bin12_TVX
683 prop_binCRF_electron2017_bin14_sm
684 prop_binSR_electron2018_bin17_TVX
685 prop_binCROS_electron2016M_bin19_WrongSign
686 prop_binCROS_electron2016M_bin18
687 prop_binCROS_muon2018_bin20_WrongSign
688 prop_binCROS_electron2017_bin15
689 prop_binCROS_electron2016M_bin20
690 prop_binCRF_muon2017_bin8_WpWpJJ_QCD
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

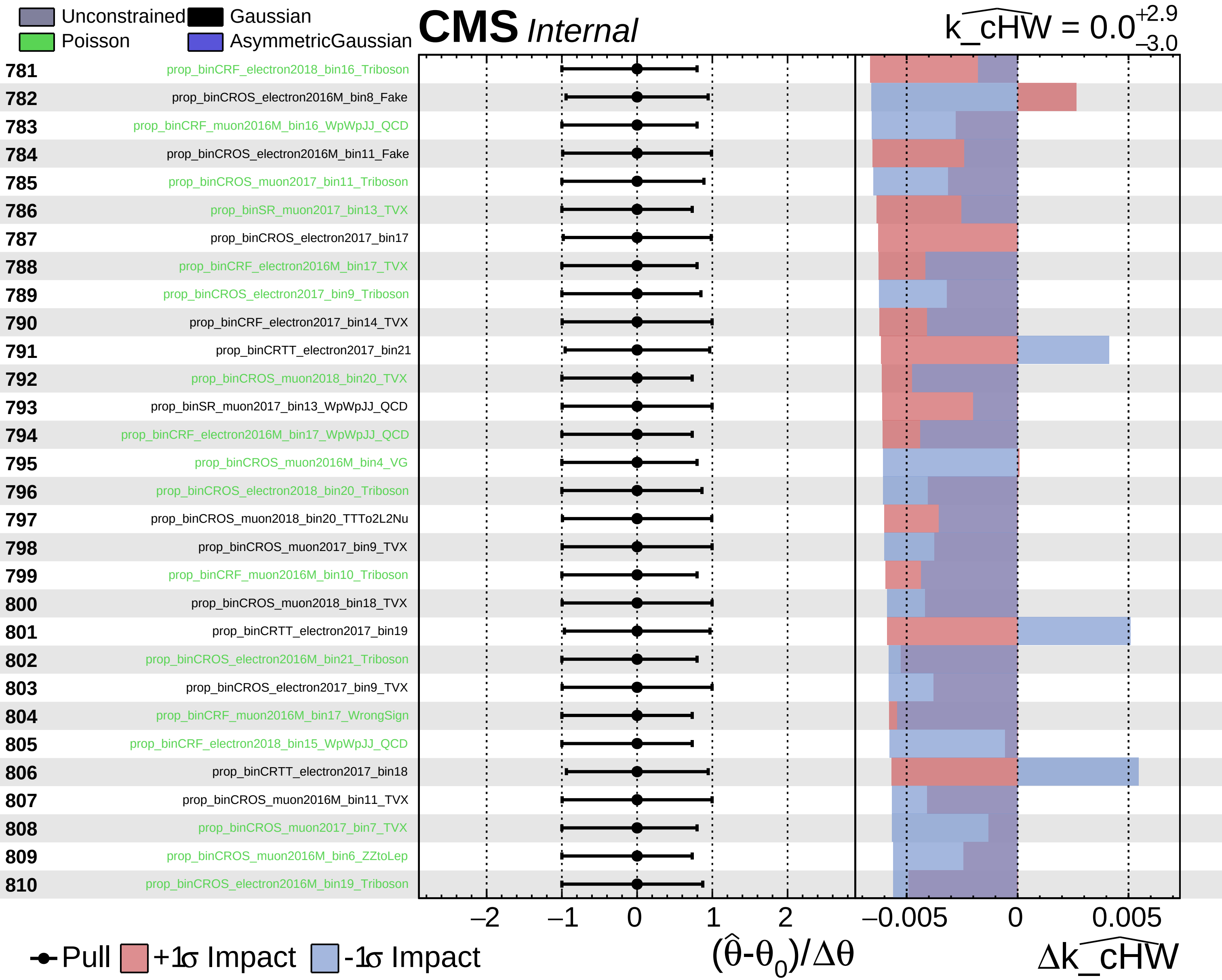


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
691 prop_binCRTT_electron2018_bin19
692 prop_binSR_electron2016M_bin1_VG
693 prop_binSR_muon2017_bin20_WrongSign
694 prop_binCROS_muon2017_bin7_TTTo2L2Nu
695 prop_binSR_electron2016M_bin12_WrongSign
696 prop_binSR_muon2016M_bin17_TVX
697 prop_binSR_electron2018_bin17_WpWpJJ_QCD
698 prop_binCROS_electron2016M_bin10_Fake
699 prop_binCROS_muon2017_bin17_Triboson
700 prop_binCRF_electron2018_bin12_ZZtoLep
701 prop_binCROS_electron2016M_bin4_WZ
702 prop_binSR_muon2016M_bin19_Triboson
703 prop_binCRF_electron2018_bin16_TVX
704 prop_binCRF_muon2018_bin12_TVX
705 prop_binCRTT_muon2018_bin13
706 prop_binCRF_muon2018_bin13
707 prop_binCRTT_electron2016M_bin18
708 prop_binCROS_electron2017_bin12_VG
709 prop_binCRF_electron2017_bin11_TVX
710 prop_binCRTT_electron2016M_bin10
711 prop_binCRF_muon2016M_bin15_WrongSign
712 prop_binSR_electron2016M_bin12_WpWpJJ_QCD
713 prop_binSR_electron2018_bin17_WrongSign
714 prop_binCROS_muon2018_bin18_Triboson
715 prop_binCROS_electron2016M_bin5_WZ
716 prop_binSR_electron2018_bin17_Triboson
717 prop_binSR_electron2016M_bin12_Triboson
718 prop_binCRF_muon2016M_bin10_WrongSign
719 prop_binCROS_muon2018_bin12_Fake
720 prop_binCROS_muon2017_bin12_Triboson
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

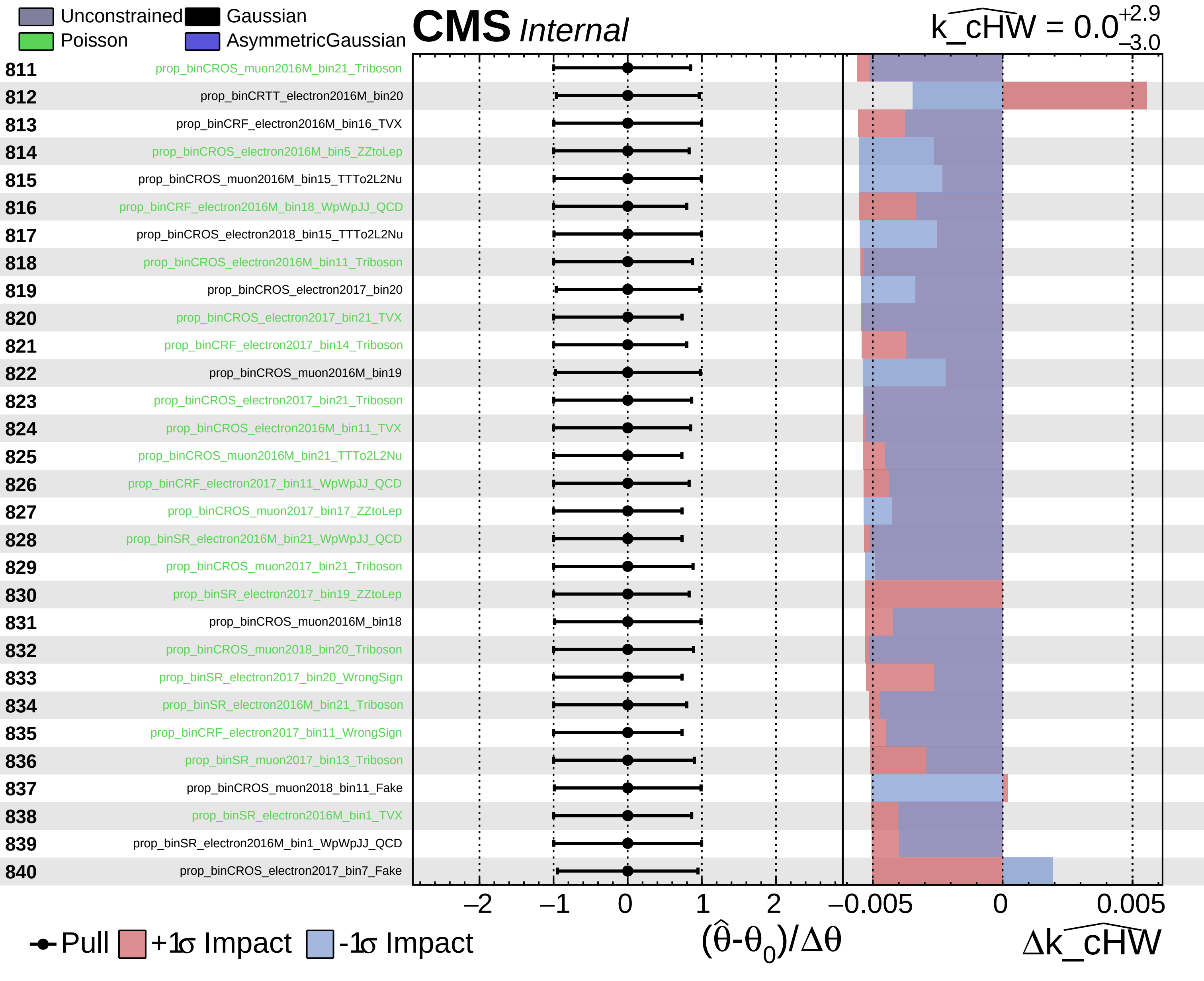


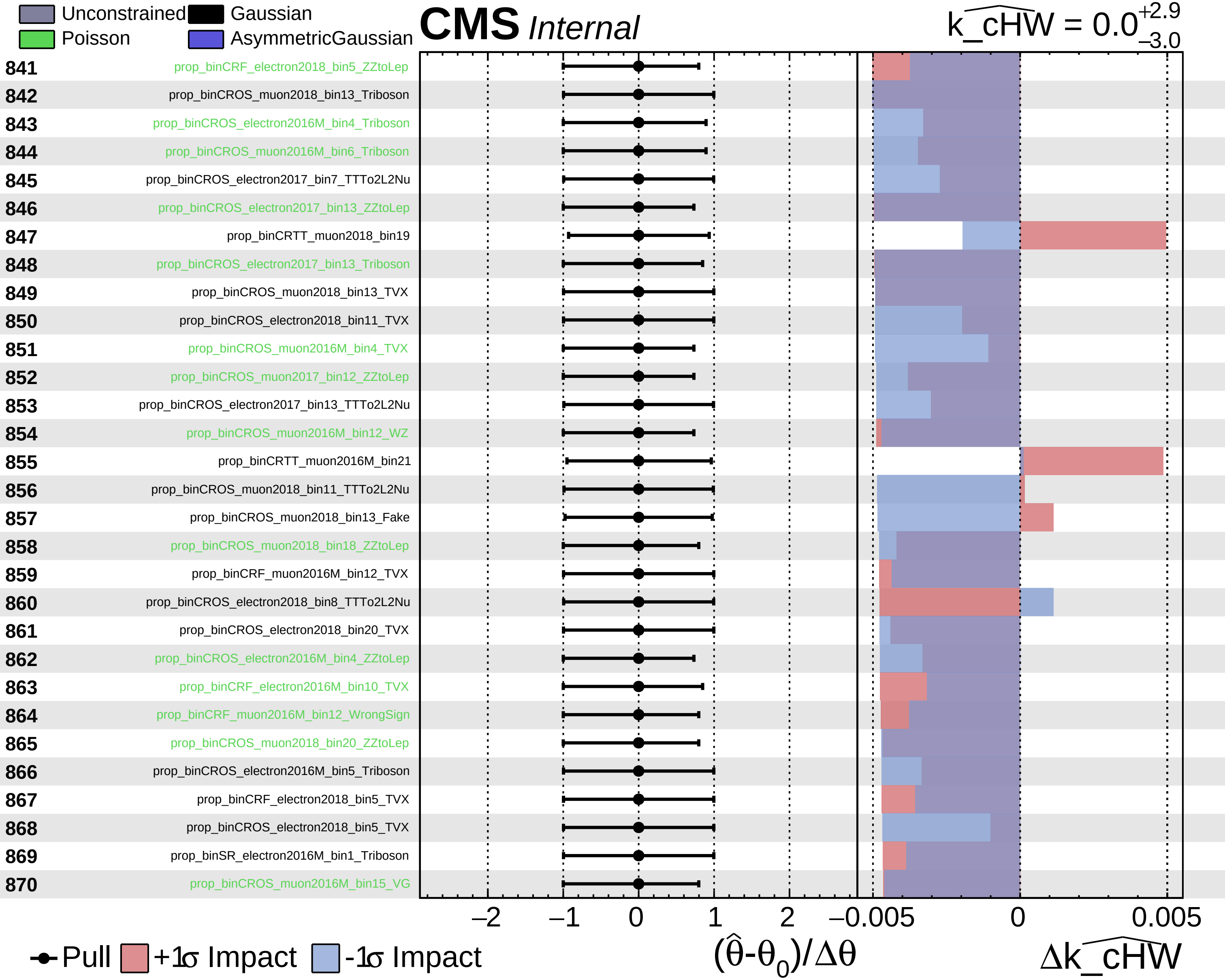
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
721 prop_binSR_muon2017_bin13_sm
722 prop_binCRF_electron2016M_bin15_WrongSign
723 prop_binCRF_electron2018_bin12_WrongSign
724 prop_binCROS_muon2018_bin6_VG
725 prop_binCROS_muon2016M_bin6_WZ
726 prop_binCRF_electron2016M_bin14_WpWpJJ_QCD
727 prop_binCROS_electron2017_bin12_Triboson
728 prop_binCRF_muon2016M_bin11_WpWpJJ_QCD
729 prop_binCRF_electron2018_bin16_WpWpJJ_QCD
730 prop_binSR_electron2017_bin20_WpWpJJ_QCD
731 prop_binCRF_electron2016M_bin14_Triboson
732 prop_binCROS_muon2016M_bin14_Fake
733 prop_binCROS_muon2016M_bin6_VG
734 prop_binCROS_electron2017_bin18
735 prop_binCROS_muon2018_bin21_ZZtoLep
736 prop_binCRF_electron2016M_bin10_TTTo2L2Nu
737 prop_binCRTT_muon2017_bin16
738 prop_binCROS_muon2016M_bin5
739 prop_binCROS_muon2018_bin16
740 prop_binCROS_muon2018_bin21_TVX
741 prop_binCROS_muon2016M_bin11_Triboson
742 prop_binCRF_electron2016M_bin15_TVX
743 prop_binCRF_electron2016M_bin14_TVX
744 prop_binCRF_muon2016M_bin10_TVX
745 prop_binCROS_electron2017_bin14
746 prop_binCROS_electron2016M_bin16
747 prop_binCRTT_muon2018_bin21
748 prop_binCROS_electron2016M_bin5_VG
749 prop_binCRF_muon2017_bin11_Triboson
750 prop_binCROS_electron2016M_bin4_VG
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
751 prop_binSR_muon2016M_bin19_TVX
752 prop_binCRF_muon2016M_bin17_ZZtoLep
753 prop_binCRTT_electron2016M_bin17
754 prop_binCROS_electron2017_bin9_ZZtoLep
755 prop_binCRF_electron2016M_bin16_WrongSign
756 prop_binSR_electron2018_bin13_ZZtoLep
757 prop_binSR_electron2017_bin20_TVX
758 prop_binCROS_electron2017_bin21_TTTo2L2Nu
759 prop_binCROS_muon2018_bin6_TTTo2L2Nu
760 prop_binCRF_electron2016M_bin15_WpWpJJ_QCD
761 prop_binCROS_muon2017_bin7_VG
762 prop_binCRF_muon2016M_bin15_TVX
763 prop_binCRF_electron2017_bin14_WrongSign
764 prop_binSR_electron2016M_bin1_ZZtoLep
765 prop_binCRF_electron2016M_bin19_sm
766 prop_binCROS_electron2016M_bin15
767 prop_binCRF_muon2016M_bin16_TVX
768 prop_binCROS_electron2018_bin17
769 prop_binCROS_electron2016M_bin21_TTTo2L2Nu
770 prop_binCRF_muon2016M_bin10_WpWpJJ_QCD
771 prop_binCRF_electron2017_bin16_WpWpJJ_QCD
772 prop_binCRF_electron2016M_bin15_Triboson
773 prop_binSR_electron2018_bin21_TVX
774 prop_binCRF_muon2016M_bin15_WpWpJJ_QCD
775 prop_binCROS_muon2016M_bin20
776 prop_binCRF_electron2016M_bin10_ZZtoLep
777 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
778 prop_binCRTT_electron2016M_bin16
779 prop_binCRF_electron2017_bin16_Triboson
780 prop_binCROS_muon2017_bin19
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW

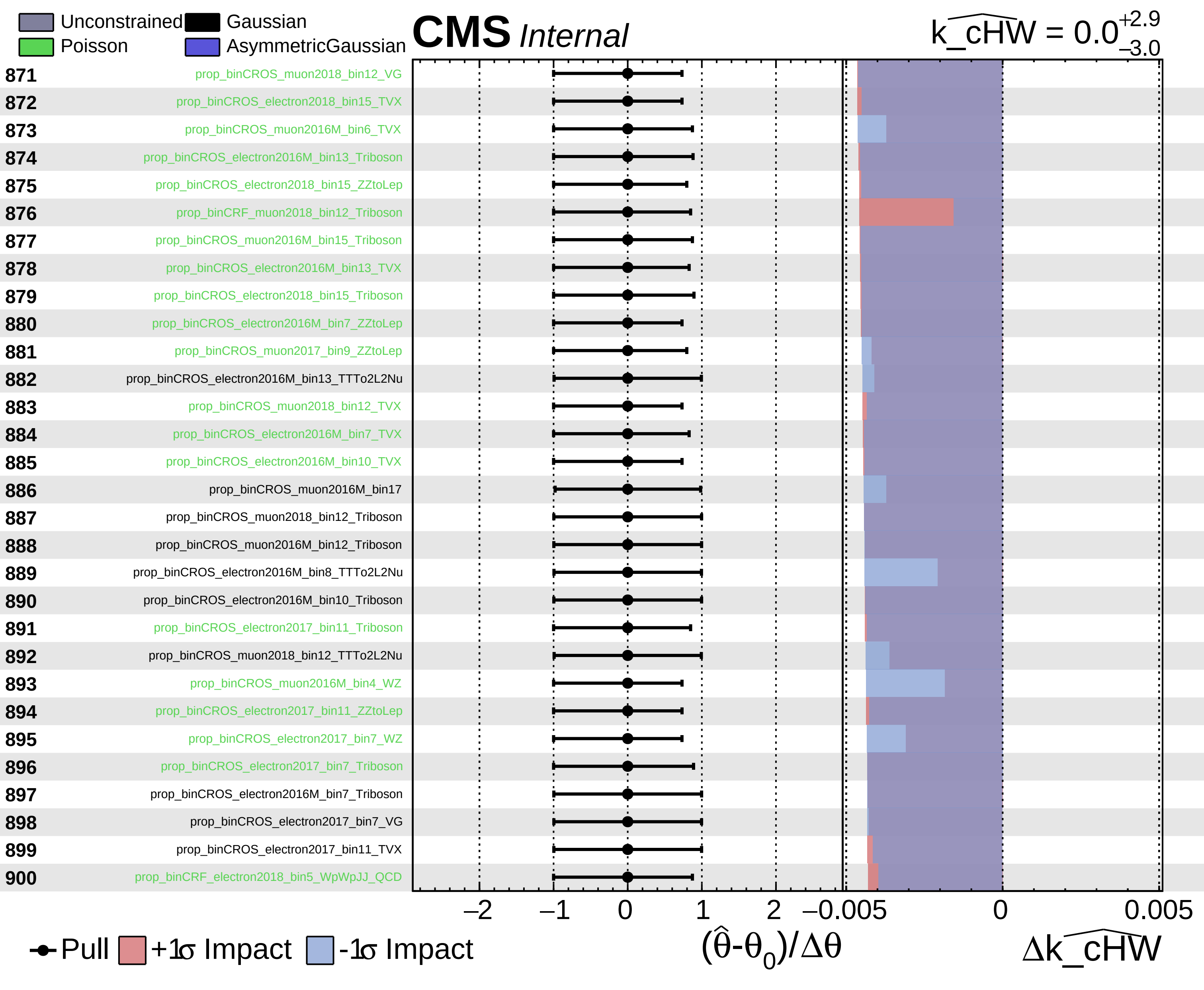


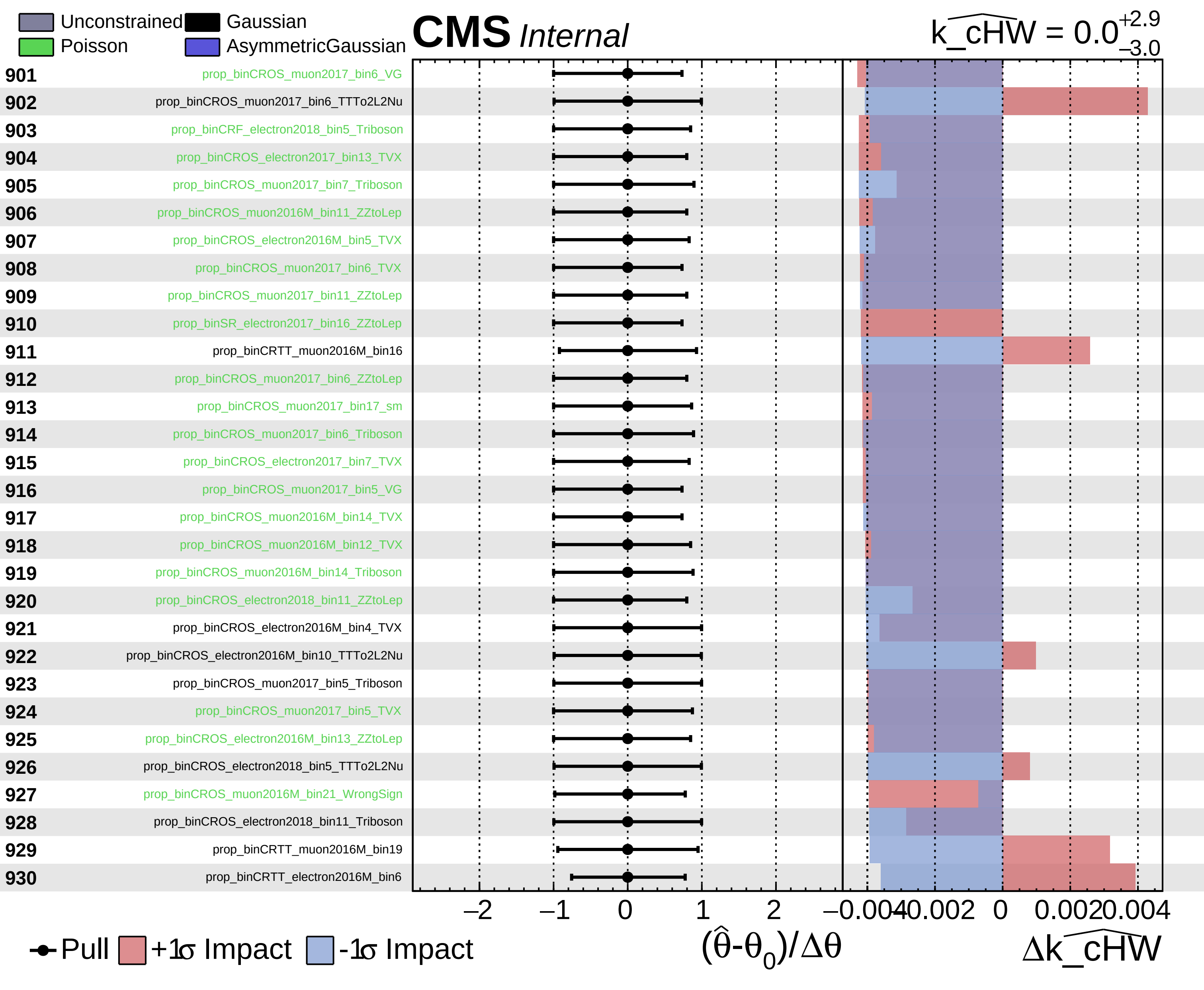
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
781 prop_binCRF_electron2018_bin16_Triboson
782 prop_binCROS_electron2016M_bin8_Fake
783 prop_binCRF_muon2016M_bin16_WpWpJJ_QCD
784 prop_binCROS_electron2016M_bin11_Fake
785 prop_binCROS_muon2017_bin11_Triboson
786 prop_binSR_muon2017_bin13_TVX
787 prop_binCROS_electron2017_bin17
788 prop_binCRF_electron2016M_bin17_TVX
789 prop_binCROS_electron2017_bin9_Triboson
790 prop_binCRF_electron2017_bin14_TVX
791 prop_binCRTT_electron2017_bin21
792 prop_binCROS_muon2018_bin20_TVX
793 prop_binSR_muon2017_bin13_WpWpJJ_QCD
794 prop_binCRF_electron2016M_bin17_WpWpJJ_QCD
795 prop_binCROS_muon2016M_bin4_VG
796 prop_binCROS_electron2018_bin20_Triboson
797 prop_binCROS_muon2018_bin20_TTTo2L2Nu
798 prop_binCROS_muon2017_bin9_TVX
799 prop_binCRF_muon2016M_bin10_Triboson
800 prop_binCROS_muon2018_bin18_TVX
801 prop_binCRTT_electron2017_bin19
802 prop_binCROS_electron2016M_bin21_Triboson
803 prop_binCROS_electron2017_bin9_TVX
804 prop_binCRF_muon2016M_bin17_WrongSign
805 prop_binCRF_electron2018_bin15_WpWpJJ_QCD
806 prop_binCRTT_electron2017_bin18
807 prop_binCROS_muon2016M_bin11_TVX
808 prop_binCROS_muon2017_bin7_TVX
809 prop_binCROS_muon2016M_bin6_ZZtoLep
810 prop_binCROS_electron2016M_bin19_Triboson
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
811 prop_binCROS_muon2016M_bin21_Triboson
812 prop_binCRTT_electron2016M_bin20
813 prop_binCRF_electron2016M_bin16_TVX
814 prop_binCROS_electron2016M_bin5_ZZtoLep
815 prop_binCROS_muon2016M_bin15_TTTo2L2Nu
816 prop_binCRF_electron2016M_bin18_WpWpJJ_QCD
817 prop_binCROS_electron2018_bin15_TTTo2L2Nu
818 prop_binCROS_electron2016M_bin11_Triboson
819 prop_binCROS_electron2017_bin20
820 prop_binCROS_electron2017_bin21_TVX
821 prop_binCRF_electron2017_bin14_Triboson
822 prop_binCROS_muon2016M_bin19
823 prop_binCROS_electron2017_bin21_Triboson
824 prop_binCROS_electron2016M_bin11_TVX
825 prop_binCROS_muon2016M_bin21_TTTo2L2Nu
826 prop_binCRF_electron2017_bin11_WpWpJJ_QCD
827 prop_binCROS_muon2017_bin17_ZZtoLep
828 prop_binSR_electron2016M_bin21_WpWpJJ_QCD
829 prop_binCROS_muon2017_bin21_Triboson
830 prop_binSR_electron2017_bin19_ZZtoLep
831 prop_binCROS_muon2016M_bin18
832 prop_binCROS_muon2018_bin20_Triboson
833 prop_binSR_electron2017_bin20_WrongSign
834 prop_binSR_electron2016M_bin21_Triboson
835 prop_binCRF_electron2017_bin11_WrongSign
836 prop_binSR_muon2017_bin13_Triboson
837 prop_binCROS_muon2018_bin11_Fake
838 prop_binSR_electron2016M_bin1_TVX
839 prop_binSR_electron2016M_bin1_WpWpJJ_QCD
840 prop_binCROS_electron2017_bin7_Fake
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

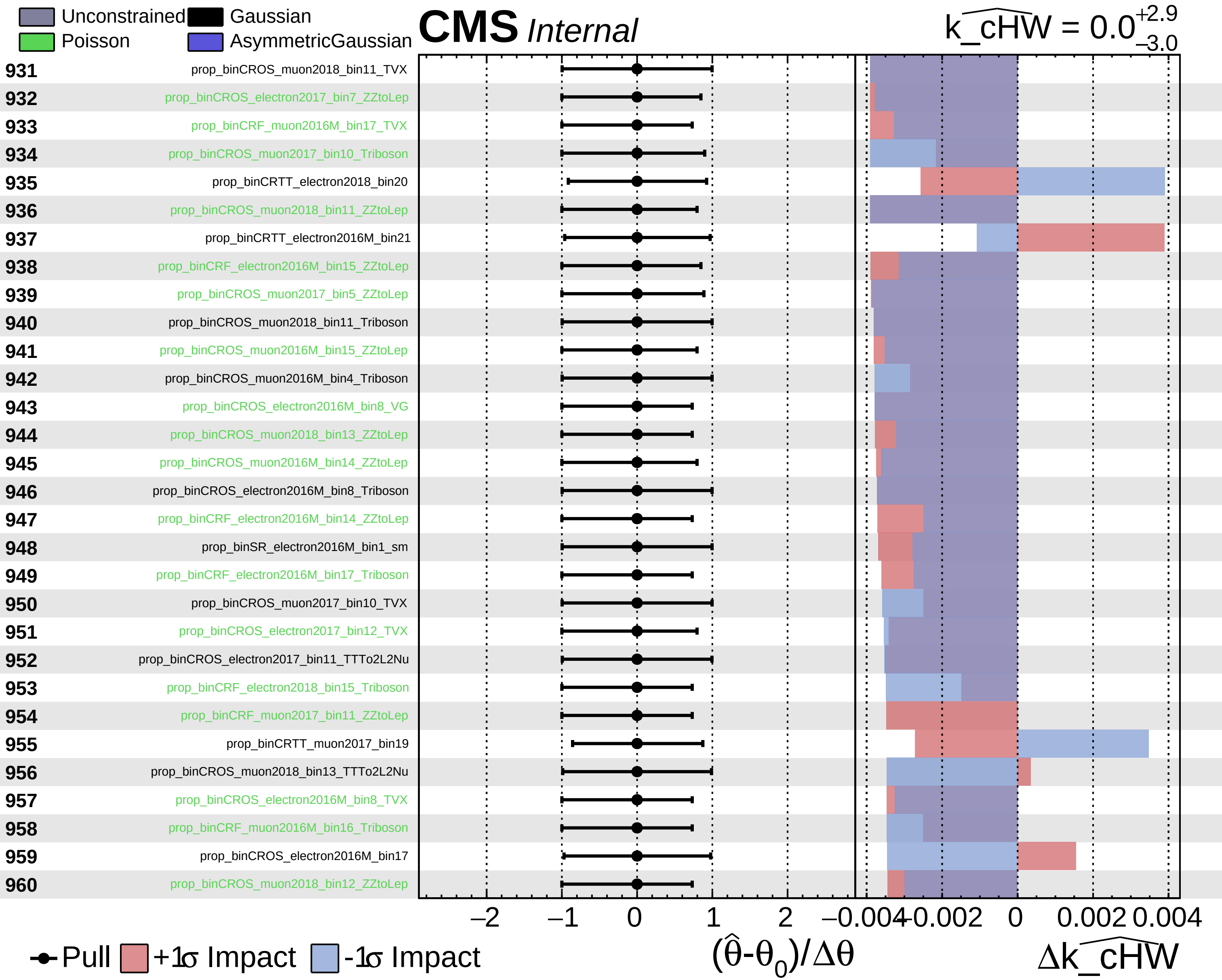


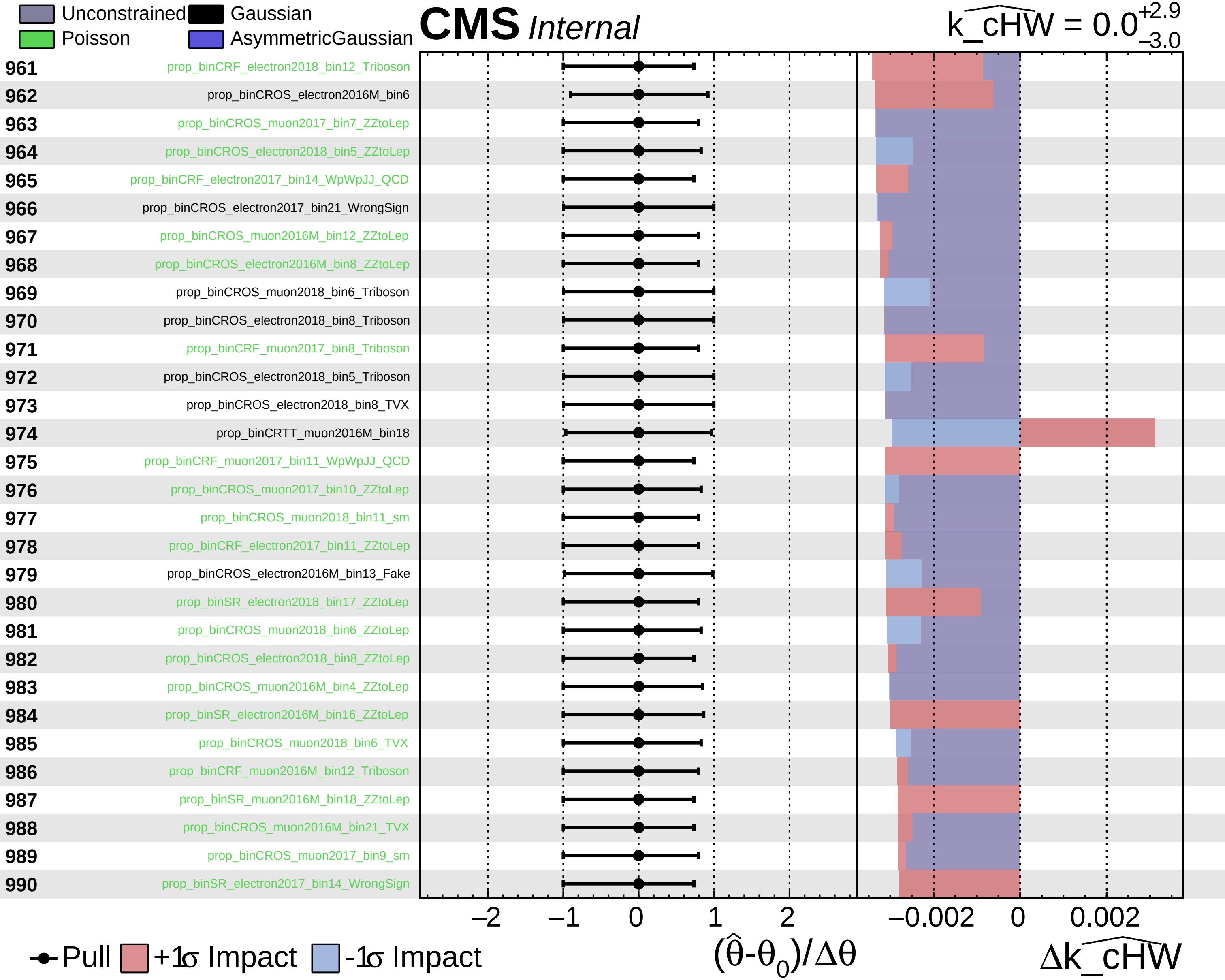
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
841 prop_binCRF_electron2018_bin5_ZZtoLep
842 prop_binCROS_muon2018_bin13_Triboson
843 prop_binCROS_electron2016M_bin4_Triboson
844 prop_binCROS_muon2016M_bin6_Triboson
845 prop_binCROS_electron2017_bin7_TTTo2L2Nu
846 prop_binCROS_electron2017_bin13_ZZtoLep
847 prop_binCRTT_muon2018_bin19
848 prop_binCROS_electron2017_bin13_Triboson
849 prop_binCROS_muon2018_bin13_TVX
850 prop_binCROS_electron2018_bin11_TVX
851 prop_binCROS_muon2016M_bin4_TVX
852 prop_binCROS_muon2017_bin12_ZZtoLep
853 prop_binCROS_electron2017_bin13_TTTo2L2Nu
854 prop_binCROS_muon2016M_bin12_WZ
855 prop_binCRTT_muon2016M_bin21
856 prop_binCROS_muon2018_bin11_TTTo2L2Nu
857 prop_binCROS_muon2018_bin13_Fake
858 prop_binCROS_muon2018_bin18_ZZtoLep
859 prop_binCRF_muon2016M_bin12_TVX
860 prop_binCROS_electron2018_bin8_TTTo2L2Nu
861 prop_binCROS_electron2018_bin20_TVX
862 prop_binCROS_electron2016M_bin4_ZZtoLep
863 prop_binCRF_electron2016M_bin10_TVX
864 prop_binCRF_muon2016M_bin12_WrongSign
865 prop_binCROS_muon2018_bin20_ZZtoLep
866 prop_binCROS_electron2016M_bin5_Triboson
867 prop_binCRF_electron2018_bin5_TVX
868 prop_binCROS_electron2018_bin5_TVX
869 prop_binSR_electron2016M_bin1_Triboson
870 prop_binCROS_muon2016M_bin15_VG
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW

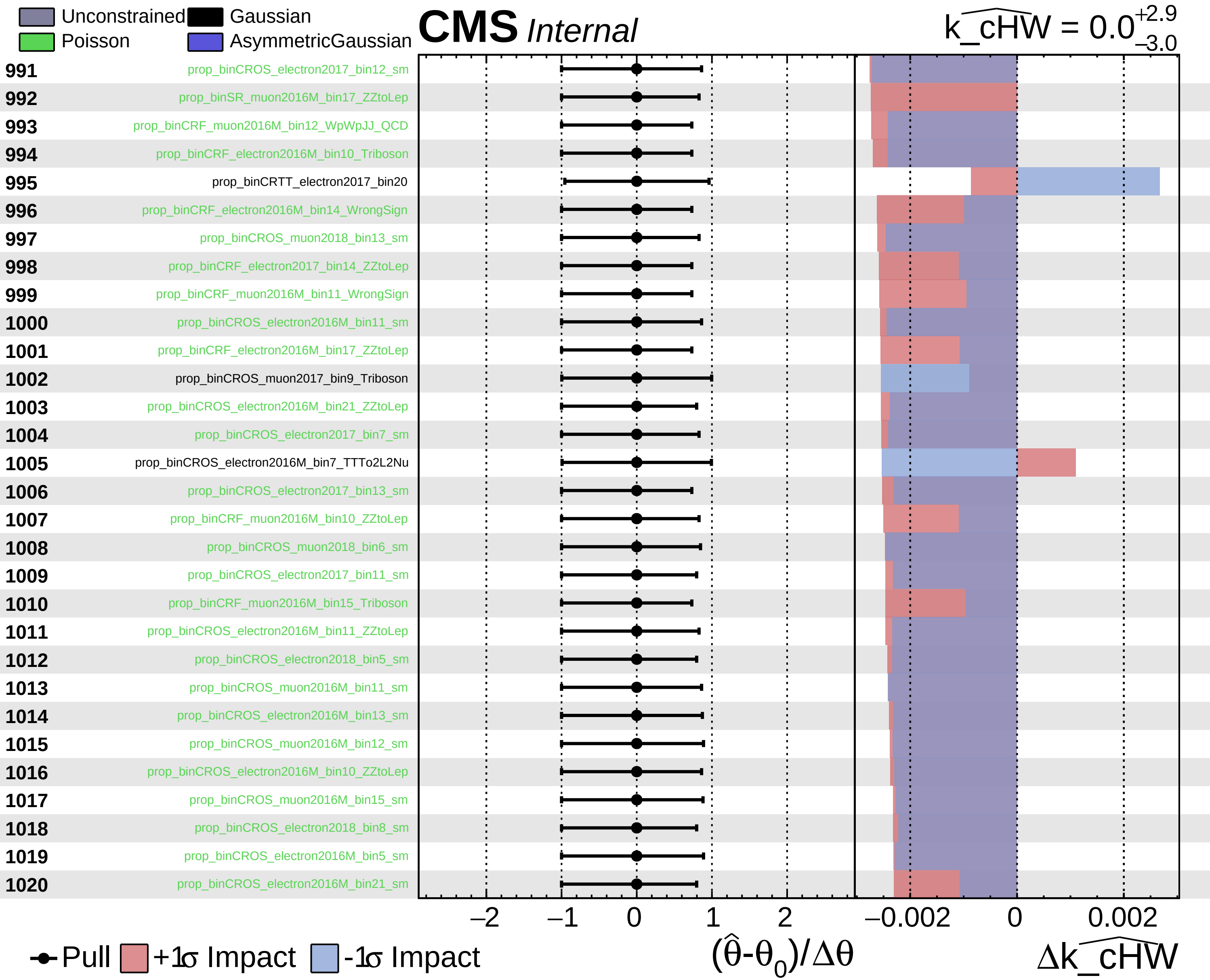

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
871 prop_binCROS_muon2018_bin12_VG
872 prop_binCROS_electron2018_bin15_TVX
873 prop_binCROS_muon2016M_bin6_TVX
874 prop_binCROS_electron2016M_bin13_Triboson
875 prop_binCROS_electron2018_bin15_ZZtoLep
876 prop_binCRF_muon2018_bin12_Triboson
877 prop_binCROS_muon2016M_bin15_Triboson
878 prop_binCROS_electron2016M_bin13_TVX
879 prop_binCROS_electron2018_bin15_Triboson
880 prop_binCROS_electron2016M_bin7_ZZtoLep
881 prop_binCROS_muon2017_bin9_ZZtoLep
882 prop_binCROS_electron2016M_bin13_TTTo2L2Nu
883 prop_binCROS_muon2018_bin12_TVX
884 prop_binCROS_electron2016M_bin7_TVX
885 prop_binCROS_electron2016M_bin10_TVX
886 prop_binCROS_muon2016M_bin17
887 prop_binCROS_muon2018_bin12_Triboson
888 prop_binCROS_muon2016M_bin12_Triboson
889 prop_binCROS_electron2016M_bin8_TTTo2L2Nu
890 prop_binCROS_electron2016M_bin10_Triboson
891 prop_binCROS_electron2017_bin11_Triboson
892 prop_binCROS_muon2018_bin12_TTTo2L2Nu
893 prop_binCROS_muon2016M_bin4_WZ
894 prop_binCROS_electron2017_bin11_ZZtoLep
895 prop_binCROS_electron2017_bin7_WZ
896 prop_binCROS_electron2017_bin7_Triboson
897 prop_binCROS_electron2016M_bin7_Triboson
898 prop_binCROS_electron2017_bin7_VG
899 prop_binCROS_electron2017_bin11_TVX
900 prop_binCRF_electron2018_bin5_WpWpJJ_QCD
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



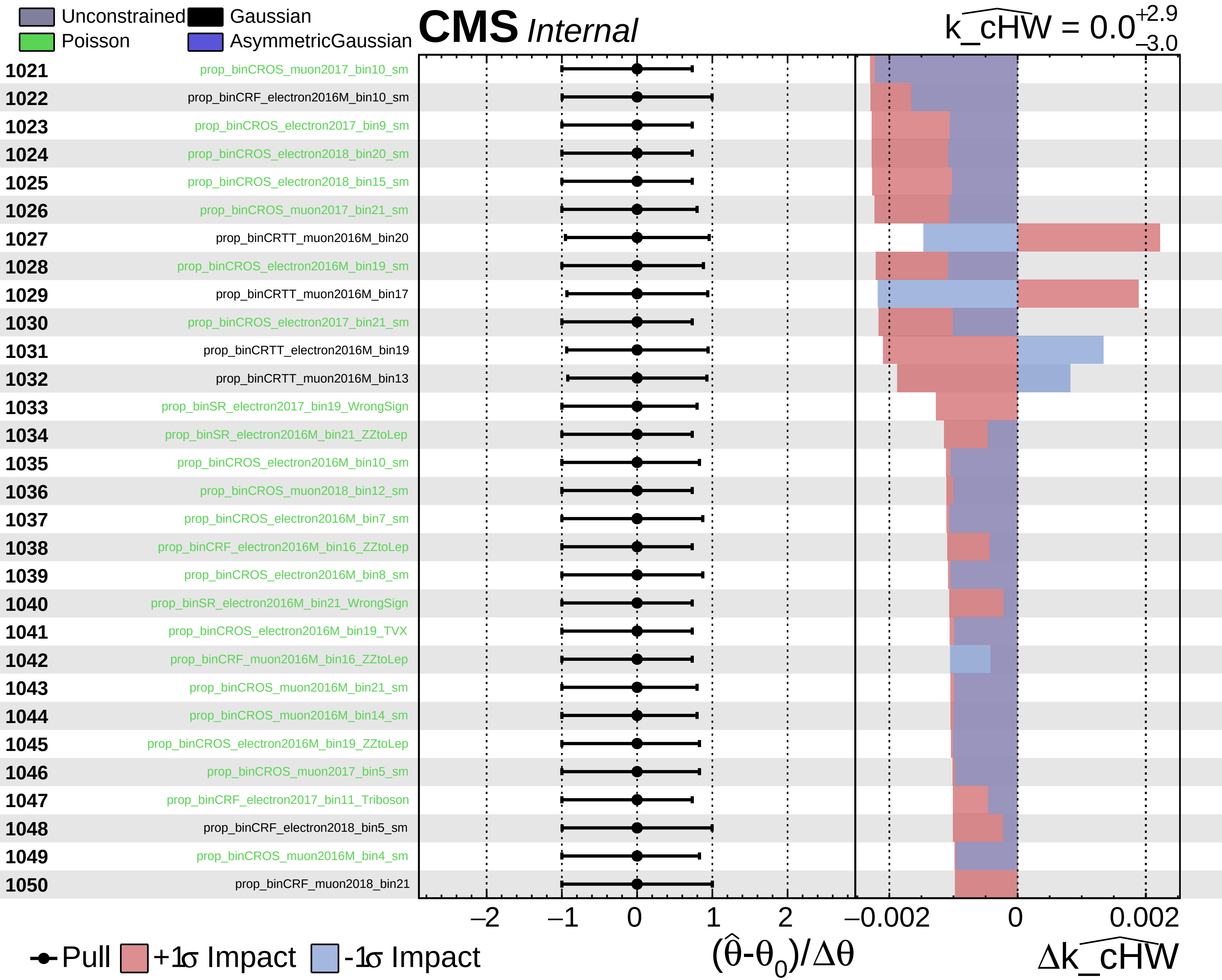
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
901 prop_binCROS_muon2017_bin6_VG
902 prop_binCROS_muon2017_bin6_TTTo2L2Nu
903 prop_binCRF_electron2018_bin5_Triboson
904 prop_binCROS_electron2017_bin13_TVX
905 prop_binCROS_muon2017_bin7_Triboson
906 prop_binCROS_muon2016M_bin11_ZZtoLep
907 prop_binCROS_electron2016M_bin5_TVX
908 prop_binCROS_muon2017_bin6_TVX
909 prop_binCROS_muon2017_bin11_ZZtoLep
910 prop_binSR_electron2017_bin16_ZZtoLep
911 prop_binCRTT_muon2016M_bin16
912 prop_binCROS_muon2017_bin6_ZZtoLep
913 prop_binCROS_muon2017_bin17_sm
914 prop_binCROS_muon2017_bin6_Triboson
915 prop_binCROS_electron2017_bin7_TVX
916 prop_binCROS_muon2017_bin5_VG
917 prop_binCROS_muon2016M_bin14_TVX
918 prop_binCROS_muon2016M_bin12_TVX
919 prop_binCROS_muon2016M_bin14_Triboson
920 prop_binCROS_electron2018_bin11_ZZtoLep
921 prop_binCROS_electron2016M_bin4_TVX
922 prop_binCROS_electron2016M_bin10_TTTo2L2Nu
923 prop_binCROS_muon2017_bin5_Triboson
924 prop_binCROS_muon2017_bin5_TVX
925 prop_binCROS_electron2016M_bin13_ZZtoLep
926 prop_binCROS_electron2018_bin5_TTTo2L2Nu
927 prop_binCROS_muon2016M_bin21_WrongSign
928 prop_binCROS_electron2018_bin11_Triboson
929 prop_binCRTT_muon2016M_bin19
930 prop_binCRTT_electron2016M_bin6
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
931 prop_binCROS_muon2018_bin11_TVX
932 prop_binCROS_electron2017_bin7_ZZtoLep
933 prop_binCRF_muon2016M_bin17_TVX
934 prop_binCROS_muon2017_bin10_Triboson
935 prop_binCRTT_electron2018_bin20
936 prop_binCROS_muon2018_bin11_ZZtoLep
937 prop_binCRTT_electron2016M_bin21
938 prop_binCRF_electron2016M_bin15_ZZtoLep
939 prop_binCROS_muon2017_bin5_ZZtoLep
940 prop_binCROS_muon2018_bin11_Triboson
941 prop_binCROS_muon2016M_bin15_ZZtoLep
942 prop_binCROS_muon2016M_bin4_Triboson
943 prop_binCROS_electron2016M_bin8_VG
944 prop_binCROS_muon2018_bin13_ZZtoLep
945 prop_binCROS_muon2016M_bin14_ZZtoLep
946 prop_binCROS_electron2016M_bin8_Triboson
947 prop_binCRF_electron2016M_bin14_ZZtoLep
948 prop_binSR_electron2016M_bin1_sm
949 prop_binCRF_electron2016M_bin17_Triboson
950 prop_binCROS_muon2017_bin10_TVX
951 prop_binCROS_electron2017_bin12_TVX
952 prop_binCROS_electron2017_bin11_TTTo2L2Nu
953 prop_binCRF_electron2018_bin15_Triboson
954 prop_binCRF_muon2017_bin11_ZZtoLep
955 prop_binCRTT_muon2017_bin19
956 prop_binCROS_muon2018_bin13_TTTo2L2Nu
957 prop_binCROS_electron2016M_bin8_TVX
958 prop_binCRF_muon2016M_bin16_Triboson
959 prop_binCROS_electron2016M_bin17
960 prop_binCROS_muon2018_bin12_ZZtoLep
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

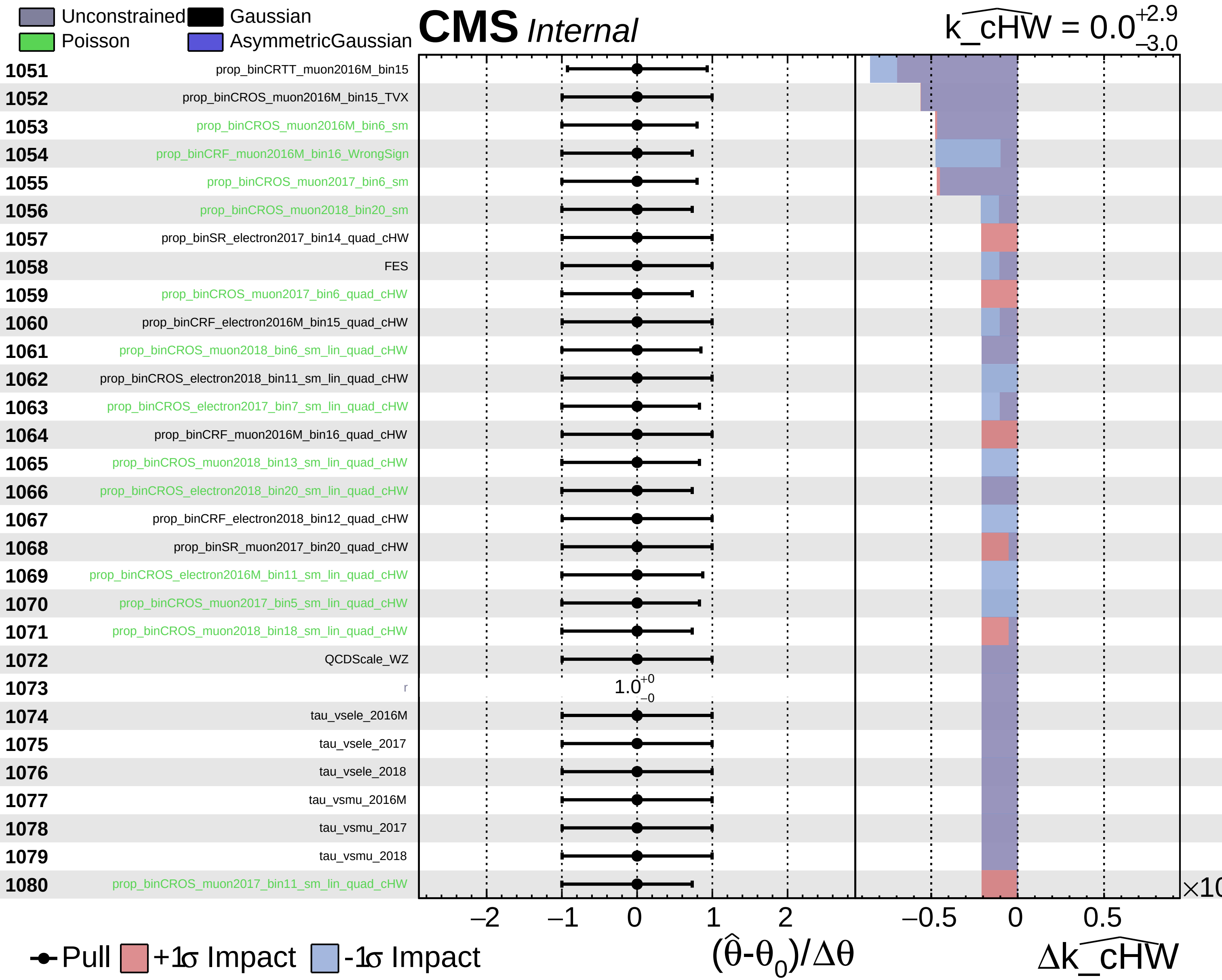

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0+2.9 −3.0
961 prop_binCRF_electron2018_bin12_Triboson
962 prop_binCROS_electron2016M_bin6
963 prop_binCROS_muon2017_bin7_ZZtoLep
964 prop_binCROS_electron2018_bin5_ZZtoLep
965 prop_binCRF_electron2017_bin14_WpWpJJ_QCD
966 prop_binCROS_electron2017_bin21_WrongSign
967 prop_binCROS_muon2016M_bin12_ZZtoLep
968 prop_binCROS_electron2016M_bin8_ZZtoLep
969 prop_binCROS_muon2018_bin6_Triboson
970 prop_binCROS_electron2018_bin8_Triboson
971 prop_binCRF_muon2017_bin8_Triboson
972 prop_binCROS_electron2018_bin5_Triboson
973 prop_binCROS_electron2018_bin8_TVX
974 prop_binCRTT_muon2016M_bin18
975 prop_binCRF_muon2017_bin11_WpWpJJ_QCD
976 prop_binCROS_muon2017_bin10_ZZtoLep
977 prop_binCROS_muon2018_bin11_sm
978 prop_binCRF_electron2017_bin11_ZZtoLep
979 prop_binCROS_electron2016M_bin13_Fake
980 prop_binSR_electron2018_bin17_ZZtoLep
981 prop_binCROS_muon2018_bin6_ZZtoLep
982 prop_binCROS_electron2018_bin8_ZZtoLep
983 prop_binCROS_muon2016M_bin4_ZZtoLep
984 prop_binSR_electron2016M_bin16_ZZtoLep
985 prop_binCROS_muon2018_bin6_TVX
986 prop_binCRF_muon2016M_bin12_Triboson
987 prop_binSR_muon2016M_bin18_ZZtoLep
988 prop_binCROS_muon2016M_bin21_TVX
989 prop_binCROS_muon2017_bin9_sm
990 prop_binSR_electron2017_bin14_WrongSign
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

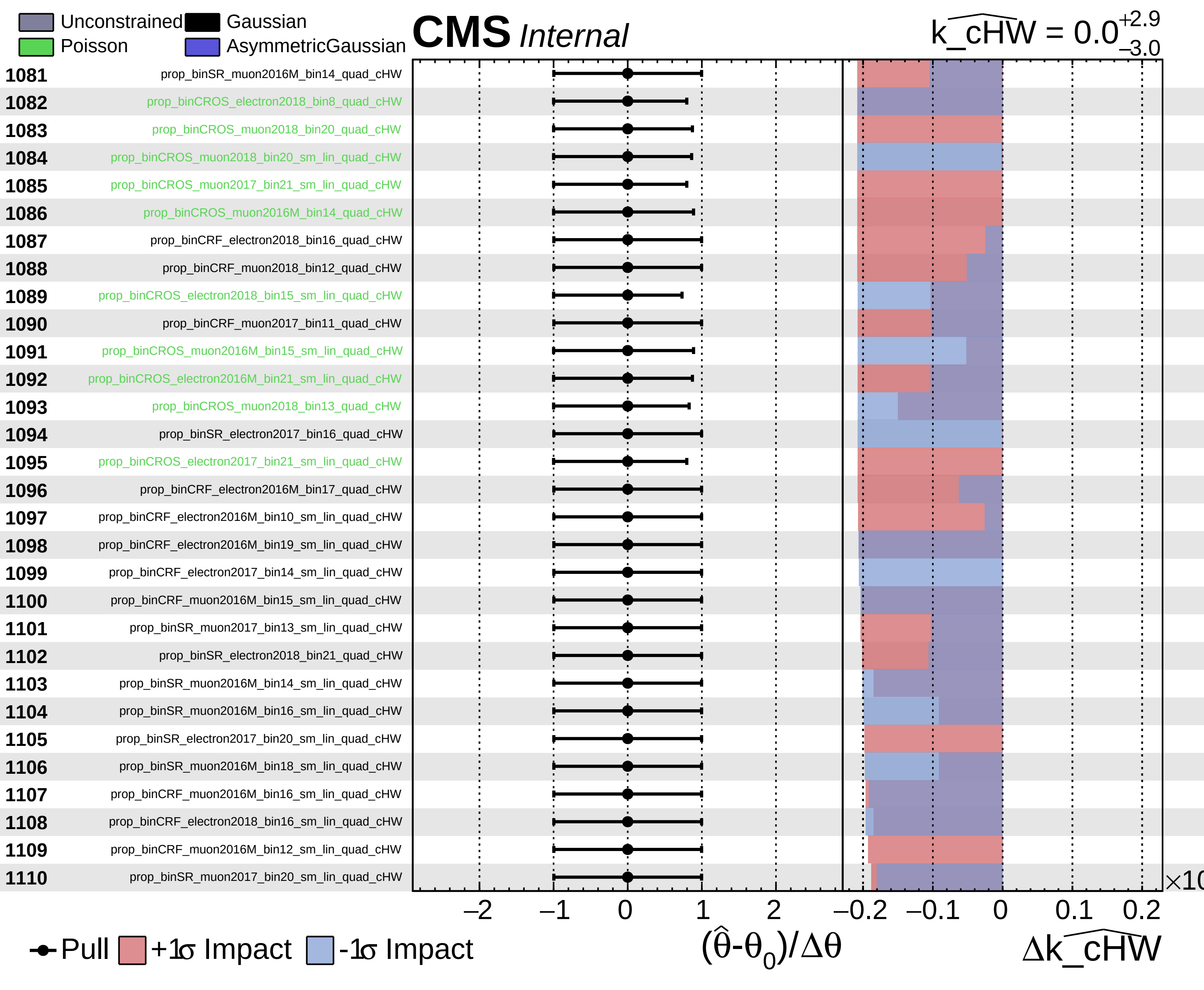


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
991 prop_binCROS_electron2017_bin12_sm
992 prop_binSR_muon2016M_bin17_ZZtoLep
993 prop_binCRF_muon2016M_bin12_WpWpJJ_QCD
994 prop_binCRF_electron2016M_bin10_Triboson
995 prop_binCRTT_electron2017_bin20
996 prop_binCRF_electron2016M_bin14_WrongSign
997 prop_binCROS_muon2018_bin13_sm
998 prop_binCRF_electron2017_bin14_ZZtoLep
999 prop_binCRF_muon2016M_bin11_WrongSign
1000 prop_binCROS_electron2016M_bin11_sm
1001 prop_binCRF_electron2016M_bin17_ZZtoLep
1002 prop_binCROS_muon2017_bin9_Triboson
1003 prop_binCROS_electron2016M_bin21_ZZtoLep
1004 prop_binCROS_electron2017_bin7_sm
1005 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
1006 prop_binCROS_electron2017_bin13_sm
1007 prop_binCRF_muon2016M_bin10_ZZtoLep
1008 prop_binCROS_muon2018_bin6_sm
1009 prop_binCROS_electron2017_bin11_sm
1010 prop_binCRF_muon2016M_bin15_Triboson
1011 prop_binCROS_electron2016M_bin11_ZZtoLep
1012 prop_binCROS_electron2018_bin5_sm
1013 prop_binCROS_muon2016M_bin11_sm
1014 prop_binCROS_electron2016M_bin13_sm
1015 prop_binCROS_muon2016M_bin12_sm
1016 prop_binCROS_electron2016M_bin10_ZZtoLep
1017 prop_binCROS_muon2016M_bin15_sm
1018 prop_binCROS_electron2018_bin8_sm
1019 prop_binCROS_electron2016M_bin5_sm
1020 prop_binCROS_electron2016M_bin21_sm
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW

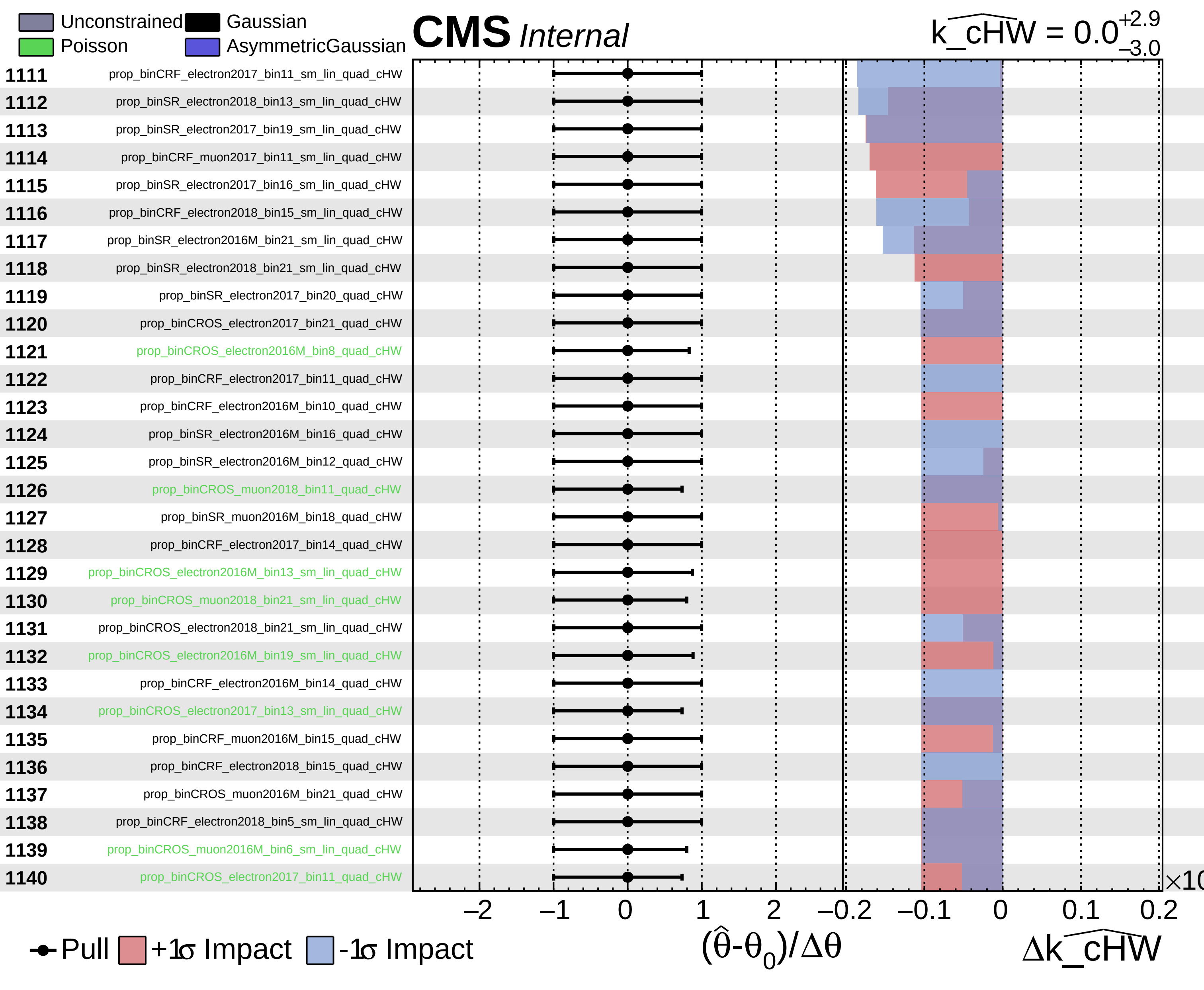


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
1021 prop_binCROS_muon2017_bin10_sm
1022 prop_binCRF_electron2016M_bin10_sm
1023 prop_binCROS_electron2017_bin9_sm
1024 prop_binCROS_electron2018_bin20_sm
1025 prop_binCROS_electron2018_bin15_sm
1026 prop_binCROS_muon2017_bin21_sm
1027 prop_binCRTT_muon2016M_bin20
1028 prop_binCROS_electron2016M_bin19_sm
1029 prop_binCRTT_muon2016M_bin17
1030 prop_binCROS_electron2017_bin21_sm
1031 prop_binCRTT_electron2016M_bin19
1032 prop_binCRTT_muon2016M_bin13
1033 prop_binSR_electron2017_bin19_WrongSign
1034 prop_binSR_electron2016M_bin21_ZZtoLep
1035 prop_binCROS_electron2016M_bin10_sm
1036 prop_binCROS_muon2018_bin12_sm
1037 prop_binCROS_electron2016M_bin7_sm
1038 prop_binCRF_electron2016M_bin16_ZZtoLep
1039 prop_binCROS_electron2016M_bin8_sm
1040 prop_binSR_electron2016M_bin21_WrongSign
1041 prop_binCROS_electron2016M_bin19_TVX
1042 prop_binCRF_muon2016M_bin16_ZZtoLep
1043 prop_binCROS_muon2016M_bin21_sm
1044 prop_binCROS_muon2016M_bin14_sm
1045 prop_binCROS_electron2016M_bin19_ZZtoLep
1046 prop_binCROS_muon2017_bin5_sm
1047 prop_binCRF_electron2017_bin11_Triboson
1048 prop_binCRF_electron2018_bin5_sm
1049 prop_binCROS_muon2016M_bin4_sm
1050 prop_binCRF_muon2018_bin21
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

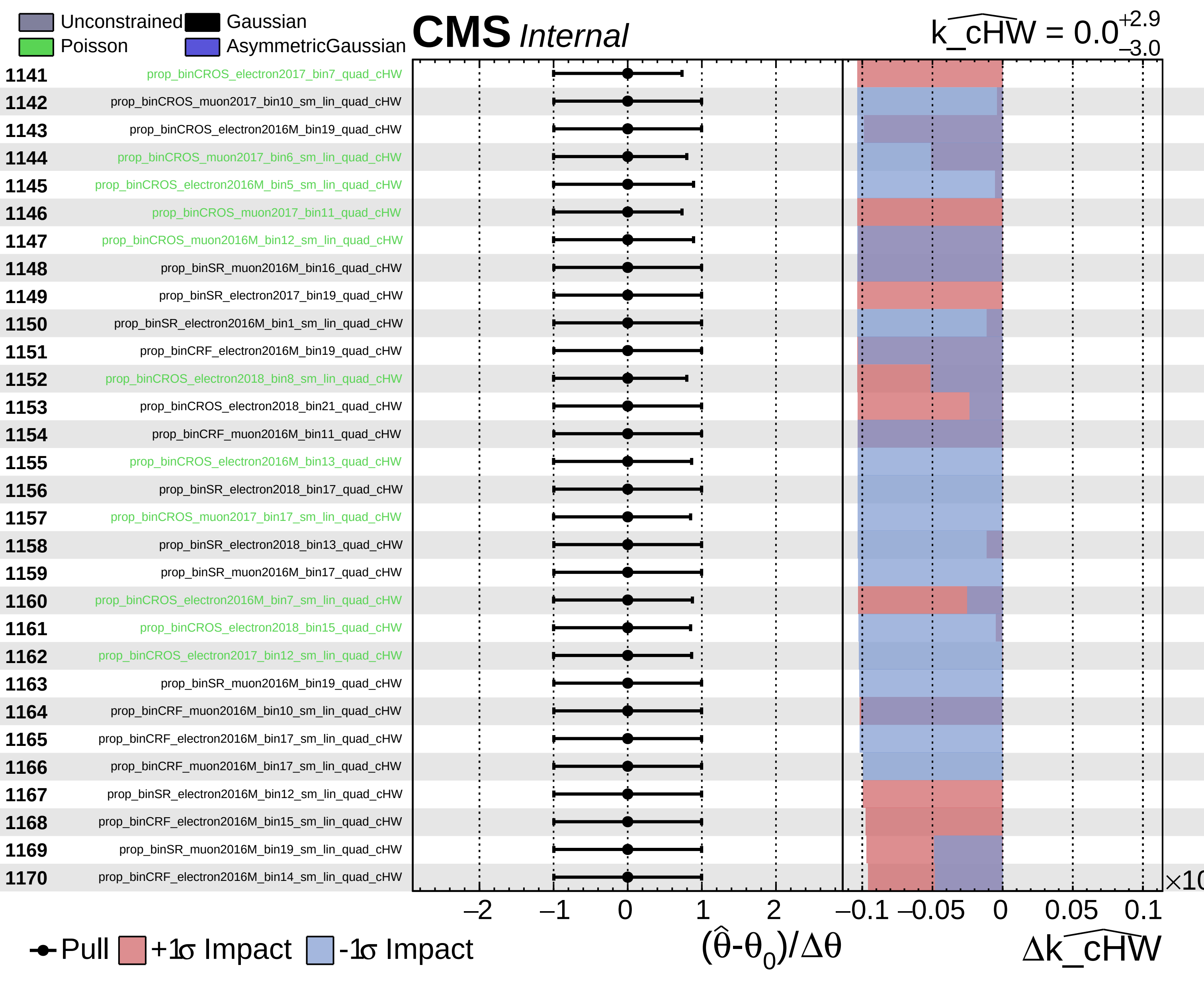

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = $0.0^{+2.9}_{-3.0}$
1051 prop_binCRTT_muon2016M_bin15
1052 prop_binCROS_muon2016M_bin15_TVX
1053 prop_binCROS_muon2016M_bin6_sm
1054 prop_binCRF_muon2016M_bin16_WrongSign
1055 prop_binCROS_muon2017_bin6_sm
1056 prop_binCROS_muon2018_bin20_sm
1057 prop_binSR_electron2017_bin14_quad_cHW
1058 FES
1059 prop_binCROS_muon2017_bin6_quad_cHW
1060 prop_binCRF_electron2016M_bin15_quad_cHW
1061 prop_binCROS_muon2018_bin6_sm_lin_quad_cHW
1062 prop_binCROS_electron2018_bin11_sm_lin_quad_cHW
1063 prop_binCROS_electron2017_bin7_sm_lin_quad_cHW
1064 prop_binCRF_muon2016M_bin16_quad_cHW
1065 prop_binCROS_muon2018_bin13_sm_lin_quad_cHW
1066 prop_binCROS_electron2018_bin20_sm_lin_quad_cHW
1067 prop_binCRF_electron2018_bin12_quad_cHW
1068 prop_binSR_muon2017_bin20_quad_cHW
1069 prop_binCROS_electron2016M_bin11_sm_lin_quad_cHW
1070 prop_binCROS_muon2017_bin5_sm_lin_quad_cHW
1071 prop_binCROS_muon2018_bin18_sm_lin_quad_cHW
1072 QCDScale_WZ
1073 r
1.0^{+0}_{-0}
1074 tau_vsele_2016M
1075 tau_vsele_2017
1076 tau_vsele_2018
1077 tau_vsmu_2016M
1078 tau_vsmu_2017
1079 tau_vsmu_2018
1080 prop_binCROS_muon2017_bin11_sm_lin_quad_cHW
−2
−1
0
1
2
−0.5
0
0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW



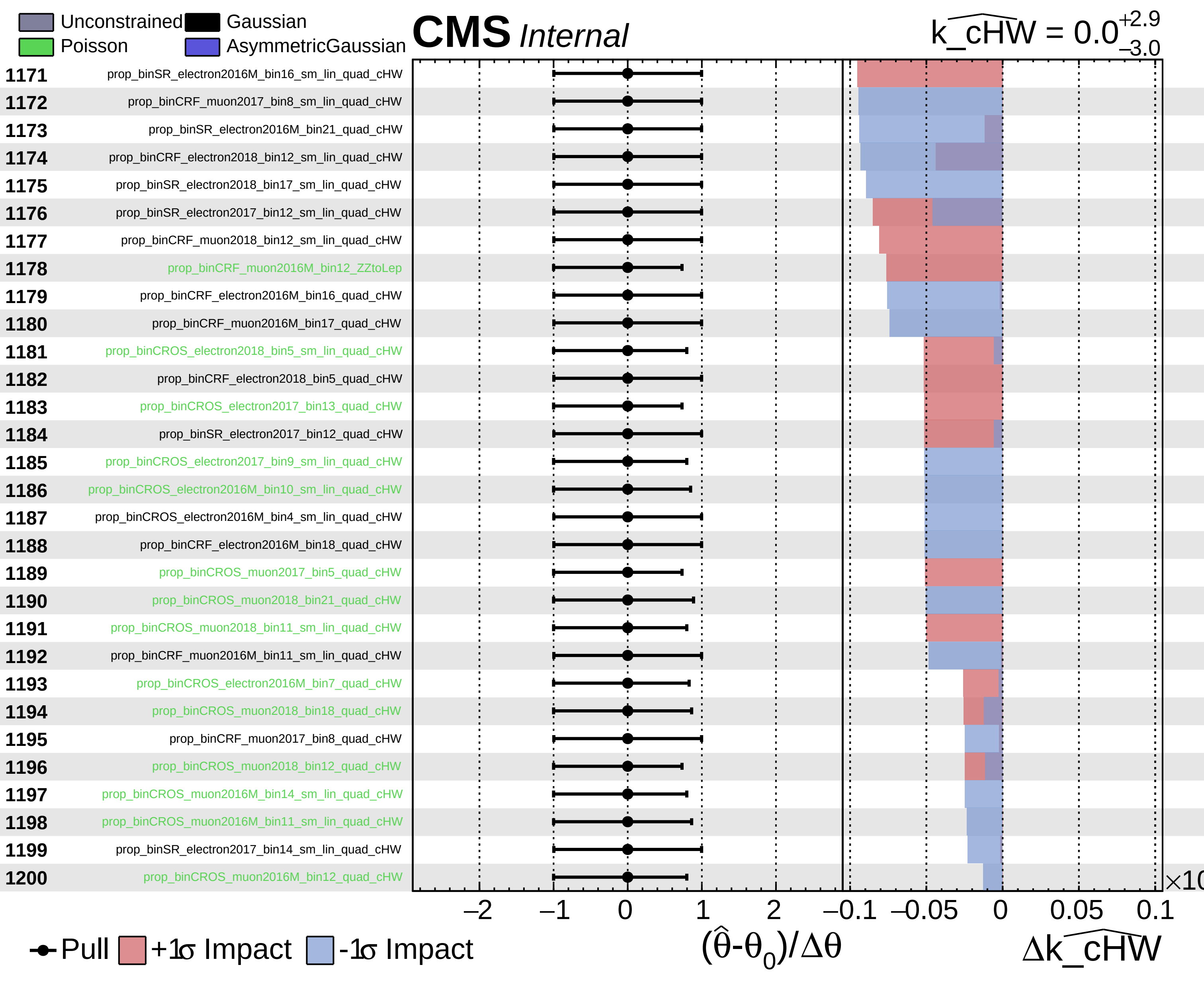
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
1081 prop_binSR_muon2016M_bin14_quad_cHW
1082 prop_binCROS_electron2018_bin8_quad_cHW
1083 prop_binCROS_muon2018_bin20_quad_cHW
1084 prop_binCROS_muon2018_bin20_sm_lin_quad_cHW
1085 prop_binCROS_muon2017_bin21_sm_lin_quad_cHW
1086 prop_binCROS_muon2016M_bin14_quad_cHW
1087 prop_binCRF_electron2018_bin16_quad_cHW
1088 prop_binCRF_muon2018_bin12_quad_cHW
1089 prop_binCROS_electron2018_bin15_sm_lin_quad_cHW
1090 prop_binCRF_muon2017_bin11_quad_cHW
1091 prop_binCROS_muon2016M_bin15_sm_lin_quad_cHW
1092 prop_binCROS_electron2016M_bin21_sm_lin_quad_cHW
1093 prop_binCROS_muon2018_bin13_quad_cHW
1094 prop_binSR_electron2017_bin16_quad_cHW
1095 prop_binCROS_electron2017_bin21_sm_lin_quad_cHW
1096 prop_binCRF_electron2016M_bin17_quad_cHW
1097 prop_binCRF_electron2016M_bin10_sm_lin_quad_cHW
1098 prop_binCRF_electron2016M_bin19_sm_lin_quad_cHW
1099 prop_binCRF_electron2017_bin14_sm_lin_quad_cHW
1100 prop_binCRF_muon2016M_bin15_sm_lin_quad_cHW
1101 prop_binSR_muon2017_bin13_sm_lin_quad_cHW
1102 prop_binSR_electron2018_bin21_quad_cHW
1103 prop_binSR_muon2016M_bin14_sm_lin_quad_cHW
1104 prop_binSR_muon2016M_bin16_sm_lin_quad_cHW
1105 prop_binSR_electron2017_bin20_sm_lin_quad_cHW
1106 prop_binSR_muon2016M_bin18_sm_lin_quad_cHW
1107 prop_binCRF_muon2016M_bin16_sm_lin_quad_cHW
1108 prop_binCRF_electron2018_bin16_sm_lin_quad_cHW
1109 prop_binCRF_muon2016M_bin12_sm_lin_quad_cHW
1110 prop_binSR_muon2017_bin20_sm_lin_quad_cHW
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

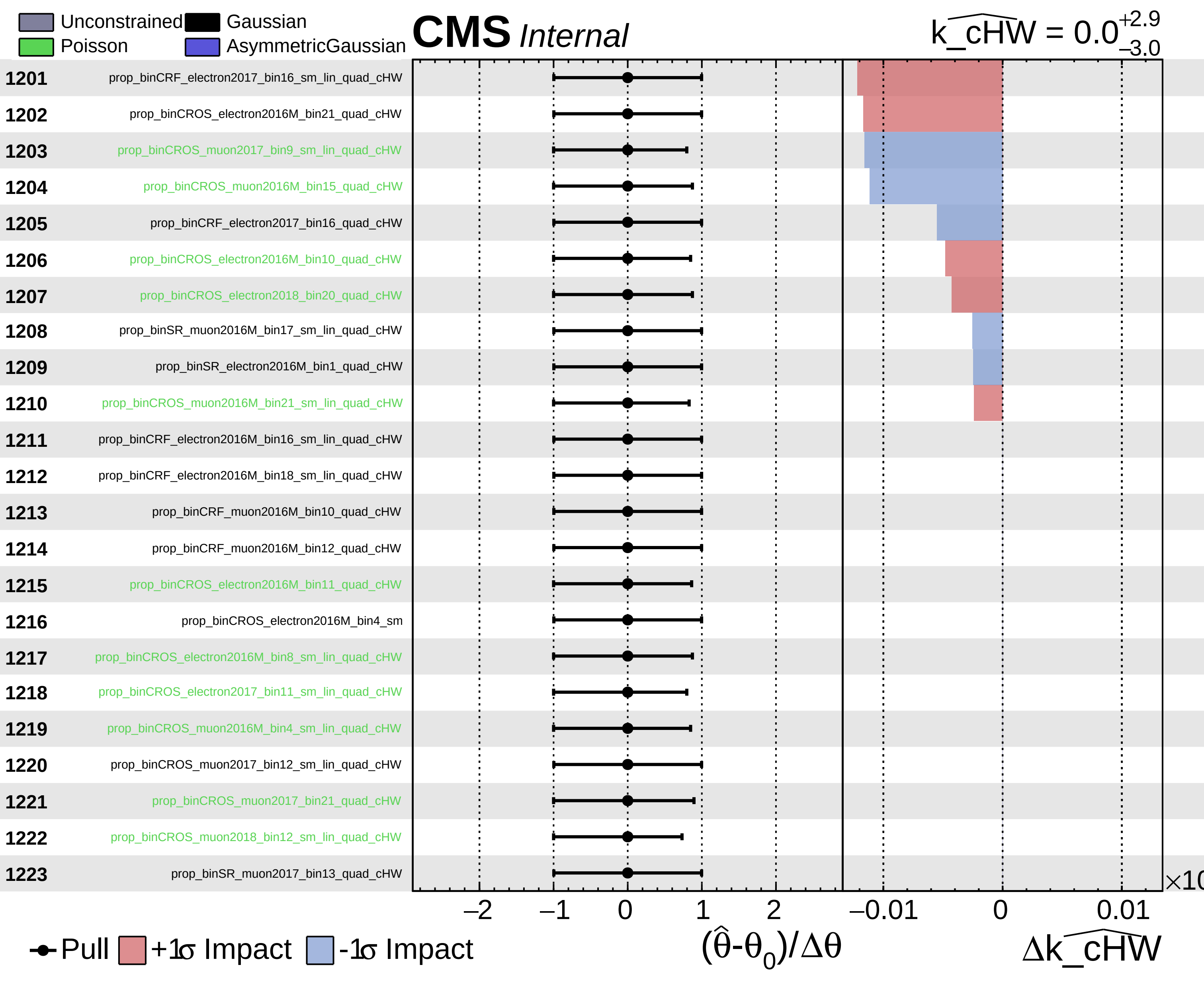


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
1111 prop_binCRF_electron2017_bin11_sm_lin_quad_cHW
1112 prop_binSR_electron2018_bin13_sm_lin_quad_cHW
1113 prop_binSR_electron2017_bin19_sm_lin_quad_cHW
1114 prop_binCRF_muon2017_bin11_sm_lin_quad_cHW
1115 prop_binSR_electron2017_bin16_sm_lin_quad_cHW
1116 prop_binCRF_electron2018_bin15_sm_lin_quad_cHW
1117 prop_binSR_electron2016M_bin21_sm_lin_quad_cHW
1118 prop_binSR_electron2018_bin21_sm_lin_quad_cHW
1119 prop_binSR_electron2017_bin20_quad_cHW
1120 prop_binCROS_electron2017_bin21_quad_cHW
1121 prop_binCROS_electron2016M_bin8_quad_cHW
1122 prop_binCRF_electron2017_bin11_quad_cHW
1123 prop_binCRF_electron2016M_bin10_quad_cHW
1124 prop_binSR_electron2016M_bin16_quad_cHW
1125 prop_binSR_electron2016M_bin12_quad_cHW
1126 prop_binCROS_muon2018_bin11_quad_cHW
1127 prop_binSR_muon2016M_bin18_quad_cHW
1128 prop_binCRF_electron2017_bin14_quad_cHW
1129 prop_binCROS_electron2016M_bin13_sm_lin_quad_cHW
1130 prop_binCROS_muon2018_bin21_sm_lin_quad_cHW
1131 prop_binCROS_electron2018_bin21_sm_lin_quad_cHW
1132 prop_binCROS_electron2016M_bin19_sm_lin_quad_cHW
1133 prop_binCRF_electron2016M_bin14_quad_cHW
1134 prop_binCROS_electron2017_bin13_sm_lin_quad_cHW
1135 prop_binCRF_muon2016M_bin15_quad_cHW
1136 prop_binCRF_electron2018_bin15_quad_cHW
1137 prop_binCROS_muon2016M_bin21_quad_cHW
1138 prop_binCRF_electron2018_bin5_sm_lin_quad_cHW
1139 prop_binCROS_muon2016M_bin6_sm_lin_quad_cHW
1140 prop_binCROS_electron2017_bin11_quad_cHW
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
1141 prop_binCROS_electron2017_bin7_quad_cHW
1142 prop_binCROS_muon2017_bin10_sm_lin_quad_cHW
1143 prop_binCROS_electron2016M_bin19_quad_cHW
1144 prop_binCROS_muon2017_bin6_sm_lin_quad_cHW
1145 prop_binCROS_electron2016M_bin5_sm_lin_quad_cHW
1146 prop_binCROS_muon2017_bin11_quad_cHW
1147 prop_binCROS_muon2016M_bin12_sm_lin_quad_cHW
1148 prop_binSR_muon2016M_bin16_quad_cHW
1149 prop_binSR_electron2017_bin19_quad_cHW
1150 prop_binSR_electron2016M_bin1_sm_lin_quad_cHW
1151 prop_binCRF_electron2016M_bin19_quad_cHW
1152 prop_binCROS_electron2018_bin8_sm_lin_quad_cHW
1153 prop_binCROS_electron2018_bin21_quad_cHW
1154 prop_binCRF_muon2016M_bin11_quad_cHW
1155 prop_binCROS_electron2016M_bin13_quad_cHW
1156 prop_binSR_electron2018_bin17_quad_cHW
1157 prop_binCROS_muon2017_bin17_sm_lin_quad_cHW
1158 prop_binSR_electron2018_bin13_quad_cHW
1159 prop_binSR_muon2016M_bin17_quad_cHW
1160 prop_binCROS_electron2016M_bin7_sm_lin_quad_cHW
1161 prop_binCROS_electron2018_bin15_quad_cHW
1162 prop_binCROS_electron2017_bin12_sm_lin_quad_cHW
1163 prop_binSR_muon2016M_bin19_quad_cHW
1164 prop_binCRF_muon2016M_bin10_sm_lin_quad_cHW
1165 prop_binCRF_electron2016M_bin17_sm_lin_quad_cHW
1166 prop_binCRF_muon2016M_bin17_sm_lin_quad_cHW
1167 prop_binSR_electron2016M_bin12_sm_lin_quad_cHW
1168 prop_binCRF_electron2016M_bin15_sm_lin_quad_cHW
1169 prop_binSR_muon2016M_bin19_sm_lin_quad_cHW
1170 prop_binCRF_electron2016M_bin14_sm_lin_quad_cHW
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +2.9 −3.0
1171 prop_binSR_electron2016M_bin16_sm_lin_quad_cHW
1172 prop_binCRF_muon2017_bin8_sm_lin_quad_cHW
1173 prop_binSR_electron2016M_bin21_quad_cHW
1174 prop_binCRF_electron2018_bin12_sm_lin_quad_cHW
1175 prop_binSR_electron2018_bin17_sm_lin_quad_cHW
1176 prop_binSR_electron2017_bin12_sm_lin_quad_cHW
1177 prop_binCRF_muon2018_bin12_sm_lin_quad_cHW
1178 prop_binCRF_muon2016M_bin12_ZZtoLep
1179 prop_binCRF_electron2016M_bin16_quad_cHW
1180 prop_binCRF_muon2016M_bin17_quad_cHW
1181 prop_binCROS_electron2018_bin5_sm_lin_quad_cHW
1182 prop_binCRF_electron2018_bin5_quad_cHW
1183 prop_binCROS_electron2017_bin13_quad_cHW
1184 prop_binSR_electron2017_bin12_quad_cHW
1185 prop_binCROS_electron2017_bin9_sm_lin_quad_cHW
1186 prop_binCROS_electron2016M_bin10_sm_lin_quad_cHW
1187 prop_binCROS_electron2016M_bin4_sm_lin_quad_cHW
1188 prop_binCRF_electron2016M_bin18_quad_cHW
1189 prop_binCROS_muon2017_bin5_quad_cHW
1190 prop_binCROS_muon2018_bin21_quad_cHW
1191 prop_binCROS_muon2018_bin11_sm_lin_quad_cHW
1192 prop_binCRF_muon2016M_bin11_sm_lin_quad_cHW
1193 prop_binCROS_electron2016M_bin7_quad_cHW
1194 prop_binCROS_muon2018_bin18_quad_cHW
1195 prop_binCRF_muon2017_bin8_quad_cHW
1196 prop_binCROS_muon2018_bin12_quad_cHW
1197 prop_binCROS_muon2016M_bin14_sm_lin_quad_cHW
1198 prop_binCROS_muon2016M_bin11_sm_lin_quad_cHW
1199 prop_binSR_electron2017_bin14_sm_lin_quad_cHW
1200 prop_binCROS_muon2016M_bin12_quad_cHW
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+2.9}_{-3.0}$
1201 prop_binCRF_electron2017_bin16_sm_lin_quad_cHW
1202 prop_binCROS_electron2016M_bin21_quad_cHW
1203 prop_binCROS_muon2017_bin9_sm_lin_quad_cHW
1204 prop_binCROS_muon2016M_bin15_quad_cHW
1205 prop_binCRF_electron2017_bin16_quad_cHW
1206 prop_binCROS_electron2016M_bin10_quad_cHW
1207 prop_binCROS_electron2018_bin20_quad_cHW
1208 prop_binSR_muon2016M_bin17_sm_lin_quad_cHW
1209 prop_binSR_electron2016M_bin1_quad_cHW
1210 prop_binCROS_muon2016M_bin21_sm_lin_quad_cHW
1211 prop_binCRF_electron2016M_bin16_sm_lin_quad_cHW
1212 prop_binCRF_electron2016M_bin18_sm_lin_quad_cHW
1213 prop_binCRF_muon2016M_bin10_quad_cHW
1214 prop_binCRF_muon2016M_bin12_quad_cHW
1215 prop_binCROS_electron2016M_bin11_quad_cHW
1216 prop_binCROS_electron2016M_bin4_sm
1217 prop_binCROS_electron2016M_bin8_sm_lin_quad_cHW
1218 prop_binCROS_electron2017_bin11_sm_lin_quad_cHW
1219 prop_binCROS_muon2016M_bin4_sm_lin_quad_cHW
1220 prop_binCROS_muon2017_bin12_sm_lin_quad_cHW
1221 prop_binCROS_muon2017_bin21_quad_cHW
1222 prop_binCROS_muon2018_bin12_sm_lin_quad_cHW
1223 prop_binSR_muon2017_bin13_quad_cHW
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$